Employment Status and Credit Cycles: An Extensive Margin Analysis in Heterogeneous Agent Life Cycle Models.

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Abstract

This paper addresses a critical gap between structural modeling conventions and empirical evidence regarding labor market adjustment margins. While most structural models emphasize the intensive margin (hours worked per agent), empirical evidence underscores the extensive margin (employment status), which remains overlooked even in models with incomplete markets. This discrepancy is addressed through an analysis of a three period Overlapping Generations (OLG) model with varying youth unemployment and with a Heterogeneous Agent (HA) Life cycle framework, featuring age dependent unemployment distributions and precautionary savings motivations with exogenous labor productivity shocks. Comparative statics in the OLG model demonstrate that variations in the extensive margin substantially reshape credit markets by lowering household debt levels and real interest rates. Within the HA model, real household debt dropped by roughly 3.3% per 1% increase in unemployment and exhibited two key mechanisms: decreased loan demand from young unemployed agents facing borrowing constraints and increased supply from households self-insuring against elevated unemployment risk through precautionary savings. Impulse response analysis reveals that negative productivity shocks induce sharper household deleveraging when labor changes through the extensive margin. The theoretical implications are empirically validated using both Vector Autoregression and Local Projection methods on European data from 1998 to 2019 and US data from 1952 to 2019. These results matter beyond methodology because higher unemployment forces borrowing constrained households to delever, which depresses real interest rates thereby narrowing monetary space, at precisely the moment when marginal utility of consumption is the highest. The findings point to income insurance and automatic stabilizers as a more direct way to mitigate the utility cost of unemployment.

Keywords: Life Cycle Models and Saving, Unemployment Rate, Precautionary Saving
JEL Classification: E21, E24, E44

1 Introduction

When the global financial crisis tore through the world economy, unemployment climbed, interest rates collapsed, and household debt fell. Greece, Italy, Portugal, and Spain bore the deepest scars. Between 2008 and 2014, real household debt fell by 30.9% in Greece, 32.4% in Spain, 8.3% in Italy, and 27.4% in Portugal. Over the same period, overall unemployment rose by 20.5, 18.3, 6.9, and 9.2 percentage points, respectively, while youth unemployment climbed by 41.2, 39.0, 24.2, and 23.6 basis points in those same countries. During this time, the European Central Bank's policy rate tested the zero lower bound. Large increases in unemployment impact households through two channels: lower incomes among the unemployed and tighter borrowing constraints, as lenders were reluctant to lend to the unemployed. By focusing on unemployment, specifically youth unemployment, this paper contributes to the macroeconomic literature. This paper demonstrates that the simultaneous loss of income and access to credit alters the loanable funds market and substantially lowers utility for unemployed youth, who are the primary borrowers.

Introducing overlapping generations dynamics and labor that fluctuates over the extensive margin poses a challenge for workhorse macroeconomic models, as they are not designed to capture these features. The RBC and New Keynesian models characterize aggregate fluctuations while abstracting from household heterogeneity and the demographic structure through which intergenerational lending operates, since they rely on representative agents who neither borrow from nor lend to one another across age groups. Heterogeneous agent and HANK models have made significant advances in linking wealth distributions to aggregate dynamics. However, their labor adjustment typically operates on the intensive margin, having hours respond continuously to productivity shocks, while the distribution of unemployment itself rarely changes. This choice rules out the very phenomenon at the heart of aggregate household debt deleveraging. Recessions hit young workers first and hardest because they are the newest entrants to the labor market and often the first to be laid off. Youth unemployment rose far more than aggregate unemployment throughout the Global Financial/Eurozone crisis, reaching 61.4%, 56.2%, 43.2%, and 42.1% in Greece, Spain, Italy, and Portugal, respectively, according to Eurostat. These dynamics matter for the loanable funds market because young people are the primary borrowers, often entering adulthood with few assets and lower incomes. The youth borrow to smooth consumption before their earnings rise with age. A young agent's transition to unemployment involves moving from a state of employment with access to credit to one of unemployment and borrowing constraints. This simultaneous loss of income and credit availability represents a double shock. Models that do not operate through the extensive margin, therefore, are not designed to generate the endogenous tightening of employment based borrowing constraints. The current literature on heterogeneous agent models that vary over the extensive margin largely focuses on precautionary savings by a single type of worker and

does not feature intergenerational household lending, unemployment based borrowing constraints, or employment that varies over the lifecycle. To capture this phenomenon, my model includes two explicit elements: an overlapping generations (OLG) demographic structure and an explicit extensive margin. The OLG framework introduces a specific age structure for intergenerational lending. This structure includes the following key features: young households, the primary borrowers, and middle aged households who save for retirement. Together, the OLG structure and extensive margin labor adjustment enable the model to capture the simultaneous loss of income and access to credit faced by unemployed youth, and how this reshapes the loanable funds market and utility.

This model builds on the endowment and intergenerational lending framework of (Eggertsson, Mehrotra, and Robbins 2019) and extends it by incorporating youth unemployment. The OLG features four agents: employed young, unemployed young, employed middle-aged, and retired old. The key departure from other models is that unemployed youth earn only a fraction of employed youth's income and cannot borrow, whereas the employed youth have a debt constraint tied to their wage income. The model analyzes the impact of exogenous increases in youth unemployment on the loanable funds market and consumer utility.

This model moves beyond OLG toward a more empirically grounded heterogeneous agent lifecycle model, calibrated to BLS income profiles and employment transition dynamics from (Shimer 2012). Its key innovation: firms decide effective labor levels instead of households, paying efficiency wages above market clearing levels, which creates involuntary unemployment. Productivity shocks trigger labor adjustment that will be compared along two distinct margins, extensive (employment) and intensive (hours), tested separately against a Modified Huggett model where agents choose hours. To offset utility losses from unemployment, countercyclical transfer payments are tested as a policy option. This isolates how the extensive margin shapes the loanable funds market and utility, and how lifecycle dynamics make the extensive margin diverge from conventional modeling.

To motivate and discipline modeling choices, I draw on evidence from Europe, where cross country variation exceeds that of aggregate US data. The analysis proceeds in two steps. First, I confirm that fluctuations in aggregate hours are driven predominantly by the extensive margin (employment) and document a positive correlation between employment and real household debt. Second, I use VARs and local projections to model the dynamic responses of weekly hours, the unemployment rate, and real household debt to labor productivity shocks. A one standard deviation productivity shock reduces real household debt significantly by 0.5% in the VAR and 0.8% in the LP, and increases the unemployment rate significantly by 0.25 % in both. The extensive margin model does a better job of explaining movements in unemployment and, subsequently, in real household debt. Countries are split into groups based on whether they are above or below the median for variation in weekly hours worked, unemployment, and real household debt. Countries

with more persistent productivity shocks experienced the larger movements in both unemployment and household debt.

This paper shows that productivity driven recessions harm households primarily through the extensive margin rather than through reduced hours. Unemployment imposes a double shock to households without assets: workers lose income and simultaneously lose the ability to borrow, because lending is tied to employment status. This collapses household debt and substantially lowers utility, with young households hurt most since they depend on borrowing to smooth consumption. Empirically, across Europe, a decline in labor productivity raises unemployment by a statistically significant margin and reduces real household debt by a significant margin, with countries that experience more persistent responses seeing the largest swings in both. The policy implications are threefold. First, demand management should prioritize reemployment over increasing hours worked, since the extensive margin is the key lever for restoring both incomes and household access to credit. Second, the finding that borrowing constraints tighten precisely when households need credit most suggests that policies should focus on increasing unemployment income to boost utility. Finally, the heterogeneity across countries in persistence implies that policy responses should be stronger in countries with longer more persistent shocks. This would require more aggressive stabilization to prevent unemployment and household deleveraging from reinforcing each other.

The paper's format is as follows. Section 2 covers the Literature Review. Section 3 covers the empirical motivation, Beyond Aggregate Data: Exploring labor market variation and debt. Section 4, Isolating the Demand Channel: A simple overlapping generations model, covers the OLG framework. Section 5, Microfoundations of Interagent Lending: Incorporating extensive margin risk into the HA framework, covers the steady state of the HA model. Section 6, Analyzing Dynamic Responses: The impact of productivity shocks in HA framework , covers HA model's impulse response functions. Section 7, Bridging Theory and Data: A VAR/LP investigation into the labor market and household debt, covers empirical IRF.

2 Literature Review

Macroeconomic modeling has undergone significant development since the introduction of the Real Business Cycle (RBC) framework by (Kydland and Prescott 1982) used a setting of fully flexible prices and rational expectations to attribute economic fluctuations primarily to technology shocks. However, this framework offered a limited account of labor market dynamics, particularly by the representative agent's hours worked as the variation in total hours worked (intensive margin). (Heckman 1984) pointed out that the RBC doesn't take into account the fact that a majority of business cycle fluctuations go through the extensive margin.

(Hansen 1985) extended the RBC model by incorporating the extensive margin of labor by modeling employment fluctuations as changes in the number of individuals working rather than just hours per worker. The indivisible labor approach aligned more closely with the observed volatility in employment data. New Keynesian (NK) models gained prominence by introducing nominal rigidities and emphasizing the role of monetary policy. (Rotemberg and Woodford 1997) introduced price stickiness into a DSGE structure, while (Clarida, Gali, and Gertler 1999) advanced this framework to analyze interest rate rules and inflation dynamics with the three equations of NK models. Building on this foundation, (Galí, Smets, and Wouters 2012) incorporated the extensive labor margin into New Keynesian models, improving their ability to capture the cyclical behavior of unemployment. These advances marked an important evolution in macroeconomic theory, integrating richer labor market structures and frictions into models used for policy analysis.

Research into the extensive margin spans both microeconomic and macroeconomic. The microeconomic literature tends to focus on the extensive margin of labor supply. While this paper's model doesn't have shifts to labor supply, the literature consistently supports the conclusion that the extensive margin, changes in employment status, is the primary channel through which labor market adjustments occur. The extensive margin of labor supply is the decision of whether to participate in the labor force and has been a central focus in microeconomic research since the foundational work of (Heckman and MaCurdy 1980). Their analysis emphasized the importance of distinguishing between participation (extensive margin) and hours worked (intensive margin) in a heterogeneous agent life cycle model of women. This contribution laid the groundwork for subsequent empirical studies, such as (Meyer 2002), who showed how policy tools like the Earned Income Tax Credit significantly affect participation decisions among low-income individuals. More recently, (Attanasio, Levell, Low, and Sánchez-Marcos 2015) and (Chetty, Guren, Manoli, and Weber 2013) have estimated labor supply elasticities at the extensive margin, finding that the extensive margin plays a dominant role in explaining overall labor supply responsiveness, particularly for women and lower-income groups. These papers provide microfoundations for the role of the extensive margin in labor market dynamics.

The foundation of heterogeneous agent (HA) macroeconomics was laid by a series of seminal works addressing individual income risk, borrowing constraints, and precautionary saving. (Bewley 1986) developed the theoretical class of models in which agents face uninsurable idiosyncratic income shocks and engage in self-insurance through precautionary saving of government bonds, which formed the basis of incomplete markets in macroeconomics. (Deaton 1991) built on precautionary saving behavior in such an environment, showing how liquidity constraints and income uncertainty drive consumption dynamics. (Imrohoroğlu 1989) quantified the welfare losses due to incomplete markets, highlighting the inefficiencies arising from a lack of full risk-sharing. (Huggett 1993) allowed the borrowing constraint to be negative. This choice meant that agents

could self-insure by saving and by borrowing against future income. (Aiyagari 1994) extended these insights to a general equilibrium setting with capital, demonstrating how precautionary savings affect aggregate outcomes and wealth inequality. (Krusell and Smith 1998) pushed the literature forward by solving a model with idiosyncratic and aggregate risk, showing that a small number of state variables can approximate the distributional dynamics effectively. More recently, the HA literature has moved towards continuous time models and away from the discrete time models that were previously used. (Kaplan and Violante 2014) introduced a two-asset HA model to explain high marginal propensities to consume among "wealthy hand-to-mouth" households. (Kaplan and Violante 2018) review how incorporating microeconomic heterogeneity, such as differences in income and wealth, into macroeconomic models significantly improves our understanding of how households react to aggregate shocks. (Auclert 2019) provided a unifying framework to decompose the transmission of monetary policy into several redistribution channels in a one-asset HANK model, emphasizing how heterogeneity amplifies aggregate demand effects. Together, these papers represent the evolution of HA macro from theoretical foundations to empirically rich models of business cycles and policy transmission.

Heterogeneous agent models have emphasized the role of the extensive margin and how it amplifies macroeconomic fluctuations, particularly through the precautionary savings motive. (Krueger, Mitman, and Perri 2016) incorporate an endogenous extensive margin that changes labor transition dynamics based on the state of the economy into a quantitative model of the U.S. labor market, showing that unemployment risk and borrowing constraints substantially affect consumption and savings decisions. (Rendahl 2016) examines fiscal policy during periods of deep unemployment, demonstrating that government spending is particularly effective when households face elevated job uncertainty. (Challe, Matheron, Ragot, and Rubio-Ramirez 2017) build a HANK model where employment status is endogenous, and aggregate demand is driven by fluctuations in precautionary savings: when uncertainty rises, agents increase precautionary savings, which lowers real interest rates, deepening recessions. (McKay and Reis 2016) highlight the stabilizing role of automatic fiscal stabilizers, like unemployment insurance, within a HANK framework where labor market frictions and the extensive margin have a critical role in dampening shocks. (Ravn and Sterk 2017) further develop this narrative by showing how job uncertainty can endogenously generate demand-driven recessions. Collectively, these papers show that the interplay between the extensive margin unemployment and precautionary motives is central to understanding macroeconomic dynamics in heterogeneous agent models.

Life cycle dynamics serve as a crucial foundation for realistic macroeconomic analysis because they capture how households' consumption and saving behavior systematically evolve with age, driven by heterogeneous precautionary motives across life cycle stages. The work (Gourinchas and Parker 2002) aligns conventional macroeconomic models with observed microdata by estab-

lishing how a life cycle model comprehensively explains early-career liquid asset management and later-life accumulation contingent upon expected income pathways. (Eggertsson, Mehrotra, and Robbins 2019) used overlapping generations to study secular stagnation and identify effective policy response. They used demographic changes such as aging populations to capture the ”secular stagnation” phenomenon more realistically than borrowers and savers models. (Kaplan, Mitman, and Violante 2020) establish that shifts in future housing demand beliefs are the principal determinant of house price appreciation, more so than credit market conditions. A life cycle model was used so to match wealth profiles and consumption saving behaviors that change significantly over a households lifespan. Life cycle modeling provides essential realism to macroeconomic frameworks by capturing how household consumption and saving behaviors evolve across different life stages.

While existing models have provided valuable insights, they often miss key features that limit their ability to match empirical labor market dynamics and household finance data. Many are formulated in discrete time, which introduces rigidity and fails to capture the nuanced timing of employment transitions as established in (Shimer 2012). Furthermore, these models frequently impose restrictive borrowing assumptions, often denying non-zero asset positions, which prevents them from realistically capturing asset accumulation, asset distribution, and consumption smoothing behavior. Moreover, many lack a fully specified life cycle framework with empirically grounded borrowing constraints, limiting their ability to reflect the economic decisions individuals make at different stages of life and based on their past employment paths. These omissions constrain the models' ability to match observed labor dynamics and household leverage patterns. This paper addresses these gaps by introducing a heterogeneous agent life cycle model with borrowing and employment dynamics that more accurately reflect real-world constraints and behaviors. These advancements contribute to the literature.

3 Beyond Aggregate Data: Exploring labor market variation and debt

3.1 Overview

The Purpose of this section is primarily to motivate the extensive margin and its correlation with household debt. It is well known that the US primarily experiences labor fluctuations over the extensive margin. The purpose of this section is to first confirm that this trend continues in both the US and Europe. In Europe there is more cross sectional variation, before examining the relationship between real household debt and employment. Looking at Europe is beneficial for a few things. First, by looking at 24 European countries, there are more observations because there

is high quality quarterly data at the country level. All 24 countries are included in the labor market decomposition, whereas the household debt analysis is restricted to 19 owing to data availability. There is limited data on quarterly household debt per state, as well as state labor markets. This means the US could only be looked at in aggregate and lacks granularity. Second, Europe is used because it provides greater cross sectional variation in labor market risk than is available in aggregate U.S. data. Furthermore, due to the common market structure and high capital mobility in the European Union, household leverage primarily changes due to country specific restricted demand for loans, specifically from high unemployment and high credit risk households. During the period of study, these European countries had different average unemployment levels, employment fluctuations, labor market flexibility, social insurance systems, government expenditures, and so on. This creates a richer panel than using only aggregate U.S. data.

3.2 Data

To address the extensive margin, I used both US and European Data. The US Data used the BLS for weekly hours worked, employment, and consumer prices while it used data from the Board of Governors of the Federal Reserve System for household debt. Eurostat data was used for employment and weekly hours worked. This included Austria, Belgium, Bulgaria, Croatia, Czechia, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Luxembourg, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, and Switzerland. It is important to note that while data were available for Cyprus, Estonia, Iceland, Latvia, Lithuania, and Malta, these countries were excluded due to their small size. The UK was also not in the data, as Eurostat data didn't have weekly hours worked. To examine the correlation among employment, the extensive margin, and real household debt, we use BIS data on household debt and Eurostat HIPC price data. This excludes Belgium, Croatia, Romania, and Slovenia, as they were not included in the BIS household debt data. This dataset is well suited to the question as it provides consistent cross-country data.

The data will first be used to decompose the variance in total hours worked into the percent deviation caused by detrended employment (extensive margin), detrended weekly hours worked (intensive margin), and the covariance between the two. This will highlight the significance of the extensive margin. The data will also be used to examine the correlation between detrended employment and detrended real household debt. The decision to focus on Eurostat/European data is deliberate. There is high quality data for a variety of countries that have faced different shocks. This provides greater granularity into trends across different countries and can be aggregated into the overall data. Later on in this paper, when we look at VAR and LPs we will be able to split the counties into groups based on weekly hour worked variance, unemployment rate variance, real

household debt variance, and household debt to GDP. This setup allows for a clearer identification of how employment outcomes relate to household borrowing behavior through different IRF splits.

3.3 Empirical Methodology

The variance decomposition of total hours worked follows the methodology outlined in (Hansen 1985). To extract cyclical components, all series are detrended using the Hodrick-Prescott (HP) filter with a smoothing parameter of 1600, which is appropriate for quarterly data. To mitigate potential bias from the HP filter, the first two and last two years of detrended data for each country were excluded. The detrended value is then calculated as the ratio of the detrended (cyclical) component to the trend, plus one, a formulation that facilitates interpretation in log-linear terms. The variances and covariances are computed using the natural log of the detrended value of each term. It is important to note that covariances can be negative if the underlying variables are negatively correlated. The variance decomposition following equation (1). T stands for total hours worked in an economy over the past year. E stands for employment. H stands for weekly hours worked. The decomposition is performed in with equal weights across all european data points to construct aggregate measures. Additionally, a within country variance decomposition is conducted for each country in the sample including the US.

$$\text{var}(\ln T) = \text{var}(\ln E) + \text{var}(\ln H) + 2\,\text{cov}(\ln E, \ln H) \tag{1}$$

To examine the relationship between labor market dynamics and household leverage, detrended employment and detrended real household debt were analyzed. Real household debt was detrended the same way as employment. The correlation is then computed in an equally weighted fashion across all available observations from European countries to construct aggregate patterns. Within-country heterogeneity was assessed to capture the heterogeneity in the employment and household debt correlation within each country, including the US.

3.4 Empirical Results

In Figure (1), the aggregate variance decomposition of total hours worked in european countries shows that fluctuations in the employment rate account for over 83.2 % of the variation across countries. The second largest contributor is the variance in weekly hours, at 18.2 %. The covariance term is negative and near zero, indicating a near-zero negative correlation between employment and weekly hours worked. This analysis underscores the pivotal role of the extensive margin, driven by fluctuations in employment, in explaining variations in total hours worked across Euro-

pean countries.

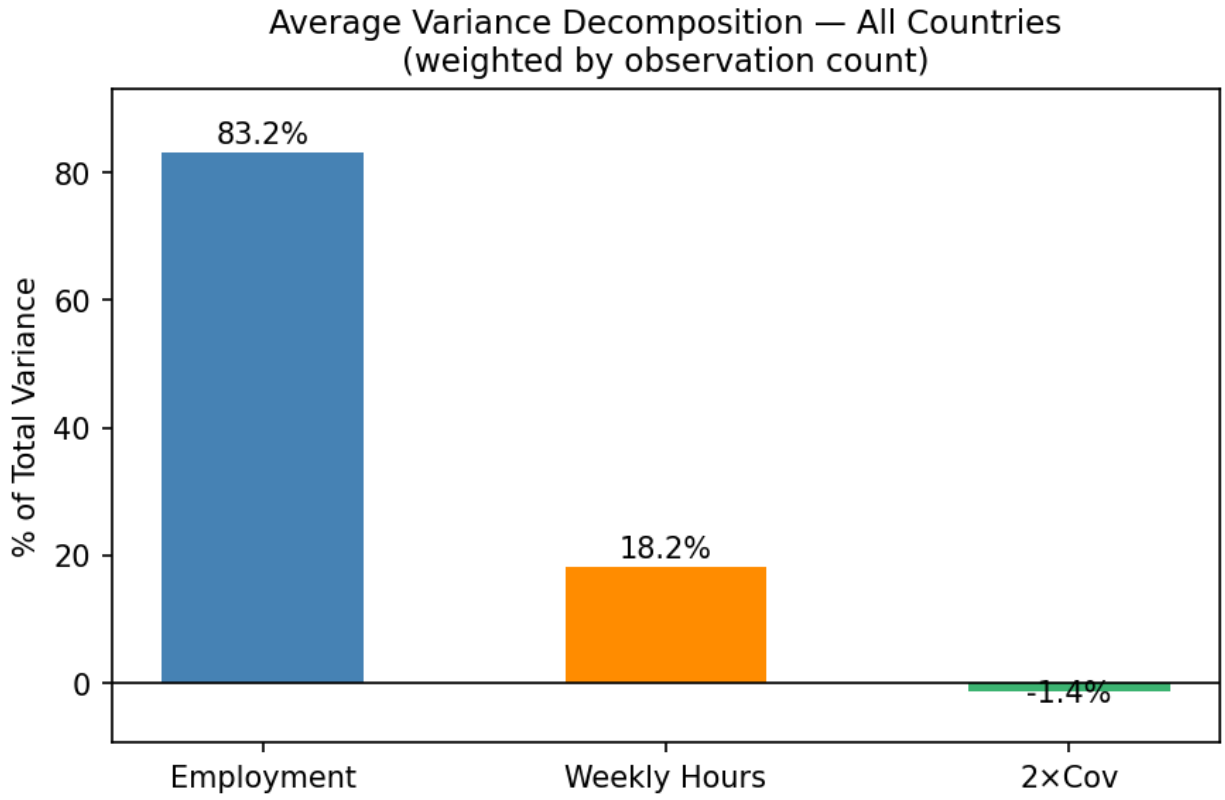


Figure 1: Aggregate Variance Decomposition

Figure (2) breaks down the variation per country. There are a few things that stand out. First, some countries: Austria, Denmark, Czechia, France, the Netherlands, Portugal, Romania, Slovenia, and Switzerland show more than 100 % share in total variance. This means that the total variance caused by employment and weekly hours worked is greater than the total variance, and the covariance term is negative because employment and weekly hours worked are negatively correlated. The green bar going back to 100 % shows a negative contribution of the covariance term. Second, Austria is the only country that has a higher percentage of total variance being caused by weekly hours worked. This is due to Austria's relatively stable employment level during the Great Recession, which rose only modestly from 4.4% in 2008 to 5.7% in 2009 and remained below 5.7% until 2014. This reflected a milder recessionary impact on the unemployment rate and employment levels overall. Third, the US shows similar patterns not only to (Hansen 1985), but also to the aggregate US data.

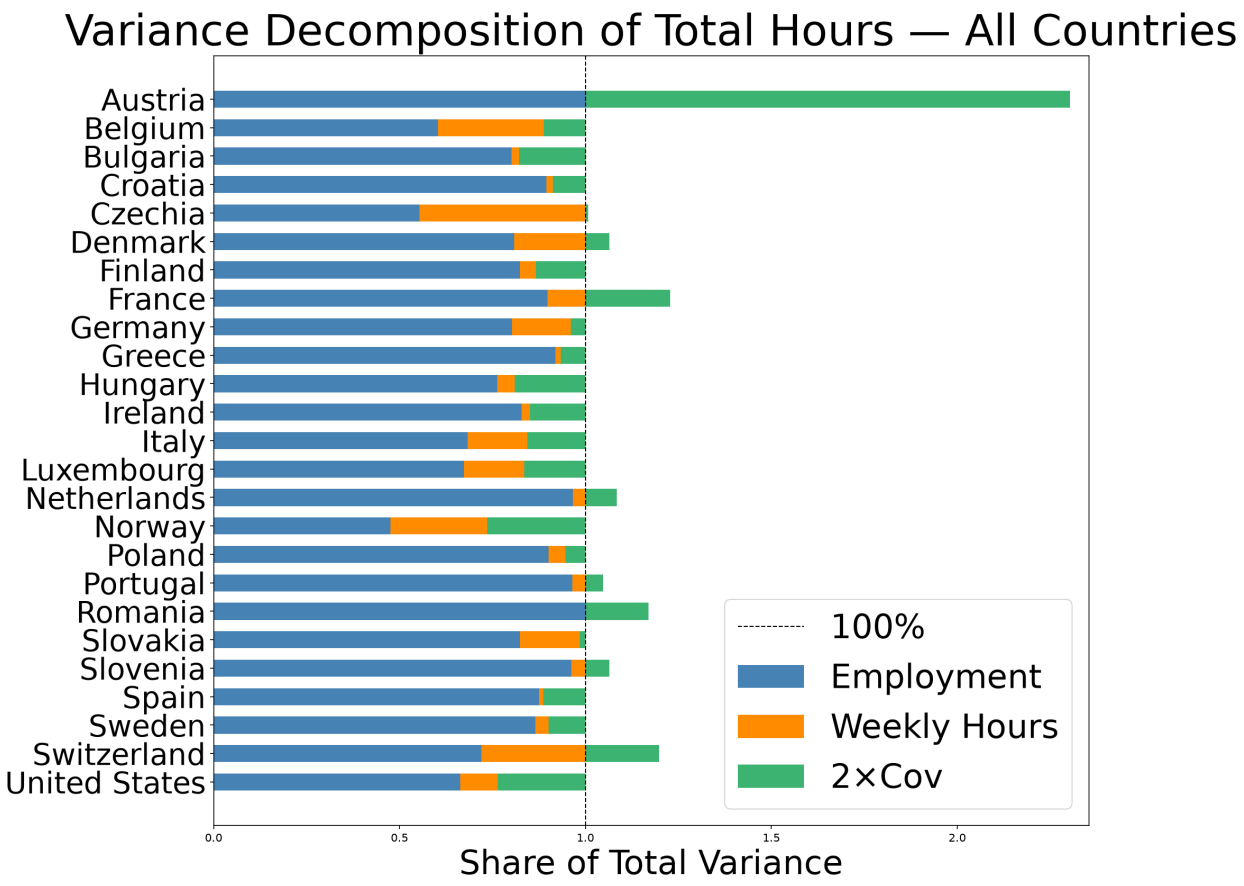


Figure 2: Within Country Variance Decomposition

Figure (**??**) illustrates the relationship between detrended real household debt and detrended employment amoung european countries and Figure (4) displays the values for the US. The two series generally display strong positive comovement, suggesting a meaningful correlation between labor market conditions and household borrowing behavior. Household debt dynamics are closely tied to labor market developments, which supports the use of the employment rate (the extensive margin) as a key driver of an economy's credit dynamics.

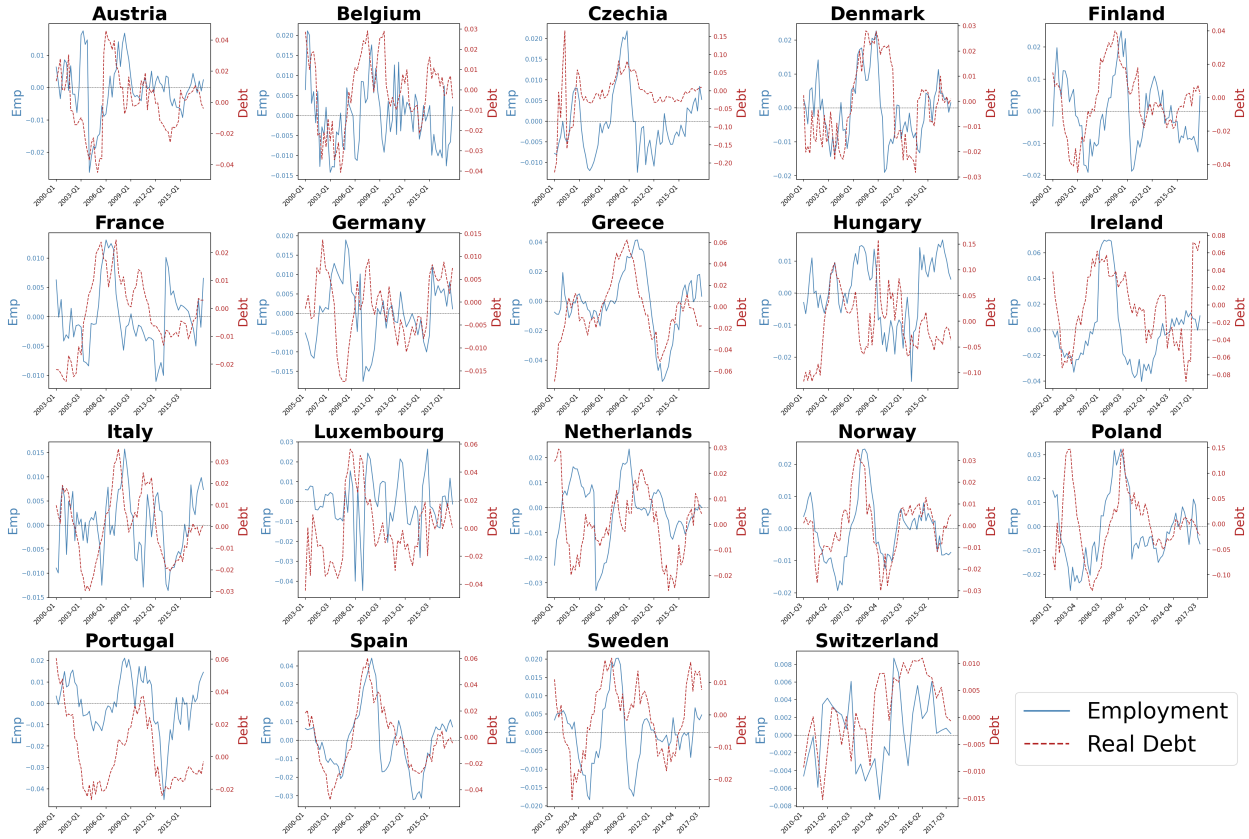


Figure 3: Detrended Household Debt and Employment

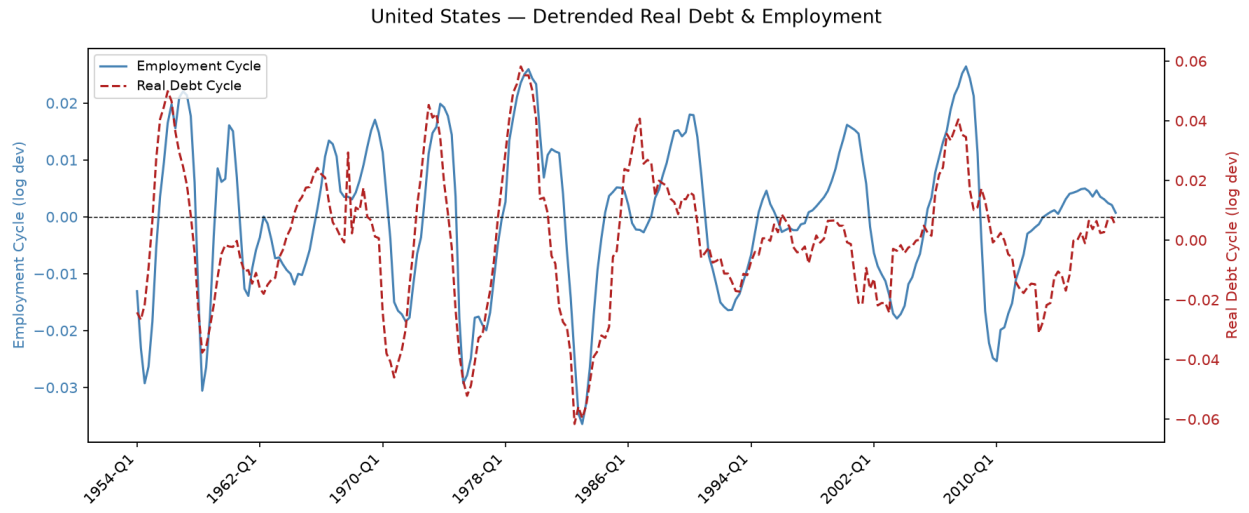


Figure 4: Detrended Household Debt and Employment US

Figure (5) presents the aggregate relationship between detrended real household debt and the

detrended employment rate within Europe. The figure shows a strong positive correlation, indicating that periods of higher employment are associated with higher household leverage, while periods of lower employment are associated with lower household leverage. This aggregate relationship provides further evidence that household debt dynamics are systematically linked to the extensive margin of labor fluctuations.

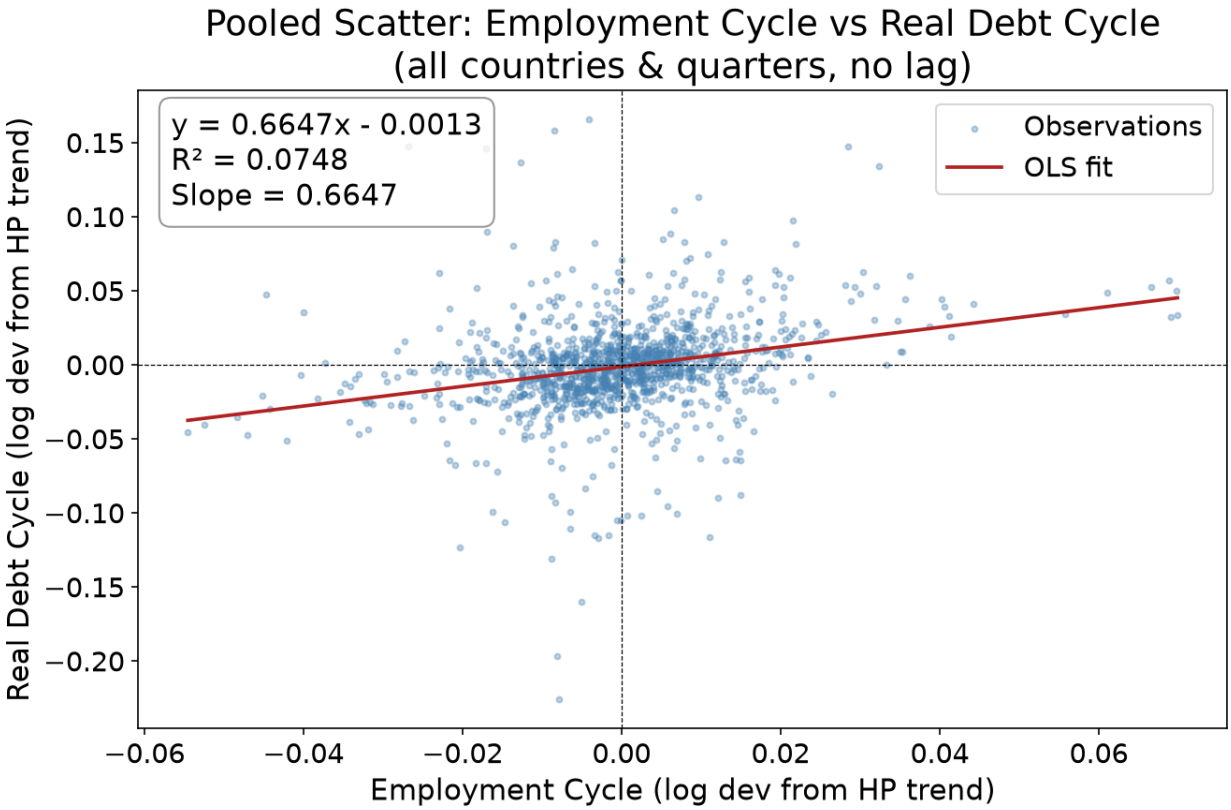


Figure 5: Aggregate Household Debt and Employment

A within-country analysis was conducted, Figure (6), to verify whether the positive relationship between lagged detrended employment and real household debt holds across individual countries. The majority of countries exhibit a positive slope, indicating a positive correlation between the two variables. Notably, Germany, Hungary, Luxembourg, and the Netherlands have a negative slope or inverse relationship, with Luxembourg and the Netherlands having an almost zero slope. This means that 16 of the 20 countries have a positive correlation. It is important to note that Czechia has a very strong relationship that could be skewed by outlier points with lower cyclical real household debt. Overall, looking at the country level confirms that there is a positive correlation between employment and household debt both in the aggregate and within country.

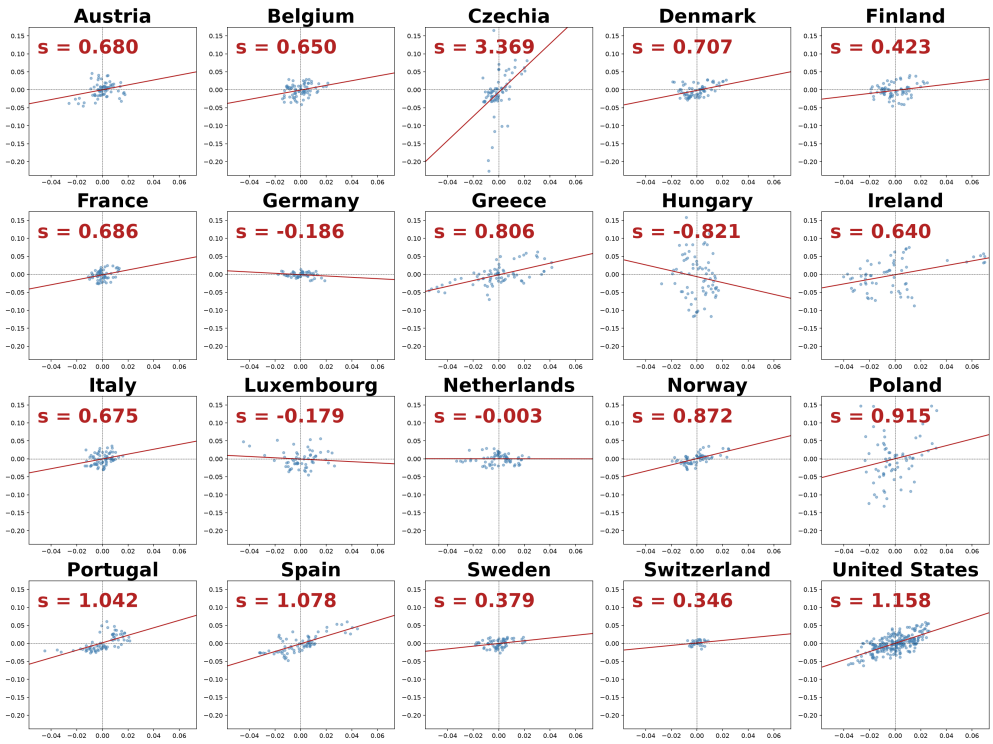


Figure 6: Within Country Household Debt and Employment

In conclusion, the empirical analysis highlights the existence and importance of the extensive margin in explaining labor market dynamics. As shown in Figure (1), the variance decomposition of total hours worked indicates that variation in employment is the dominant contributor to the total hours worked variance, surpassing both weekly hours worked and the covariance term. This finding is further supported by the within-country decompositions Figure (2), which reinforces the prominence of the extensive margin despite some country-specific deviations. Furthermore, Figures (**??**), (4), (5), and (6) illustrate a significant relationship between employment and real household debt. A positive correlation is observed at both the aggregate and country levels. This suggests that improvements in employment conditions are associated with increased household borrowing, whereas downturns coincide with deleveraging. These results collectively support the hypothesis that labor market variation through the extensive margin plays a key role in shaping both labor and loan dynamics.

4 Isolating the Demand Channel: A simple overlapping generations model

4.1 Overview

The purpose of the simple overlapping generation (OLG) model is to demonstrate how changes in youth unemployment, the extensive margin of youth labor, affect loan demand. Loan supply will be analyzed more thoroughly in a heterogeneous agent (HA) framework. The OLG model is used to clearly and concisely isolate the demand channel. It builds on the endowment-based OLG framework in (Eggertsson, Mehrotra, and Robbins 2019) and extends it by incorporating youth unemployment. Households are divided into three age groups: the young (20–39), the middle-aged (40–59), and the old (60–79). There are four agent types: employed young, unemployed young, employed middle-aged, and retired old. The model does not include physical capital. The model instead focuses on intergenerational borrowing, where young agents borrow from middle-aged savers. Young households are born during the current period and consume from their low current income and borrowing, while middle-aged households use their higher earnings to repay debt accumulated in the previous period and save for retirement. Retired households consume out of their now lower current income and accumulated savings. Borrowing is constrained by an exogenous loan-to-income limit. This borrowing constraint is assumed to be an external solution, so it is utility maximizing for young agents to borrow up to their limit. This feature simplifies the math and reflects realistic credit behavior. Critically, unemployed youth receive lower unemployment income than employed youth. The unemployed youth cannot borrow against unemployment income. As youth unemployment rises aggregate loan demand drops. This mechanism highlights how labor market conditions among the young translate into macroeconomic credit fluctuations.

4.2 Setup

The household utility function eq.(2) of the young depends on the current consumption when the household is young C_t^y, middle-aged consumption C_{t+1}^m in one period from now, and old age consumption C_{t+2}^o in two periods from now. The future utility is discounted by a rate of time preference β in one time period and by β^2 in two time periods.

$$U_t^y = ln(C_t^y) + \beta ln(C_{t+1}^m) + \beta^2 ln(C_{t+2}^o) \tag{2}$$

The debt constraint, eq.(3), is given as a fraction of youth income. It is important to note that the debt limit will form an external solution as the young households will borrow the full debt limit because it is utility maximizing to do so. Later, we will adjust the youth's income based

on employment status. This will tighten the debt limit to zero for the unemployed youth. This will lower loan demand and decrease the total quantity of loans. Borrowings B_t is less than the debt limit because your debt limit includes the interest on borrowings. The amount borrowed is consumed, not the debt limit. This allows young households to consume more when the real interest rate is lower, as that lets them borrow more with the same debt constraint. The Debt limit becomes a ratio, γ, of the young agents wage income.

$$D_t = B_t(1+r_t) = \gamma Y_t^y \tag{3}$$

The young budget constraint eq.(4), middle budget constraint eq.(5),and the old budget constraint eq.(6) govern the consumption that the young, middle, and old households. The young households consume all of their income and borrowings. Youth borrowings can be expressed as a function of their income and the real interest rates. The middle-aged households consume out of their current income less the their debt limit from last period when they were young, plus borrowings they take out in the current period. They will have negative borrowings (savings) in the current period, so saving in the current period reduce consumption. Old-age households will consume out of old-age income less middle-aged borrowing (plus middle-aged savings).

$$C_t^y = Y_t^y + B_t^y = Y_t^y \left(1 + \frac{\gamma}{1+r_t}\right) \tag{4}$$

$$C_{t+1}^m = Y_{t+1}^m - (1+r_t)B_t^y + B_{t+1}^m \tag{5}$$

$$C_{t+2}^o = Y_{t+2}^o - (1+r_{t+1})B_{t+1}^m \tag{6}$$

The Euler equation, eq.(7), is derived from eq.(2), 5, and 6.

$$\frac{1}{C_t^m} = \beta E_t(1+r_t)\frac{1}{C_{t+1}^o} \tag{7}$$

By combining the middle-aged budget constraint eq.(5) and the old-age budget constraint eq.(6), and applying the Euler equation eq.(7), we derive the consumption function for middle-aged agents, shown in eq.(8).

$$C_{t+1}^m = \left(\frac{1}{1+\beta}\right)\left(Y_{t+1}^m + \frac{Y_{t+2}^o}{1+r_{t+1}} - (1+r_t)B_t^y\right) \tag{8}$$

4.3 Adding Youth Unemployment

To capture unemployment's effect on loan demand, the young are divided into two employment states: employed and unemployed. This split captures the extensive margin of employment, with $1-\omega_t$ percent of the young employed and ω_t unemployed. This differs from the empirical section: instead of focusing on the number of employed individuals, we focus on the unemployment rate because it indicates economic distress and scales across economies. It is assumed that middle-aged households are fully employed. This assumption reflects the observation that youth unemployment tends to be more volatile than overall unemployment.

Young individuals disproportionately bear the brunt of labor market fluctuations for two main reasons: they are more susceptible to layoffs because of their shorter tenure, and they are more affected by hiring slowdowns because they typically enter the labor market from unemployment. (Shimer 2012) highlighted that the primary driver of rising unemployment is reduced hiring rather than increased layoffs. For simplicity, the model assumes that all of the middle-aged are employed. The focus is placed on the extensive margin of youth unemployment, as the young are the primary drivers of loan demand both empirically and in OLG literature. The unemployed youth earn 1/3 of the employed youth's income. This setup will demonstrates that fluctuations in the real interest rate are primarily driven by changes in youth unemployment, consistent with the empirical evidence.

Loan demand eq.(9) is now endogenously determined by the level of youth unemployment. This impacts loan demand because the unemployed cannot borrow as much: their debt constraint, D^u, is zero and significantly lower than the employed's, D^e. The extensive margin of young unemployment now directly lowers loan demand due to the lower debt limit. The equation for loan demand is now the fraction of employed youth times the employed debt constraint plus the fraction of unemployed youth times the unemployed debt constraint divided by the gross real interest rate. The equation can be simplified further because D^u is zero. This equation shows that increasing the unemployment rate reduces loan demand.

$$L^d = \frac{(1-\omega_t)D^e + \omega_t D^u}{1+r_t} = \frac{(1-\omega_t)D^e}{1+r_t} \tag{9}$$

Loan supply, on the other hand, is an interior solution determined by the Euler equation (eq.).(7) and the middle-aged budget constraint eq.(5) . Loan supply is positively affected by middle-aged income and real interest rates. Loan supply is negatively affected by income in old age and by the debt the middle-aged accumulated in the last period when they were young. This implies that households that experienced unemployment when they were young accumulated less debt and smoothed their higher wealth into old age consumption. Since income during the middle-aged period is exogenously fixed and independent of earlier employment history, these households adjust their intertemporal consumption accordingly. However, this behavior is not the model's

central focus; it is primarily designed to illustrate loan demand dynamics on the extensive margin of youth employment.

$$L^s = \left(\frac{\beta}{1+\beta}\right)(Y_t^m - (1-\omega_{t-1})D^e) - \left(\frac{1}{1+\beta}\right)\frac{Y_{t+1}^o}{1+r_t} \tag{10}$$

4.4 Results

Loan demand and loan supply are shown in Figure (7). The youth unemployment in the steady state is assumed to be 5%. The youth unemployment rate is assumed to be the same in the current and past periods. The figure shows that loan demand is more inelastic than loan supply. When the youth unemployment rate increases, the black line shifts to the dashed black line, and the quantity of loans decreases more than the real interest rate. Conversely, youth unemployment led to a marginal increase in credit supply during the transition from the magenta line to the dashed magenta line. This effect is driven by reduced initial borrowing, which leads to higher wealth levels that are intertemporally smoothed to old age. This means that the impact of youth unemployment on loan demand has a greater influence on loan quantity and interest rates than changes in loan supply. Adjustments over the extensive margin, youth unemployment, play a central role in shaping aggregate loan demand, as fluctuations in youth employment status directly impact borrowing behavior, making it a critical channel through which labor market dynamics influence loan quantity and real interest rates.

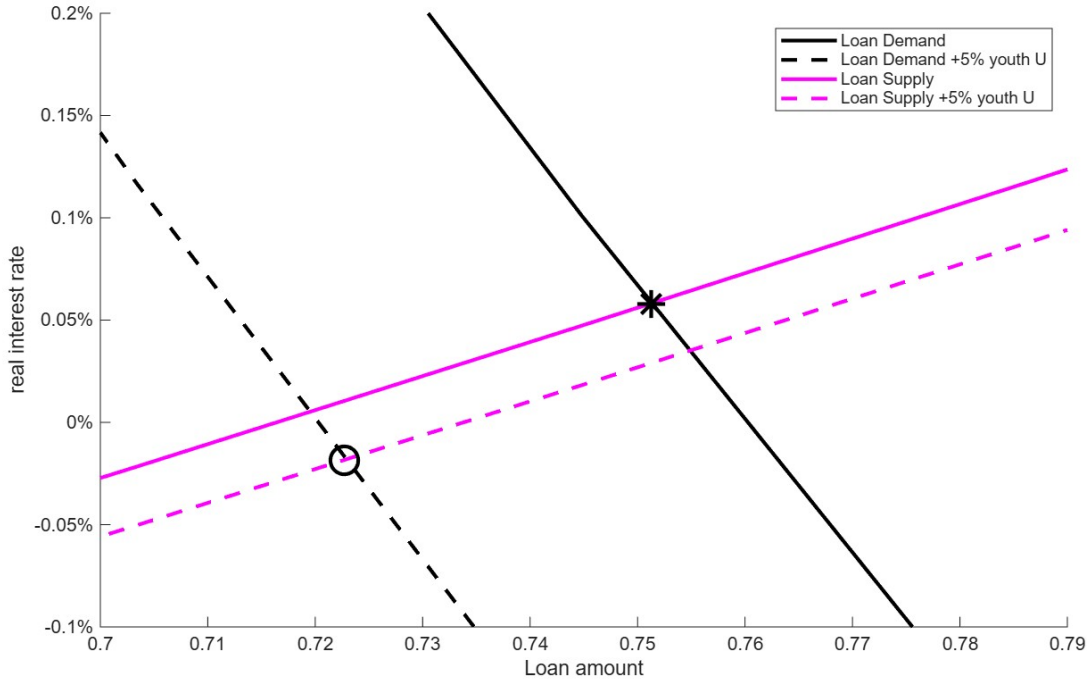


Figure 7: Loan Demand and Loan Supply

Equating loan demand eq.(9) with loan supply eq.(10) yields equilibrium real interest rate eq.(11). Youth unemployment in the current period lowers the real interest rate by tightening borrowing constraints, as a greater share of credit-constrained unemployed youth reduces aggregate loan demand, exerting downward pressure on equilibrium rates, as observed in Figure (8). This relationship arises because unemployed youth borrow less, leading to lower indebtedness when they enter middle age. Consequently, middle-aged households that experienced youth unemployment retain a larger share of their income after debt repayment. Given their desire to smooth consumption over time, as captured by the Euler equation, these households increase their loan supply in the present period. Thus, higher past youth unemployment raises current loan supply, placing downward pressure on the equilibrium interest rate.

$$1 + r_t = \frac{(1+\beta)(1-\omega_t)D^e + Y^o_{t+1}}{\beta(Y^m_t - (1-\omega_{t-1})D^e} \tag{11}$$

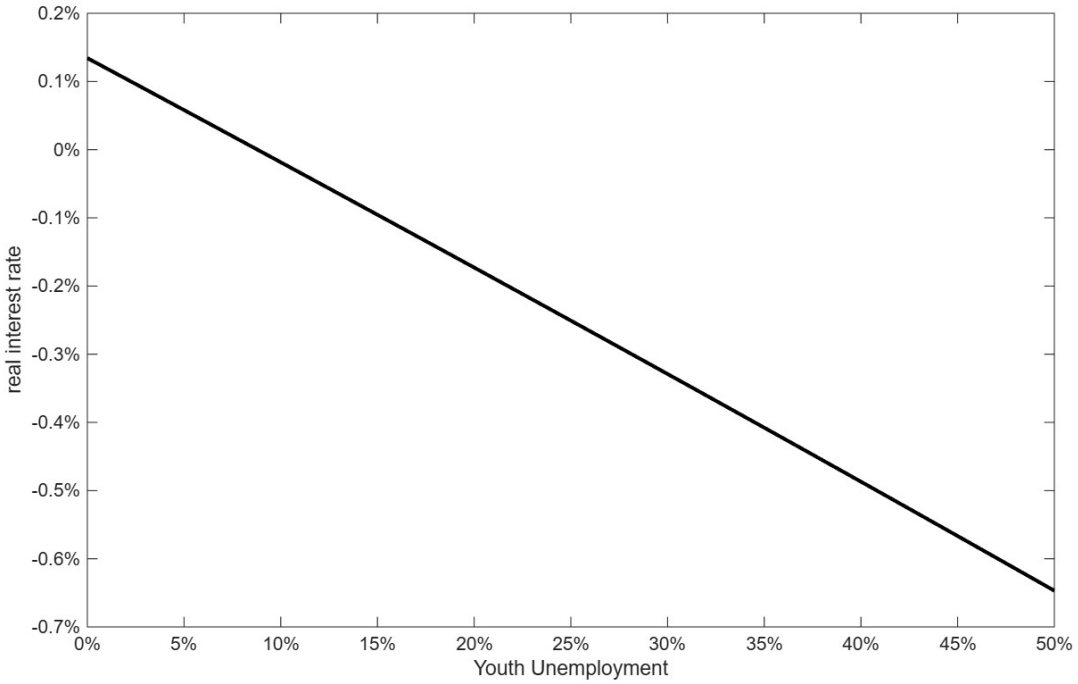


Figure 8: Real Interest Rate based on Youth Unemployment

Figure (9) illustrates the total loan quantity as a function of steady-state youth unemployment. As youth unemployment increases, the total quantity of loans in the economy declines. Equation (12) represents loan quantity as a function of the unemployment rate ω.

$$L_t = (\frac{\beta}{1+\beta})\frac{(1-\omega_t)D^e(Y_t^m - (1-\omega_{t-1})D^e)}{(1-\omega_t)D^e + \frac{1}{1+\beta}Y_{t+1}^o} \tag{12}$$

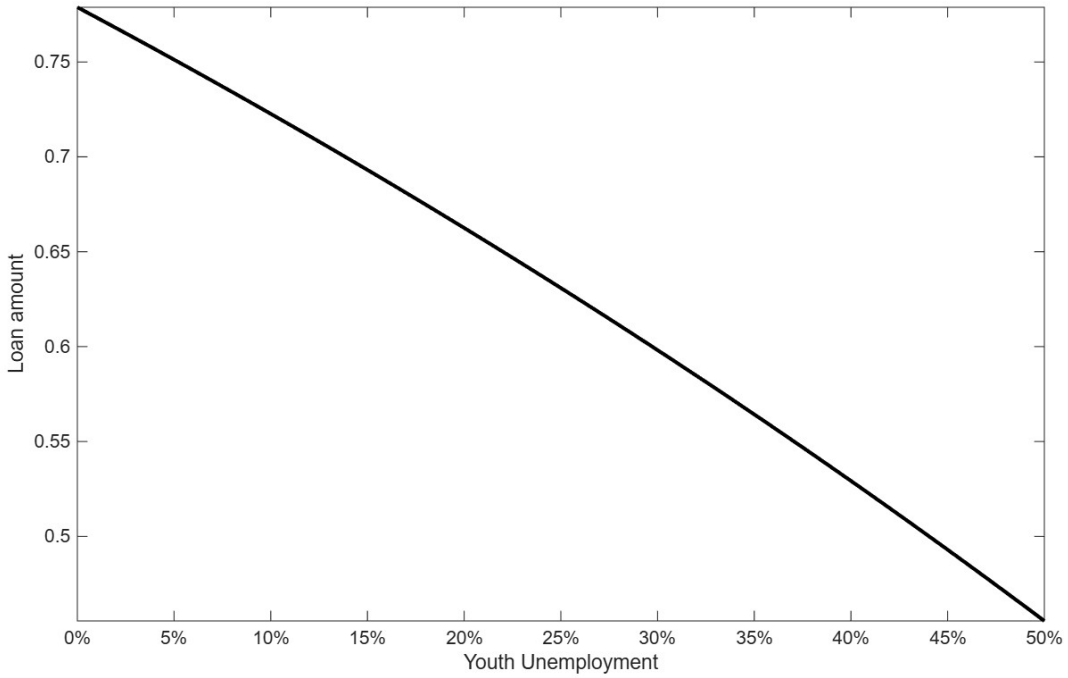


Figure 9: Loan Quantity based on Youth Unemployment

The Utility of the Youth is captured based on eq. (2). It is based on the current period utility from consumption, a discounted value, β, of the utility of next period's consumption, and finally the doubly discounted, β^2, value of utility from consumption when old. The consumption and utility paths of the unemployed and employed differ throughout their lives, not just because of differences in consumption when young. This is because the unemployed don't borrow and have a higher asset level when middle aged and old.

Figure (10) displays the utility of young agents based on the unemployment rate. The utility of the unemployed youth is shown by the black line, which decreases slightly as unemployment increases. This is because young unemployed consumption is constant, so current period utility doesn't change with real interest rates. Lower real interest rates now reduce the income this agent receives from savings accumulated in middle age, lowering the young agent's future and overall utility. The utility of the employed youth is shown by the blue line. As unemployment picks up, their utility increases because they can borrow more at a lower interest rate. This effect outweighs the negative impact of lower rates on the savings the agent accumulates in middle age. The red line represents the average utility of the young agents. It is a weighted average of unemployed utility times the probability of being unemployed plus the employed utility times the probability of being employed. This graph highlights the stark difference in utility between employed and unemployed individuals, as well as the average utility of young agents as the unemployment rate increases,

bringing a focus on utility that is missed by standard models.

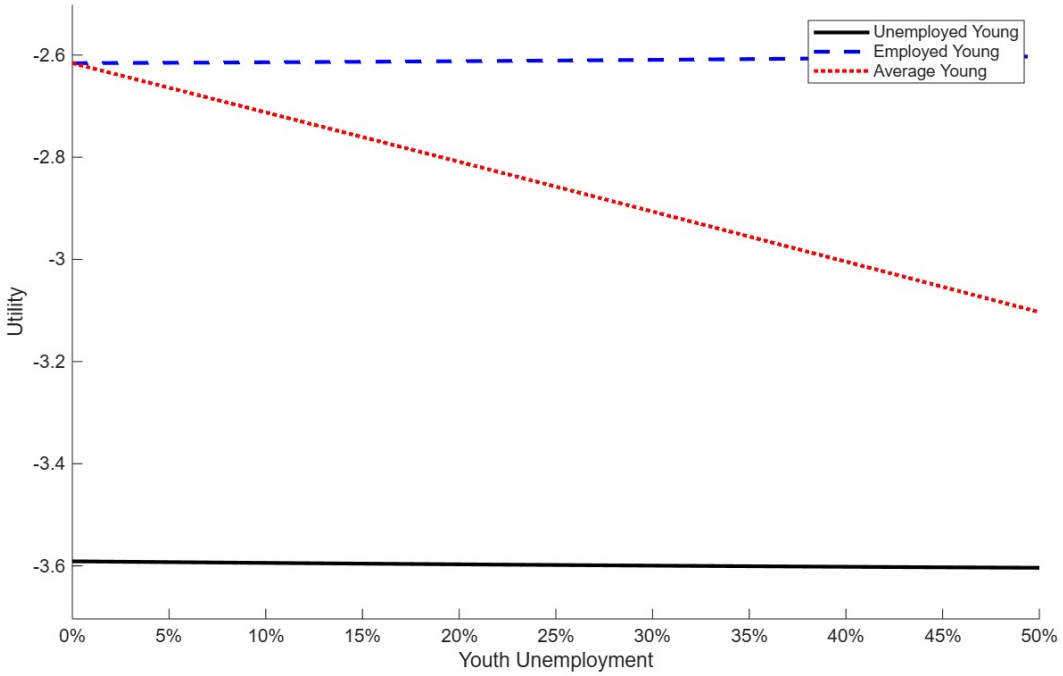


Figure 10: Youth Utility

4.5 Section Conclusion

In conclusion, this paper develops an OLG model to analyze how youth unemployment, the extensive margin, influences loan demand, loan supply, aggregate outcomes for interest rates, the quantity of debt outstanding, and utility levels. By introducing heterogeneity in unemployment among young households, the model captures how borrowing constraints and income differences across employment states affect consumption and savings behavior. The results demonstrate that youth unemployment significantly depresses aggregate loan demand by tightening borrowing constraints on the unemployed. Loan supply increases modestly when middle-aged households experience unemployment in the previous period, as this reduces prior borrowing. The dominant effect arises on the demand side, and loan supply is not a focus of this model. The real interest rate is shown to fall with higher youth unemployment, reflecting the dominant shift in loan demand caused by the borrowing constraint. The reduction in borrowing is accompanied by a decline in overall utility. This drop in utility is often overlooked and is only weakly captured in representative-agent economies. Because the borrowing was utility maximizing for the agents as (Auclert 2019) argued, not being able to borrow decreased utility further than the drop in income

alone. Overall, the analysis highlights the central role that fluctuations in youth employment play in shaping credit market dynamics and agent utility. The following section extends this framework from an endowment economy to a heterogeneous agent model in general equilibrium with dynamic job finding and separation rates, thereby assessing how the present results hold under more realistic labor market dynamics.

5 Microfoundations of Interagent Lending: Incorporating extensive margin risk into the HA framework

5.1 Overview

This section employs a heterogeneous agent (HA) model with incomplete markets to examine how unemployment risk along the extensive margin shapes household consumption and saving behavior. Unlike the overlapping-generations (OLG) framework, which incorporates only intertemporal substitution and consumption-smoothing motives, the HA approach captures the precautionary savings motive and therefore more accurately replicates equilibrium shifts in loan supply. The analysis identifies how changes in unemployment resulting from a negative productivity shock affect the quantity of household loans, real interest rates, and utility. To isolate unemployment risk, the intensive margin model, the extensive margin model, and a benchmark Huggett model are compared using comparative statics. This decomposition separates the effect of extensive margin risk from the life cycle savings motives embedded in both HA models.

Several results stand out. First, the HA model's steady state differs markedly from that of the benchmark Huggett model, particularly in asset distribution and the savings policy function, reflecting the interaction between the life cycle motive and unemployment risk at equilibrium. Second, when both models are hit by a negative productivity shock, in the HA model, effective labor adjusts only over the extensive margin in one case and only over the intensive margin in the other. Equilibrium household loans fall more sharply with the extensive margin adjustment, and real interest rates drop in the extensive margin adjustment, while they rise slightly in the intensive margin adjustment. The Huggett model's outcomes after the productivity shock are almost indistinguishable from steady state productivity. Third, across a range of productivity shocks, agents' utility and borrowing decline more steeply as productivity falls when labor adjusts over the extensive margin. Fourth, under a countercyclical transfer policy, higher transfer payments weaken the precautionary motive because utility increases in the unemployed state for both young and old agents.

5.2 Setup

5.2.1 Utility Function and Budget Constraint

The model has a standard Constant Relative Risk-Aversion (CRRA) utility curve eq. (13). There is no disutility for work. This assumption will make all households under the retirement age want to work regardless of asset level, and no agents above retirement age will work.

$$U_t = \frac{c_t^{1-\sigma}}{1-\sigma} \tag{13}$$

Eq. (14) represents the budget constraint. $wz_l h_e$ is wage income earned. Wage income earned is based on the wage rate w times z_l the age productivity profile times hours worked h_e. The income profile is parabolic, peaking when agents are in their 50's. For the steady state example, hours worked will be 0 or 1. Unemployment/social security u_e is given to all the unemployed individuals, $(1-e)$, including the retired. π_l is the dividend income from firms. The dividend profile matches the z_l income profile. Interest payments are assumed to be untaxed in order to isolate the real rate of return. Taxing interest income would introduce a wedge between the gross and net return, affecting the equilibrium real interest rate. Taxation is omitted from interest income since the focus of this analysis is on the real return absent tax distortions.

$$c(a,e,l) = (wz_l h_e + u_e(1-e) + \pi_l)(1-\tau) + ra - s(a,e,l) \tag{14}$$

Eq.(15) is the recursive value function known as the Hamilton-Jacobi-Bellman (HJB). The rate of time preference is 4% in line with the findings from (Gourinchas and Parker 2002). The solution to the HJB gives a value function $v(a,e,l)$, a policy function for consumption $c(a,e,l)$, and a policy function for savings $s(a,e,l)$ for each discrete agent based on asset level, employment status, and age. Life cycle and employment flows are exogenous to the value function, whereas the 3 policy functions are endogenous.

$$\rho v(a,e,l) = \max_c \; u(c) + v_a'(a)\dot{a} + \sum_{e' \neq e} \lambda_{ee'}(v(a,e',l) - v(a,e,l)) + \sum_{l' \neq l} \lambda_{ll'}(v(a,e,l') - v(a,e,l)) \tag{15}$$

The first-order condition of the HJB, eq.(16), states that the marginal utility of consumption will equal the partial derivative of the value function with respect to assets. The FOC is necessary to convert the HJB into a viscosity solution, enabling numerical methods to be performed.

$$u'(c) = v_a'(a,e,l) \tag{16}$$

To solve the function computationally, the inverse of the marginal utility function of the derivative of the asset grid $u^{'-1}(v_a^{'}(a,e,l))$ is substituted into where consumption c was. This yields the viscosity solution eq.(17). This eliminates consumption from the equation, allowing it to be solved recursively based on grid location and exogenous transition dynamics.

$$\begin{aligned}\rho v(a,e,l) = \max_{c} \; & u(u^{'-1}(v_a^{'}(a,e,l)))+ \\ & v_a^{'}(a)((wz_l h_e + u_e(1-e) + \pi_l)\tau_1 + ra - u^{'-1}(v_a^{'}(a,e,l)))+ \\ & \sum_{e' \neq e} \lambda_{ee'}(v(a,e',l) - v(a,e,l)) + \sum_{l' \neq l} \lambda_{ll'}(v(a,e,l') - v(a,e,l)) \end{aligned} \tag{17}$$

5.2.2 Asset, Employment, and Life cycle Dynamics

Income profiles are calibrated using age-earnings data from the 2013 Bureau of Labor Statistics (BLS) report. The data is fitted using ordinary least squares (OLS), with age and age squared as covariates. All employed individuals in the same age group are homogeneous, implying that wage income only varies by age and employment status. While incorporating income heterogeneity within age groups would better match real-world dynamics, it would also introduce significant computational complexity at a stage where lending and borrowing dynamics are already captured by life cycle income dynamics. This specification results in a parabolic income profile over the life cycle, with earnings peaking in their 50's. The estimated income curve is then scaled so that aggregate productivity equals 1 at the natural rate of employment among working age individuals. The scaling accounts for full retirement at age 65, with all individuals 65 or older being retired. By matching the shape of the life cycle income profile, the HA model better captures life cycle dynamics than the OLG framework. The model retains the standard borrowing motive for the young and a retirement savings motive for the middle-aged. Consistent with (Guerrieri and Lorenzoni 2017), unemployment benefits and Social Security income are both modeled as 33 percent of average labor income. Dividend income is modeled as age-dependent and follows the same profile as labor income. It is assumed that both employed and unemployed individuals receive the same dividend income. Dividend income is scaled differently, as it is distributed across all age groups, not just the working population.

Agents enter the workforce at age 20 and exit the model upon death at age 75. The life cycle is discretized into 11 five-year age groups: 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, and 70. Because the model is in continuous time, agents age according to a Poisson process. Each quarter, an agent has a probability of $\frac{1}{20}$ of transitioning to the next age group or dying if the agent is 70. The distribution of agents across age is assumed to be uniform. For every agent that reaches age 75 and exits the model, a new agent is introduced at age 20. New entrants inherit the agent's asset holdings upon

death, similar to a bequest from a grandparent. This includes whether the exiting agent holds a negative asset position. Due to the stochastic nature of the Poisson aging process, some agents may age faster than they pay down their debt, as they may be randomly selected to advance in age without a corresponding opportunity to pay down debt. In rare instances, the structure permits agents to pass away with a negative asset level.

The borrowing constraint is modeled using a maximum payment-to-income (PTI) ratio eq.(18). Following the methodology of (Kaplan, Mitman, and Violante 2020), the minimum loan payment is set so the loan fully amortizes by the time the agent reaches retirement age. The PTI limit is set to 28% of current income, consistent with the front-end PTI ratio documented in (Quercia, McCarthy, and Wachter 2003). The borrowing constraint is further strengthened by restricting borrowing to employed households. While this assumption may initially appear overly restrictive, real-world evidence shows that many households have limited consumption-smoothing capabilities. According to a Forbes article (McGrath 2016), "63% of Americans Don't Have Enough Savings to Cover a $500 Emergency," indicating that a significant portion of the population lacks adequate financial buffers. This means that while individuals are in debt, a good portion do not have adequate savings in their savings and checking accounts. Furthermore, research from (Haughwout et al. 2024) highlights that approximately 18% of credit cards are utilized up to 90% of their limits. These findings suggest that individuals with negative or near-zero net assets face substantial limitations in accessing credit and therefore have limited means to smooth consumption during periods of financial stress or unemployment. These findings provide empirical support for the assumption that a single asset spectrum does a decent job of matching reality without adding the computational complexity of two distinct spectra for debt and savings.

$$pmt_t = .28 y_{e,l} e_e \tag{18}$$

The minimum debt payment is calculated using a fixed annuity formula eq.(19), reflecting how most household debt is structured in the U.S. financial system. The borrowing limit per each age group is fixed and will remain that way when studying the impulse response. As agents age and gradually repay their debt, the required minimum payment adjusts over time. While this implies the payment profile is not constant, maintaining a fixed payment schedule would introduce additional complexity, requiring households to maintain their payment schedule despite changes in debt repayment or age, which follow a Poisson distribution. Such an extension would be problematic, particularly in continuous time, because agents who age up more quickly (due to the Poisson process) might retire while still holding significant debt. Although this is not ideal for modeling debt repayment, continuous time modeling is essential for accurately capturing employment transition dynamics.

$$pmt_t = \frac{-a * r}{1 - (1 + r)^{65 - age}} \tag{19}$$

An important assumption is that agents cannot default on their debt. This assumption is required because loans are unsecured by collateral and must be backed only by future income. If an agent is constrained and unable to borrow further, they must make the minimum debt payment, provided their consumption remains above the minimum threshold of 0.1. Otherwise, the payment equals income minus the minimum consumption threshold. The imposed saving policy function sets both the consumption policy function and the value function. Parts of the distribution grid close to the forced savings regime must incorporate this constraint into the value function because they could switch to unemployed or age up. This adjusts agents' optimal consumption and saving behavior as they approach retirement. This forced savings rule ensures that nearly all agents avoid entering retirement with zero or negative net assets. The minimum consumption threshold only binds for the high borrowing, old, unemployed agents. In general, agents voluntarily save above the minimum requirement to pay down debt and accumulate wealth for retirement.

5.2.3 Firm and Employment Dynamics

Firms pay real wages above the level needed for full employment in accordance with efficiency wage theory. As discussed in (Katz 1986), firms may pay higher wages to reduce turnover, motivate greater worker effort, discourage shirking, deter unionization, and attract higher-ability employees. Unemployment is introduced into this model because the wage is higher than the wage that would constitute full employment. In this model, the unemployment rate is 6% at equilibrium. Total wage income at steady state equilibrium is set to be (1 - α) or the labor share.

The labor supply curve is calibrated so that its elasticity matches the Macroeconomic Frisch elasticity of 2.84, based on the findings in (Chetty, Guren, Manoli, and Weber 2011). This includes .54 for the intensive margin and 2.50 for the extensive margin. For simplicity, the aggregate is used across both margins to compare model dynamics between them. The macro calculation is much larger than the micro calculations they found because the micro estimate was based on a distribution of reservation wages and quasi-experimental evidence. This is purely a focus on labor supply choices. The participation only extensive margins treats the labor market as Walrasian, which is exactly what the efficiency wage setup fixes. In my model, the labor supply curve determines the wage as a function of the quantity of labor for all effective labor values below full employment. This is based on empirical data rather than a Nash bargaining approach. Nash bargaining can get very complex in an environment with such a large distribution of agents across age, wealth, and employment status. The graph of the labor market can be seen in Figure (16)

The firm's production function is modeled as a labor only neoclassical function with dimin-

ishing returns. Costs are real wage income, denoted as total effective labor times the wage per productivity. Together, the profit function is the production function less the real labor cost as seen in eq.(21). Total effective labor is computed in eq.(20) as the aggregate sum of employed individuals, each weighted by their age-specific productivity z_l and individual hours worked $h_{e,t}$. In calibration, age-specific productivity is rescaled by a common factor across all age groups so that at the natural rate of employment, total effective labor input is normalized to 1. In the steady state, hours worked are treated as binary because individuals either work and provide one full unit of labor or do not work and provide zero units of labor. The firm can adjust total labor input along two margins: the extensive margin, by varying the number of employed agents, and the intensive margin, by adjusting the hours worked by employed agents, from 1 to a different level. This provides a flexible framework for analyzing the difference between the intensive and extensive margins in response to a productivity shock.

$$N_t = \sum_e \int_l (z_l h_{e,t}) \tag{20}$$

$$\pi_t = A_t (N_t)^{1-\alpha} - wN_t \tag{21}$$

Employment dynamics play a pivotal role in this model's framework. Rather than incorporating a full job search model, job flow parameters are calibrated based on the empirical findings of (Shimer 2012). Specifically, the Shimer assumes that hiring rates are determined ex-ante. This model allows individuals who enter the labor force to be already employed, particularly those aged 20. This assumption reflects real-world patterns, such as college students securing employment offers before graduation and before formally entering the labor market. To replicate observed labor market behavior, a portion of retiring agents' jobs are given to new 20 year old agents. This calibration ensures the model achieves the desired natural spread, as observed among 20-24 year olds, with a higher unemployment rate in the steady state. Unlike some models that assume uniform job separation probabilities across workers, this framework introduces age dependent exit probabilities for two reasons. First, all agents retire at age 65. Second, to align the age profile of unemployment rates with empirical data, the job separation rate for young workers aged 20 and 25 is increased by 22.7% to match the data during recessions. This is consistent with labor market dynamics, as younger workers are more likely to be laid off due to lower tenure and job security. The higher youth job exit probability helps calibrate the model to match targeted unemployment rates across the life cycle. The overall separation rate is calibrated to 4.6% to ensure that the model's steady-state equilibrium matches the natural rate of unemployment given the job finding probability.

(Shimer 2012) observed that the average monthly hire rate was 44 %. Assuming homogeneous

hiring rates, this corresponds to a quarterly hire rate of 82.4 %. This means it takes the average agent a little over two months, or 2/3 of a period, to find a job once unemployed. The job separation rates are set so that the model achieves the targeted natural unemployment rate at the steady state. (Shimer 2012) finds that 77 % of fluctuations in the unemployment rate since 1948 were because of changes to the hiring rates. For simplicity, this model assumes 75 % of the fluctuations in unemployment were caused by a reduced hiring rate. To incorporate this finding into the model, for every 2.5% reduction in the hiring rate, the separation rate increased by 1% due to the standard job finding probability $\frac{s}{s+f}$.

5.2.4 Market Clearing Conditions

The Kolmogorov Forward Equation eq.(22) ensures that the distribution of agents remains stationary because the model is analyzed in steady state. This condition implies that, across all three dimensions: savings, employment status, and age, the sum of inflows must equal the sum of outflows at each discrete point of the distribution.

$$\begin{aligned}\partial_t g_t(a,e,l) = \partial_a[s(a,e,l)g_t(a,e,l)] - g_t(a,e,l)\sum_{e'\in E}\lambda_{ee'} + \\ \sum_{x\in E}\lambda_{xe}g_t(a,e,l) - g_t(a,e,l)\sum_{l'\in L}\lambda_{ll'} + \sum_{x\in L}\lambda_{xl}g_t(a,e,l)\end{aligned} \tag{22}$$

The government's budget must balance, as expressed in eq.(23). Government revenues are generated entirely through wage and dividend taxation eq. (24), while expenditures consist of transfer expenses eq.(25).

$$gov_surplus = 0 = revIncome - expTransf \tag{23}$$

The income tax base includes labor income, unemployment benefits, and dividends. Notably, interest income is excluded from the tax base. This is intentional and ensures that tax policy does not distort the real interest rate. During periods of negative productivity shocks, tax rates would be elevated to balance the budget. This would increase the real interest rate because the model would account for the tax rate and require a higher interest rate to offset higher taxes. Having interest income taxed potentially biases the model's core interest rate results.

$$revIncome = \sum_e \int_l \int_a \tau_1(w(z_l h_{e,t}) + \pi_l) \tag{24}$$

The transfer expense is the sum of all unemployed individuals times the unemployment/social security payment.

$$expTransf = \sum_{e} \int_{l} \int_{a} u_e(1-e) \tag{25}$$

The aggregate level of assets in the economy must sum to zero in the steady state equilibrium of the Huggett model eq.(26). This assumes a closed economy assumption, where every asset held by one agent corresponds to a liability held by another. In other words, for every loan extended, there must be a corresponding borrower and vise versa. The real interest rate adjusts to ensure the asset market clears.

$$A(r) = 0 = \sum_{e} \int_{l} \int_{a} ag(a,e,l;r)da \tag{26}$$

5.3 Typical Huggett Model Setup

A standard Huggett Model will be used to compare with the heterogeneous-agent model specified above so that the differentiation of the HA model is clear. The Hugget model will be referred to as the Hugget model, and the Heterogeneous agent overlapping generations model specified above will be referred to as the HA model.

The Huggett model has the same CRRA utility curve eq. (13). The Huggett model differs in that it has endogenous labor supply, allowing agents to choose their own hours, h, based on their asset level and idiosyncratic productivity. Labor disutility is added in eq. (27) so that the agents can pick their hours. The scale of labor disutility is denoted as χ. The Frisch Elasticity is $\frac{1}{\phi_h}$.

$$d = \chi \frac{h^{(1+1/\phi_h)})}{1+1/\phi_h}; \tag{27}$$

The budget constraint eq. (28) is simpler than the HA model, as it does not require a government equation. This is because there is no unemployment, so all agents have income to make interest payments. The lowest ideosyncratic productivity level, z, can fully cover interest payments on the full amount of debt, leaving money over. This makes consumption equal to income from work plus interest income, less savings.

$$c(a,z) = zh + ra - s(a,z) \tag{28}$$

The value function eq.(29) at steady state is now based on the utility of the consumption less dissutility from work, plus the value of the next asset grid level times the saving rate, plus the transition probability of moving to a new position times the difference in value between the current and future position.

$$\rho V(a,z) = \max_{c,h} \; u(c) - d(h) + v'_a(a)\dot{a} + \sum_{z' \neq z} \lambda_{zz'}(v(a,z') - v(a,z)) \tag{29}$$

Decision on FOC and velocity equation The FOC is now expanded to include the marginal disutility of work so that the marginal utility of consumption equals the marginal value of saving/disaving and the marginal disutility of labor. This is seen in eq. (30). The c and h can be removed with $u^{'-1}(v'_a(a,z))$ and $(d^{'-1}(v'_a(a,z)$. This forms a viscosity solution eq. (31) that allows the value function to be computed.

$$u^{'}(c) = v^{'}_a(a,z) = d^{'}(h) \tag{30}$$

$$\begin{aligned} \rho v(a,z) = \max_{c} \; & u(u^{'-1}(v^{'}_a(a,z))) - d(d^{'-1}(v^{'}_a(a,z)) + \\ & v^{'}_a(a)(zd^{'-1}(v^{'}_a(a,z)) + ra - u^{'-1}(v^{'}_a(a,z))) + \\ & \sum_{e' \neq e} \lambda_{ee'}(v(a,z') - v(a,z')) \end{aligned} \tag{31}$$

The borrowing limit rate of time preference was set to match the household debt level and real interest rates in the HA model. This enables comparisons between the models. This model is now much simpler as we have a simpler KFE eq. (32) as well as just the asset market clearing condition equation, eq. (26). The KFE equation now only focuses on agents moving in and out of each point of the distribution by saving in and out of each point, as well as the agents moving in and out based on changes to their idiosyncratic productivity.

$$\begin{aligned} \partial_t g_t(a,z) = \partial_a[s(a,z)g_t(a,z)] - g_t(a,z) \sum_{z' \in Z} \lambda_{zz'} + \\ \sum_{x \in Z} \lambda_{xz} g_t(a,z) \end{aligned} \tag{32}$$

5.4 Saving Motives

Each agent in the model faces three primary savings motives. First, there is the intertemporal savings motive, which depends on the relationship between the real interest rate r and the rate of time preference ρ. When $r < \rho$, agents are incentivized to borrow, as saving for future consumption is relatively less valuable. Conversely, when $r > \rho$, agents are incentivized to save as the return on the asset is greater than their rate of time preference. In the case where $r = \rho$, agents' desired consumption growth equals zero. In this model, the steady-state equilibrium satisfies $r < \rho$, en-

couraging young agents to borrow more and save less. Second, agents save to smooth consumption over the life cycle because the utility function is concave. This leads to the typical life cycle pattern of borrowing when young, saving during middle age for retirement, and dissaving in retirement. Third, agents engage in precautionary saving behavior in response to income uncertainty, which in this model arises from unemployment risk. This motive arises because the model employs a Constant Relative Risk Aversion (CRRA) utility function, which implies convex marginal utility $u''' > 0$. Uncertainty related to future employment status drives this behavior, particularly over the extensive margin. As unemployment increases, the probability and duration of unemployment also rise. The reduction in total hours worked along the extensive margin lowers the value function not only for unemployed agents, but also for employed individuals with limited asset holdings, who face heightened unemployment risk and lack sufficient savings to self-insure against periods of unemployment. As a result, agents exhibit a stronger precautionary savings motive, aiming to build positive asset positions from which they can draw down resources during future unemployment spells, thereby avoiding borrowing constraints. As a result, the precautionary motive becomes a key driver of the differences observed between the two labor adjustment margins.

The Huggett Model has lower idiosyncratic productivity levels that can mimic the low income associated with unemployment, along with the precautionary motive that accompanies it. The distribution of ideosyncratic productivity doesn't change, and this lacks the increased precautionary motive that will be felt in aggregate when total factor productivity is down, and more agents are unemployed. In summary, my model will have a precautionary savings motive that varies with unemployment.

5.5 Steady State

Figure (11) presents the steady state consumption and saving functions for four representative groups: employed 20 year olds, employed 45 year olds, retired 70 year olds, and unemployed 20 year olds. These functions are further analyzed using a mesh diagram that depicts savings behavior as a function of all of the age and asset grid levels for both the employed and unemployed. The mean asset holdings by age closely replicate the empirical profile of asset accumulation over the life cycle. Similarly, the model's unemployment rates are calibrated to match observed patterns. For instance, the unemployment rate for 20–24 year olds is approximately 1.7 times the overall unemployment rate, consistent with real world observations. This calibrates the unemployment rate assigned to 20 year olds in the model. The 25-34 age group is modeled to reflect the overall unemployment rate when combined, with both 25 and 30 year olds assigned values aligned with the aggregate data. For all remaining age groups, unemployment rates are relatively constant and below the overall unemployment rate.

The model also produces a parabolic before tax income profile by age, making all agents within age and employment status homogeneous, with all individuals aged 65 and older classified as unemployed/retired. Although the income gap between employed and unemployed 20 year olds is not substantial in absolute terms, the value of employment is significantly higher. This stems from the fact that employed individuals are permitted to borrow against future income, which enables them to increase current consumption and enhances their overall utility.

The asset distribution curve displays a large mass of individuals clustered around zero assets. This concentration is driven by both precautionary saving and life-cycle consumption-smoothing motives. Young employed agents tend to borrow to smooth consumption across their lifetime. Two distinct borrowing kinks emerge in the distribution: one near -3, where the 20 year olds become borrowing constrained, and another near -5, where 30 year olds hit the final grid constraint of -5. It is important to note that the majority of 20, 25 and 30 year olds do not hit these borrowing limits, indicating that only a subset of employed young agents are severely liquidity constrained.

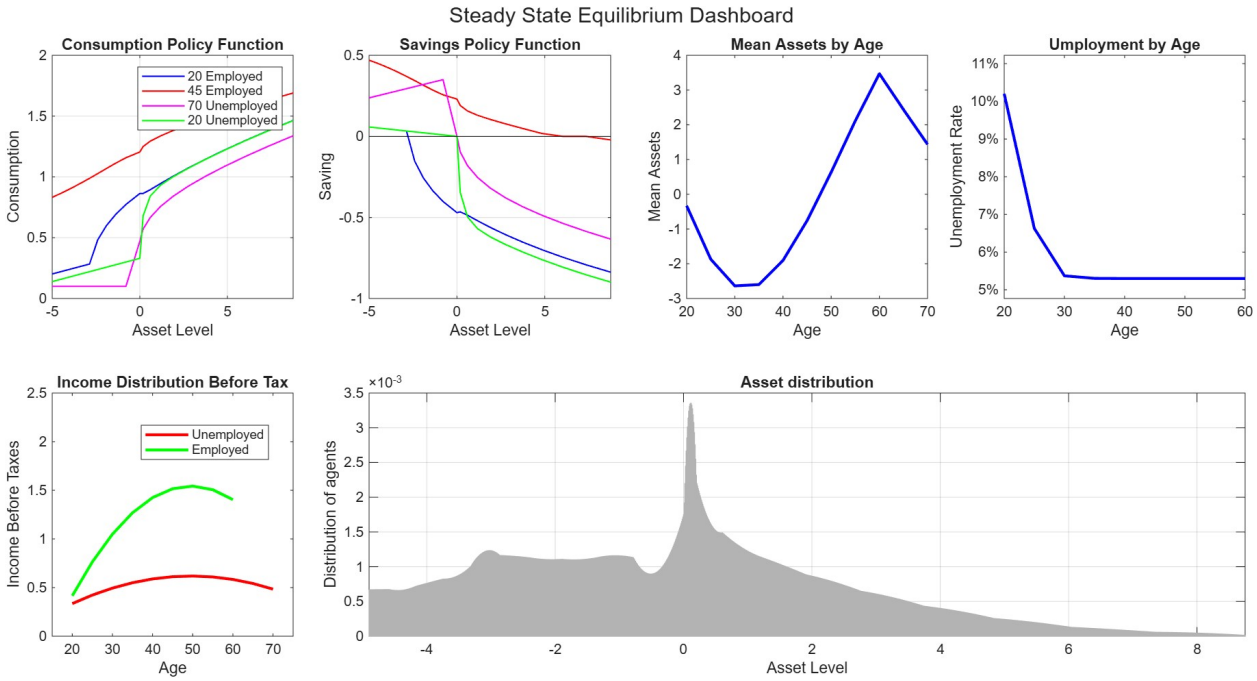


Figure 11: Steady State Dashboard

Figure (12) shows the dashboard for the Huggett model used for comparison against the HA model. The Huggett economy is calibrated to reproduce the HA model's steady state aggregate household debt and real interest rate, so that any differences can be attributed to model structure rather than to the starting calibration. The aggregate debt target is hit by shifting the minimum asset grid, which acts as the borrowing limit for all agents. The real interest rate is then pinned down by adjusting the rate of time preference. This means there is a different rate of time preference

between the HA and Huggett models. The Huggett benchmark departs from the HA model in two structural ways: households choose their own hours through the hours policy function, and idiosyncratic productivity follows a normal distribution rather than the life cycle process used in the HA model.

The consumption and savings policy functions show that households with higher idiosyncratic productivity tend to spend and save more than those with lower idiosyncratic productivity. Households with higher wealth levels tend to spend more and save less than households with less wealth at the same productivity level. A big departure from this paper's HA model is that agents pick their own hours. Agents with higher idiosyncratic productivity work more to take advantage of the temporary productivity boost, whereas wealthier agents work less because they already have large assets to smooth future idiosyncratic productivity shocks. The main exception is at the bottom of the income distribution, where hours rise sharply near the borrowing constraint. These households cut consumption to reduce the probability of hitting the debt limit. A large share of their income would be absorbed by interest on existing debt once they hit the debt limit and could no longer borrow to consume. They would rather consume less now so that the odds of hitting the debt limit will drop. The income distribution has 11 income levels to match the 11 age groups in the HA model. They are distributed normally. The asset distribution reveals the well known missing middle of heterogeneous-agent economies. The distribution resembles a water drop, with a higher concentration of agents at negative asset levels, which decreases as wealth increases. In reality, the highest concentration of assets is around zero, which matches the distribution of the HA model.

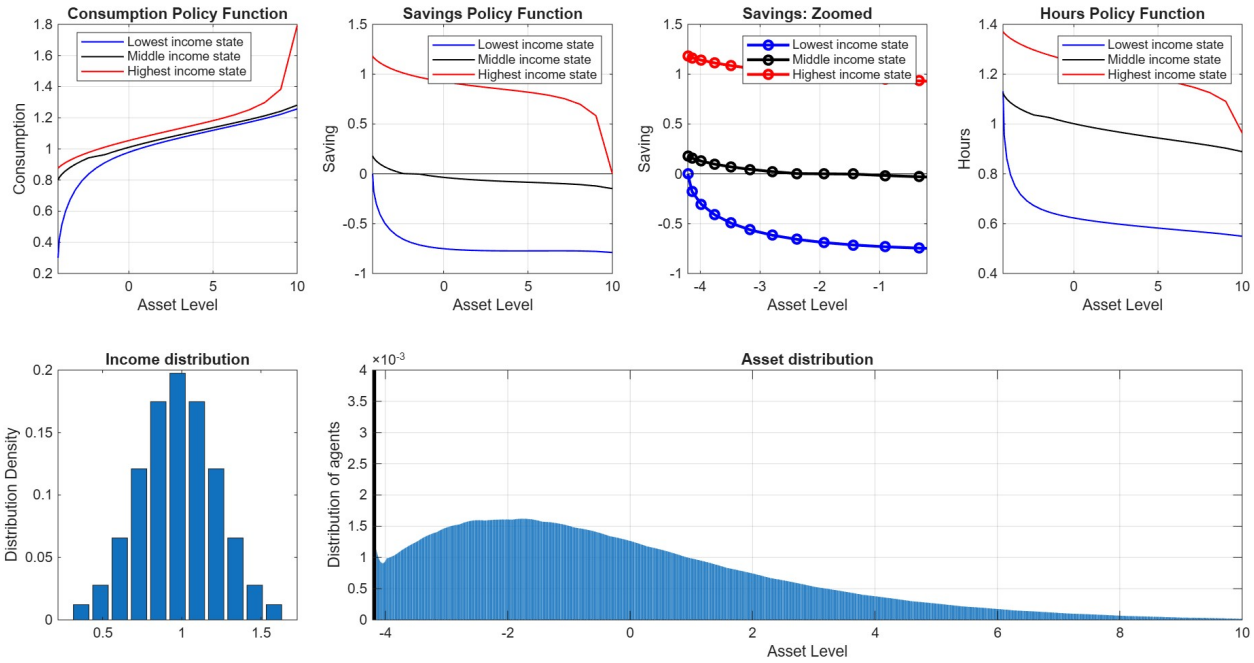


Figure 12: Huggett Steady State Dashboard

Figure (13) presents the equilibrium savings behavior for all employed agents. This does not include agents 65 or older because they have exited the labor force and thus do not have savings policy function. Notably, the asset axis in the figure is ordered from high to low. This display choice facilitates clearer visualization of the savings function's structure.

One key observation is that younger households tend to exhibit negative savings, reflecting borrowing. The points at which dissaving transitions from negative to zero indicate where borrowing constraints bind. A visible ridge forms around zero assets for young agents, representing agents who deliberately spend less to maintain positive asset balances for longer periods, thereby smoothing consumption in the event of future unemployment. The precautionary motive for saving drives this phenomenon. Another distinct feature occurs among agents aged approximately 35 to 60. Within this age range, individuals with lower asset holdings tend to save more aggressively. These middle aged households face a life cycle incentive to reduce debt and accumulate savings for retirement because their labor income remains high relative to other periods of their life cycle. Additionally, some wealthy 60 year old agents begin to dissave as their income drops before formally retiring, particularly those with substantial asset holdings. In contrast, 60 year olds with large negative asset balances are subject to higher savings rates that increase the chance of debt being paid off before retirement. This modeling choice reduces the likelihood that agents retire and ultimately die with negative asset positions. This preserves the model's integrity by forcing agents to repay their debt.

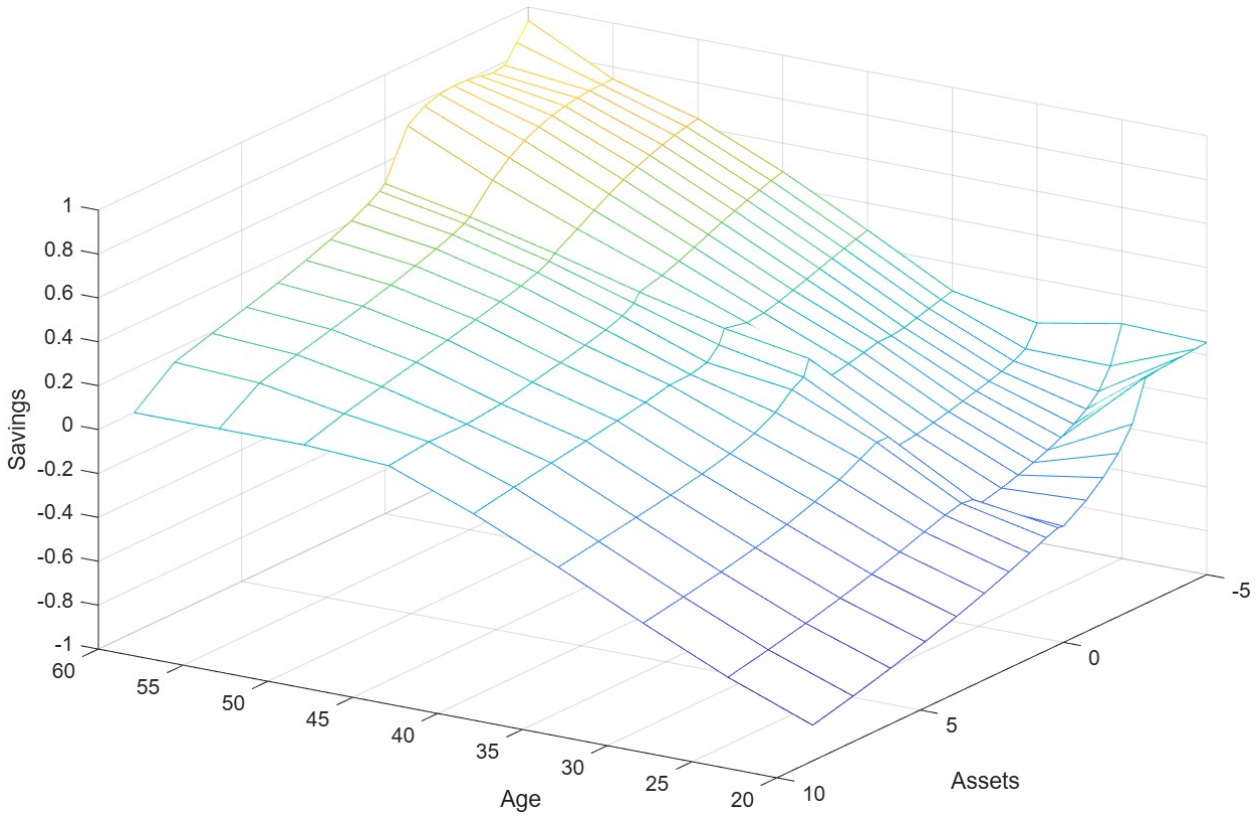


Figure 13: Employed Equm Savings Graph

In figure (14), the borrowing constraint tied to employment status is made clear by the savings behavior: once the constraint binds, savings are strictly non-negative. This reflects the model's assumption that unemployed individuals are not permitted to borrow and must make payments large enough to amortize the debt by retirement. Unemployed agents exhibit positive savings. This is due to the amortizing nature of their debt obligations, which require not only interest payments, but principal repayment. These payments increase with loan size and the agent's age, leading to higher required savings for older agents with outstanding debt. A notable pattern emerges among older agents with large negative asset balances. Agents with larger debts save less than their counterparts with smaller debt positions. This is driven by the minimum consumption constraint of 0.1. These agents consume the minimum amount and allocate the remaining income to savings after covering interest payments. However, agents with greater debt levels experience lower cash flows after interest obligations, reducing their capacity to save.

In contrast, agents tend to dissave until they approach the zero-asset constraint. A spike in dissaving is observed at age 70, reflecting a $\frac{1}{20}$ probability of death and the limited bequest motive. The bequest motive is the value of being 20 with the same asset level, representing grandparents capturing utility from the bequest they leave to their grandchildren. The value difference between asset grids is much smaller for 20 year olds because they can borrow to spend if they have a job. It

is important to note, however, that the average 70 year old holds approximately 0.5 units of assets. While dissaving behavior intensifies at this age, only a small proportion of agents have sufficient assets to sustain high levels of consumption.

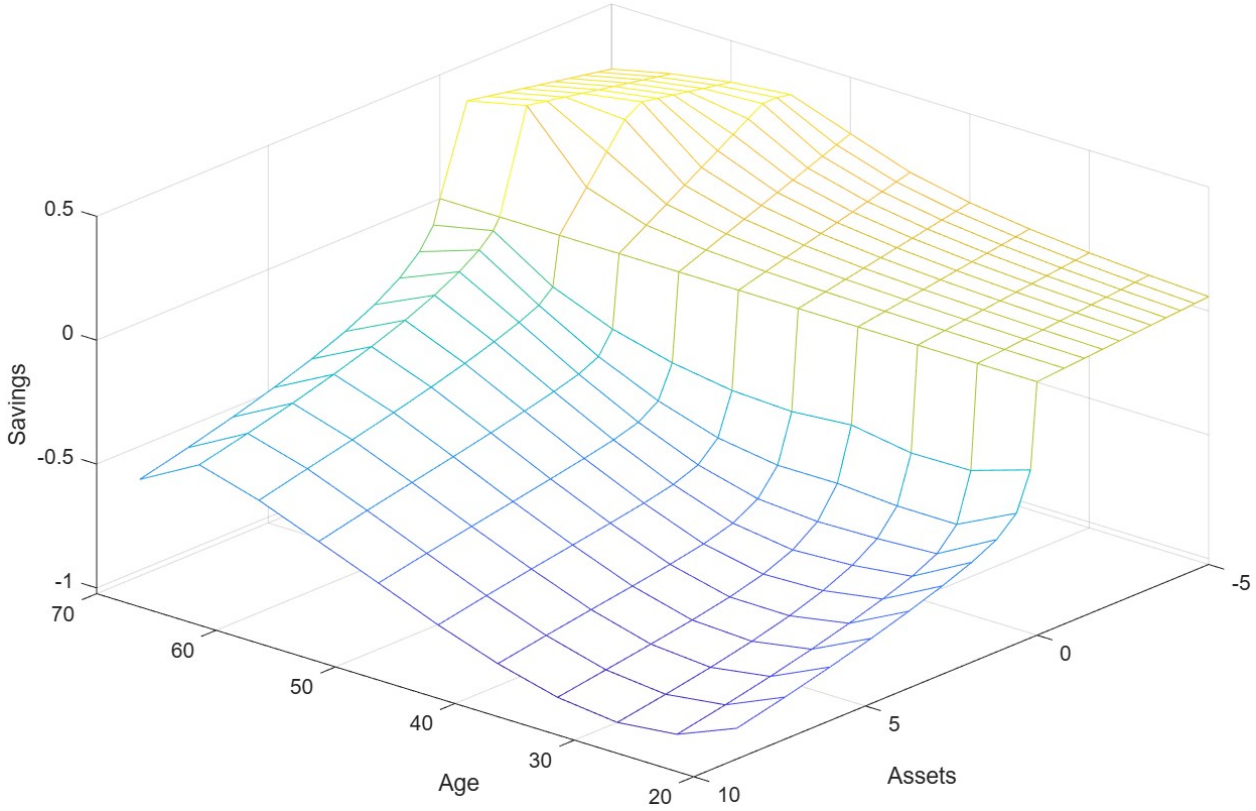


Figure 14: Unemployed Equm Savings Graph

Figure (15) shows the equilibrium savings behavior for all agents in the Huggett model. Only one graph is needed, as there are only 2 dimensions: idiosyncratic productivity and asset level. At the highest asset level, savings cannot be positive because you can't save beyond the maximum asset grid level. Savings drops with lower idiosyncratic productivity because it decreases income. Savings increase slowly as the asset level declines until the saving rate kinks upward for every productivity state as the borrowing limit is approached. For high-productivity agents, the savings are higher, pushing them further away from the borrowing limit in the event of a negative productivity shock. The low income borrowing constrained agent is forced to save zero at the very least.

The HA model has many more complex shapes, based on borrowing limits and the zero asset borrowing limit for the unemployed. The key difference is that you see savings for older agents increase dramatically around the zero asset line, and for young agents, the savings increase dramatically near the borrowing limit. The saving of the unemployed increases drastically around zero. Because of these differences and life cycle dynamics, the mass of the asset distribution is

concentrated near zero.

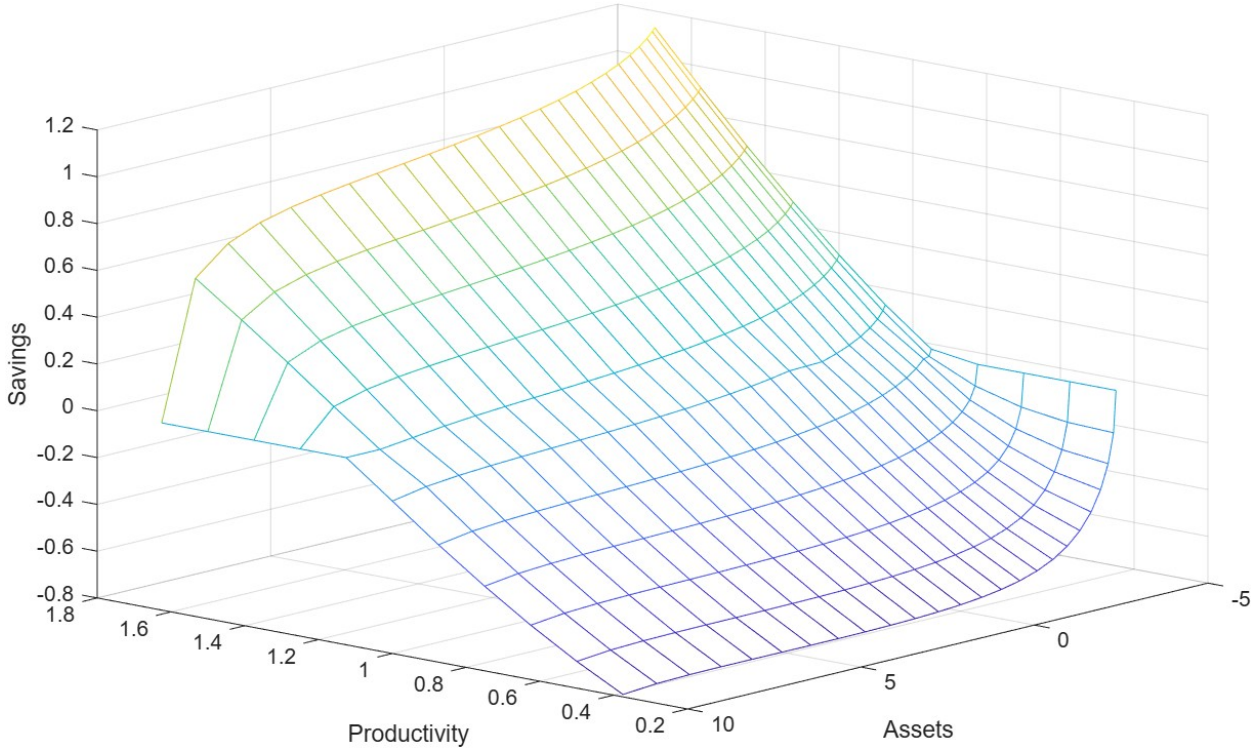


Figure 15: Huggett Equm Savings Graph

5.6 Market Shock

I compare the effects of negative shocks to total factor productivity, A_t, over the intensive and extensive margins of the HA model to identify how changes to loan quantity, real interest rates, and utility differ between margins. Productivity shocks reduce the marginal product of labor (MPN), thereby decreasing labor demand. The impact on employment depends on whether the adjustment occurs at the intensive margin (hours worked by agents) or the extensive margin (the number of employed agents).

To understand these two adjustments, we must first look at the labor market. Figure (16) illustrates the structure of the labor market in the model. Full employment is represented by a vertical line, reflecting the assumption that all individuals below age 65 want to work one unit of time, while those aged 65 and above are retired and do not work. For simplicity, there is either no disutility associated with working or the disutility of working is smaller than the utility gained from working, so that all agents under 65 will choose to work. The labor supply slope is set to equal the empirical macroeconomic Frisch elasticity. Labor demand eq.(33) is determined by the marginal product of labor (MPN), which depends on total factor productivity. In this setting, firms take the wage curve as given and adjust the quantity of total labor demanded in response to productivity

shocks, either by reducing hours worked by agents (the intensive margin) or by reducing the number of employed agents (the extensive margin). The model assumes that labor demand always lies to the left of the full employment line, resulting in unemployment. The intersection of the labor supply and MPN determines total employment, with the gap between full employment and labor demand representing involuntary unemployment because the wage set by the labor supply is higher than the wage that would clear the market at full employment.

$$N_t = \left(\frac{(1-\alpha)A_t}{w_t} \right)^{\frac{1}{\alpha}} \tag{33}$$

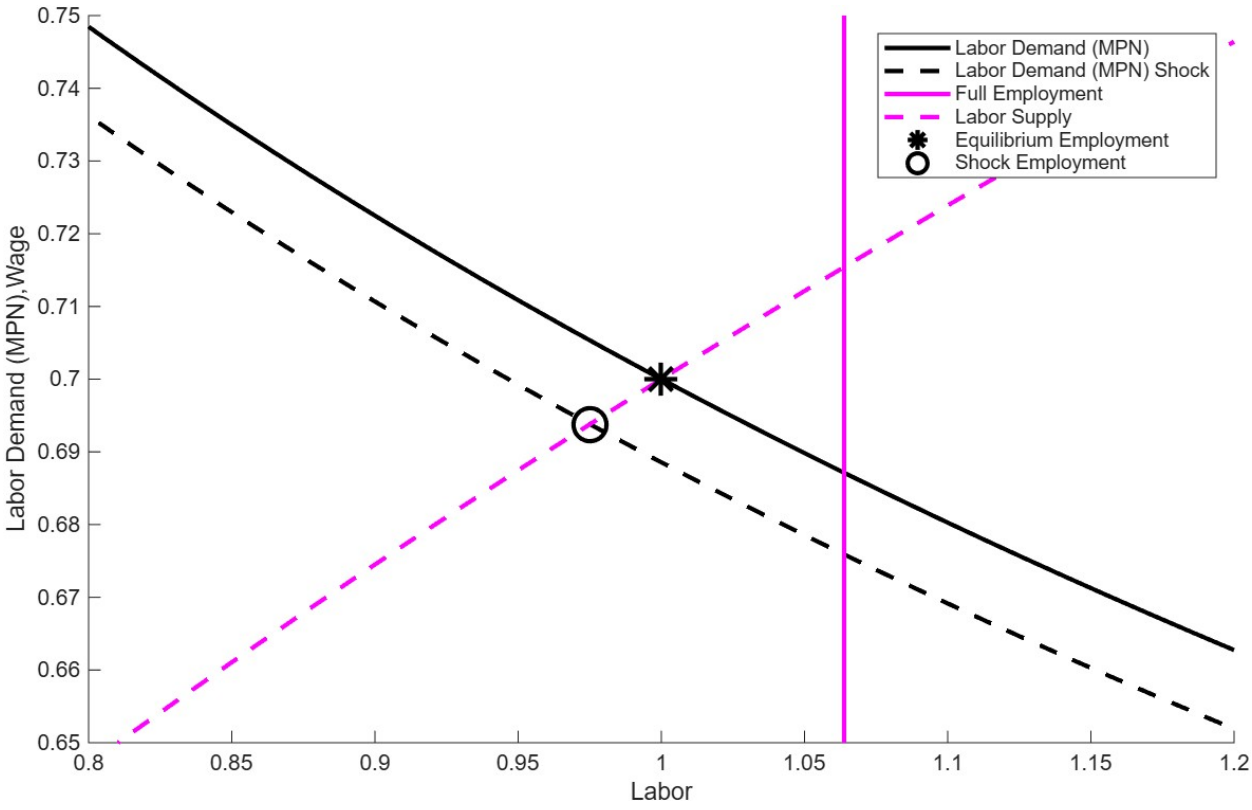


Figure 16: Labor Market

Figure (17) illustrates how utility responds to labor market shocks in a simplified incomplete markets environment. This simplified setting assumes agents earn an income of 1 when employed (working one unit of labor) and 0.4 when unemployed. The agent consumes their entire income. This hand-to-mouth behavior abstracts from saving but will highlight the implications of labor market uncertainty.

Referencing the labor market framework in Figure (16), a negative productivity shock reduces labor demand, forcing firms to either cut hours or reduce employment. If all agents reduce their

hours uniformly, or if all agents share unemployment risk completely, consumption for all agents declines smoothly along the black line toward the complete market utility (represented by the black circle). This is represented by equation (34). In contrast, in an incomplete market, agents face unemployment risk without the ability to insure; utility falls to the incomplete market outcome (magenta star) along the dashed magenta line, as expected utility is now based on the weighted values of being employed or unemployed, given uninsured income risk. The utility of the incomplete market is represented by equation (**??**). This reflects heightened exposure to income volatility caused by unemployment and a lower utility level. In asset based models, this utility difference will lead agents to have a stronger precautionary savings motive.

$$U = \ln((1-u)*1+u*.4) \tag{34}$$

$$U = (1-u)*\ln(1)+u*\ln(.4) \tag{35}$$

In a more realistic setting, when agents anticipate rising unemployment, their precautionary motives increase, leading to changes in saving behavior. Despite this increased precautionary motive, younger individuals continue to borrow for several reasons. First, they earn less and aim to smooth consumption over their life cycle. So it is still utility maximizing for them to borrow. Second, their income is closer to the level of unemployment benefits, reducing the value difference between employed and unemployed among the young compared with older age groups. Third, intertemporal preferences encourage borrowing because the real interest rate is below the discount rate. Lastly, they typically have negative net wealth and must repay debt before they can accumulate precautionary savings, leaving them unable to self-insure and exposed to consumption risk. The reduction in borrowing among young individuals is driven primarily by their inability to borrow during periods of unemployment, as highlighted in the OLG model, not by a reduction in borrowing from the young employed households.

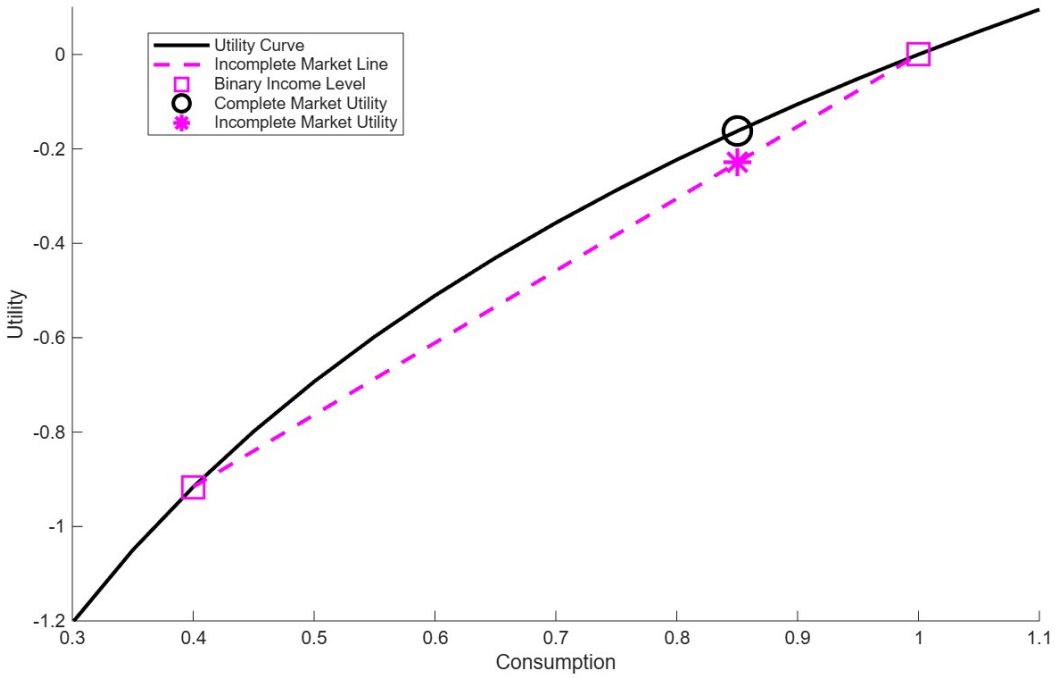


Figure 17: Incomplete Market

Figure (18) shows what happens when there is a permanent productivity shock to a simplified hand to mouth Model. This is done to demonstrate, in a simple setup, why hours worked remain roughly unchanged in the Huggett model regardless of changes in productivity. A decrease in productivity shifts Labor Demand (MPN) down because $w_t - A_t$. The labor supply schedule is given by eq. (36). With log preferences, σ = 1, and hand to mouth, $C_t = A_t * N_t$, the labor supply simplifies to $N_t = \psi^{\frac{-1}{1+\phi}}$. The new intersection of labor supply and labor demand intersects at the same hours worked at a lower wage.

$$w_t = \psi C_t^{\sigma} N_t^{\phi} = \psi A_t N_t^{\phi+1} \tag{36}$$

This point deserves emphasis despite the hand to mouth because it explains how labor stays relatively unaffected in the complete Huggett model. The HA model in this paper differs in that the quantity of labor can drop permanently on both the extensive and intensive margins, whereas the Huggett model does not have meaningful changes in hours worked. In the extensive HA model, a permanent productivity shock can lower employment on the extensive margin, causing the resulting rise in unemployment risk to shift the precautionary savings motive even in the new steady state.

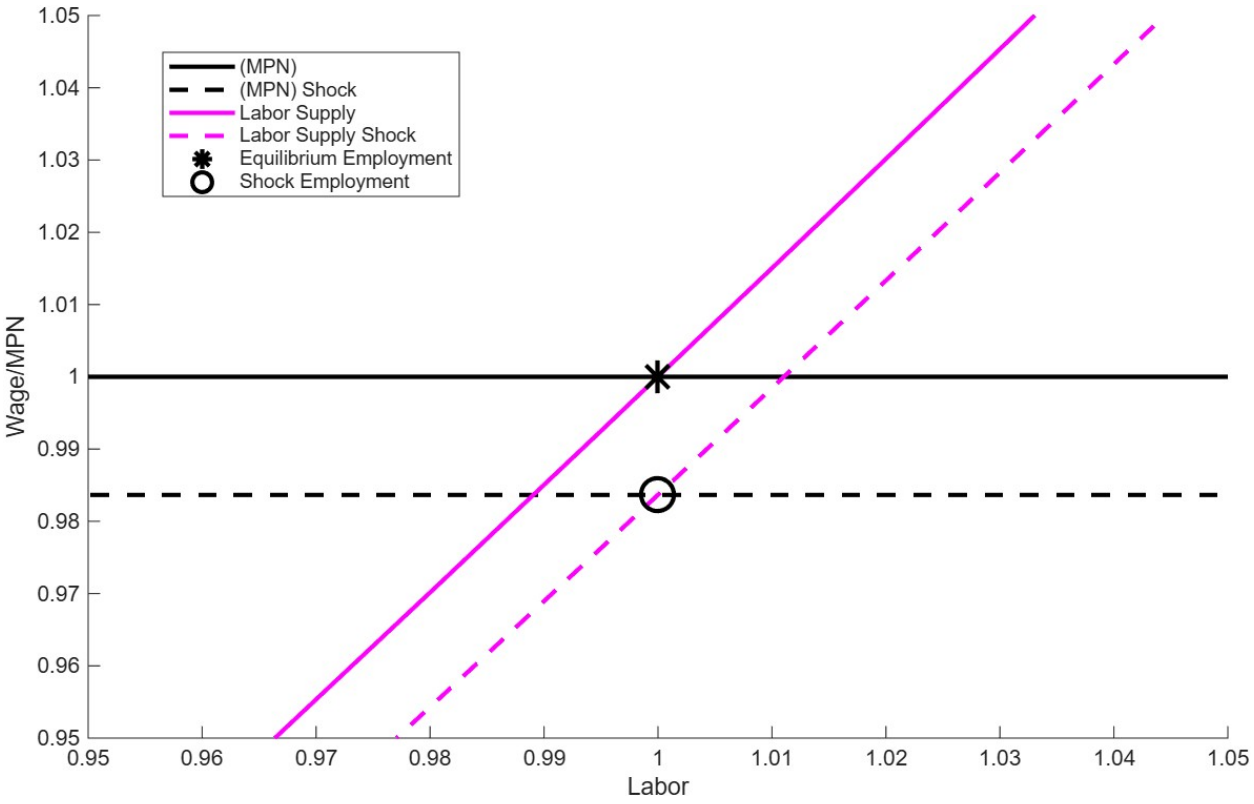


Figure 18: Huggett Labor Supply Decision in SS

5.7 Intensive vs Extensive Steady State

This section conducts a comparative statics analysis of the intensive and extensive margins at lower productivity levels. This serves as the primary identification strategy for isolating the extensive margin's impact relative to that captured by the intensive margin, given the life-cycle component. Both margins use the same calibrations from the same steady state, incorporating the same wage levels, age specific productivity, the employment flow methodology, effective labor levels, profit function, borrowing constraints, and market clearing conditions. Thereby providing a realistic benchmark for evaluating the response along each margin. The distinct effects of each margin on real interest rates, quantity of loans, utility levels, and broader macroeconomic outcomes will be compared.

Both margins are studied as deviations from the steady state. A negative productivity shock lowers the marginal product of labor (MPN) and reduces labor demand. Under the intensive margin, hours worked fall uniformly across all employed agents. At the extensive margin, the number of employed agents declines through simultaneous changes in hiring and separation rates. Unemployed agents continue to supply zero hours.

$$N_t = A_t^{\frac{1}{\phi+\alpha}} \tag{37}$$

Effective labor declines in line with aggregate productivity, as shown in Eq. (37). Figure (19) displays the model's key outcomes following the intensive margin adjustment. Although consumption and saving profiles remain broadly consistent with the steady state, the consumption policy function declines modestly and the saving function shifts toward zero. Age specific employment transition dynamics are unchanged. The asset distribution is largely preserved, but fewer young employed workers reach the borrowing constraint because desired borrowing decreases with lifetime income. Asset holders likewise accumulate slightly less wealth. Dividend income declines, lowering pre-tax income for both employed and unemployed agents, with a larger decline for the employed due to the direct reduction in hours worked and, hence, earnings.

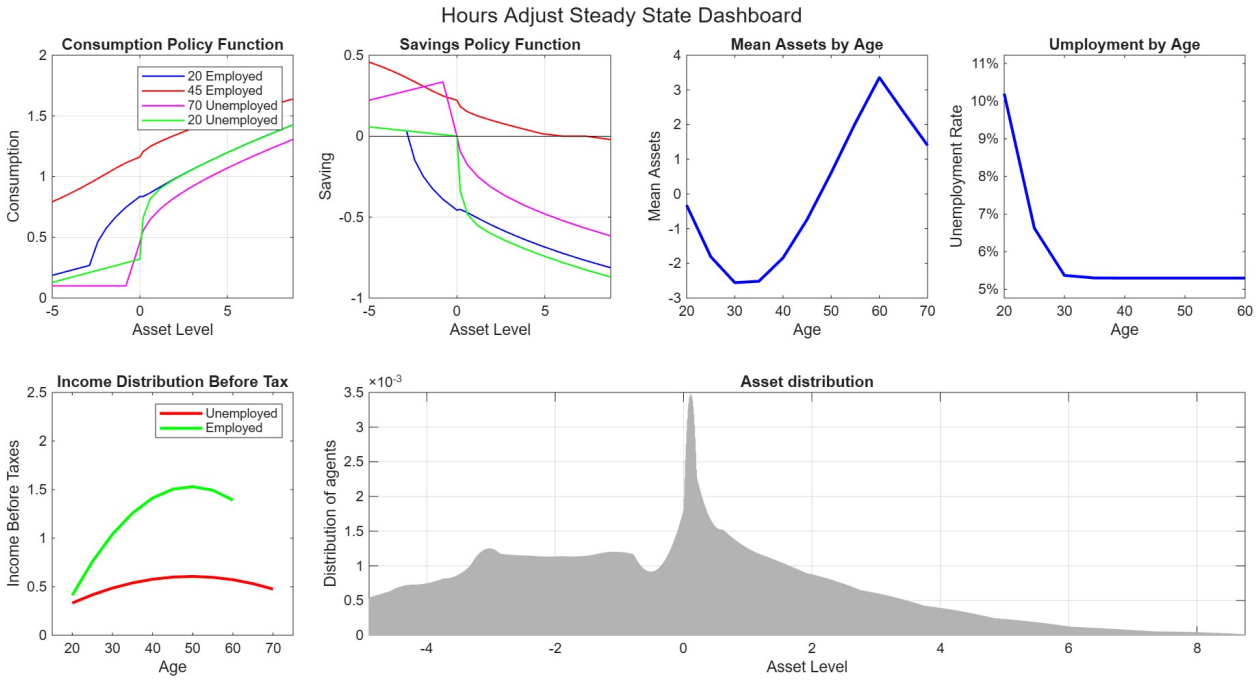


Figure 19: Intensive Adjust Dashboard

Adjustments along the extensive margin are implemented by modifying the hiring and separation rates. These rates are interlinked such that a 2.5% reduction in the hiring rate corresponds to roughly a 1% increase in the separation rate, consistent with the notion from (Shimer 2012) that 77% of employment fluctuations are caused by changes in the hiring rate. Separation rates for 20 and 25 year olds maintain the same mark up over regular separation rates, as does the proportion of retiring agents whose positions are filled by younger workers. The hiring and separation rate parameters are calibrated to ensure that effective labor aligns with the firm's desired labor demand.

Figure (20) displays the model's outcomes following the extensive margin adjustment described above. Both the separation probability and the duration of unemployment increase, driven by reduced hiring activity and a larger pool of unemployed agents awaiting hire. As a result, the overall unemployment rate increases, particularly among younger cohorts, with 20 and 25 year olds disproportionately affected. While pretax wage income decreases for employed individuals due to lower wages, dividend payouts decline as firm profits decline. The consumption and saving policy functions experience mild shifts, including a noticeable increase in savings for employed 20 and 45 year olds, reflecting the intensified precautionary savings motive.

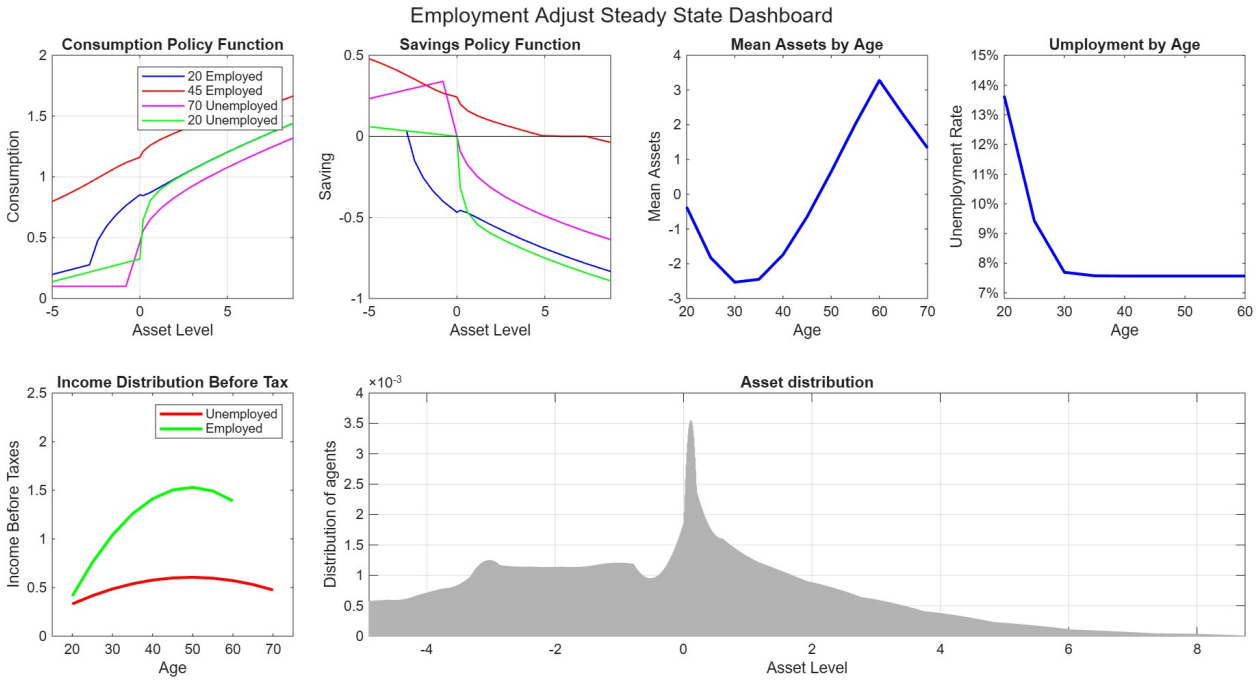


Figure 20: Extensive Adjust Dashboard

Figure (21) illustrates the shifts in loan demand and loan supply in response to a permanent aggregate productivity shock. The initial loan supply is represented by the magenta line and loan demand is represented by the black line, with the black star indicating the initial equilibrium. Loan quantities are expressed relative to the real GDP in the steady state. The real interest rate is initially below the rate of time preference, ρ, due to the strength of the precautionary savings and retirement planning motives outweighing young agents' incentives to borrow and smooth consumption. At equilibrium, the quantity of loans is roughly 1.1 times annual GDP, or 110% of annual GDP. This is significantly higher than the roughly 60 % to 100 % range that the US household debt has been in since the 1990's. This is because the model has no other forms of debt, and government expenditures equal zero. First, government debt is set to zero for simplicity, and there is no corporate debt in this model. Adding corporate and government debt would crowd out household borrow-

ing, leading to higher overall borrowing but lower household borrowing. Additionally, if we set government expenditures to a non-zero number, this will lower after tax income throughout the life cycle as taxes increase to pay for spending. This will reduce the extent of income smoothing and lower borrowing from the young and saving from the middle class. While this will change the loan amount, it does not affect the direction of the shifts in the loan supply. Because our Huggett Model does not include government spending or debt, the HA comparison will set both to zero to better compare the baseline model rather than the effect of adding government debt and spending. An example of a loan amount graph, Figure (46), is placed in the appendix to show that there is no major change.

The age specific borrowing limits are fixed at their steady state values. This assumption matters because the asset grid is discrete: a borrowing limit does not change continuously but jumps only when the interest rate shifts enough to reach an adjacent grid point. This discreteness creates two difficulties. First, it produces jagged figures. When a change in the interest rate triggers a discrete jump in the borrowing limit, the associated loan amount also jumps, so the plotted response shows visible kinks. This is purely cosmetic, not a problem with theory, but it makes the figures harder to read. Second, it complicates the computational methods. The model is solved with a quasi-Newton method, which presumes that small changes in the interest rate induce proportionally small changes in the loan amount. When the interest rate crosses the relevant threshold, the loan amount instead jumps discretely, causing the solver to overshoot and requiring additional iterations to converge. Finally, because the borrowing limits remain fixed at their steady state values when computing the impulse response functions (IRFs), this source of computational problem does not propagate into the IRF computation.

The intensive margin results are the circle. This represents higher interest rates and lower quantity of loans compared to the original steady state represented by the star. The dashed black line shows the intensive margin loan demand following a -1% productivity shock, with firms responding by reducing hours worked for all employed agents (an intensive margin adjustment). This results in a modest leftward shift in the demand curve. With reduced productivity, agents have lower incomes across the life cycle, diminishing the need for intertemporal consumption smoothing among young agents. The dashed magenta line shows a decrease in loan supply under the same productivity shock and intensive margin adjustment. Here, the loan supply shifts slightly to the left due to two primary effects. First, a reduction in income lowers the quantity of lending by middle aged agents, as they have less income available to save for retirement consumption. The second reason is that there is a weaker incentive to save for precautionary reasons. With unchanged transfers and lower employment income, the difference between the value of becoming unemployed and the value of being employed diminishes. This weakens the precautionary motive. However, one offsetting effect is that agents accumulate less debt. This additional wealth creates a motive to

smooth some of this extra consumption potential forward.

The square marks the new equilibrium when the productivity shock is absorbed through the extensive margin, with firms maintaining hours while reducing the number of workers employed. This scenario results in a larger contraction in the total quantity of loans and real interest rates that have instead decreased. The model has a similar intertemporal consumption smoothing effect on both loan supply and loan demand, but now has two additional factors to consider. First, there is a sharper decline in loan demand (dotted black line) when aggregate hours adjust over the extensive margin rather than the intensive margin. The rise in youth unemployment primarily drives the decline in loan demand, as unemployed agents are unable to borrow. Upon transitioning to employment, these agents do not accelerate their borrowing sufficiently to offset the borrowing shortfall incurred during their period of unemployment. Second, loan supply(dotted magenta line) (dotted magenta line) increases when total hours are adjusted on the extensive margin, rather than contracting when total hours are adjusted on the intensive margin. The heightened desire to save as a precaution is driven by increased unemployment risk. Standard models often overlook this outcome, typically treating hours worked as the primary margin of risk, even though changes in unemployment risk are empirically more common and substantially more significant.

These findings underscore a key insight: even modest declines in productivity can generate substantial shifts in the loanable funds market, thereby affecting the quantities of loans and real interest rates. The adjustment of employment between the intensive margin (hours worked) and the extensive margin (number of employed agents) has significant implications, both empirically and within this model.

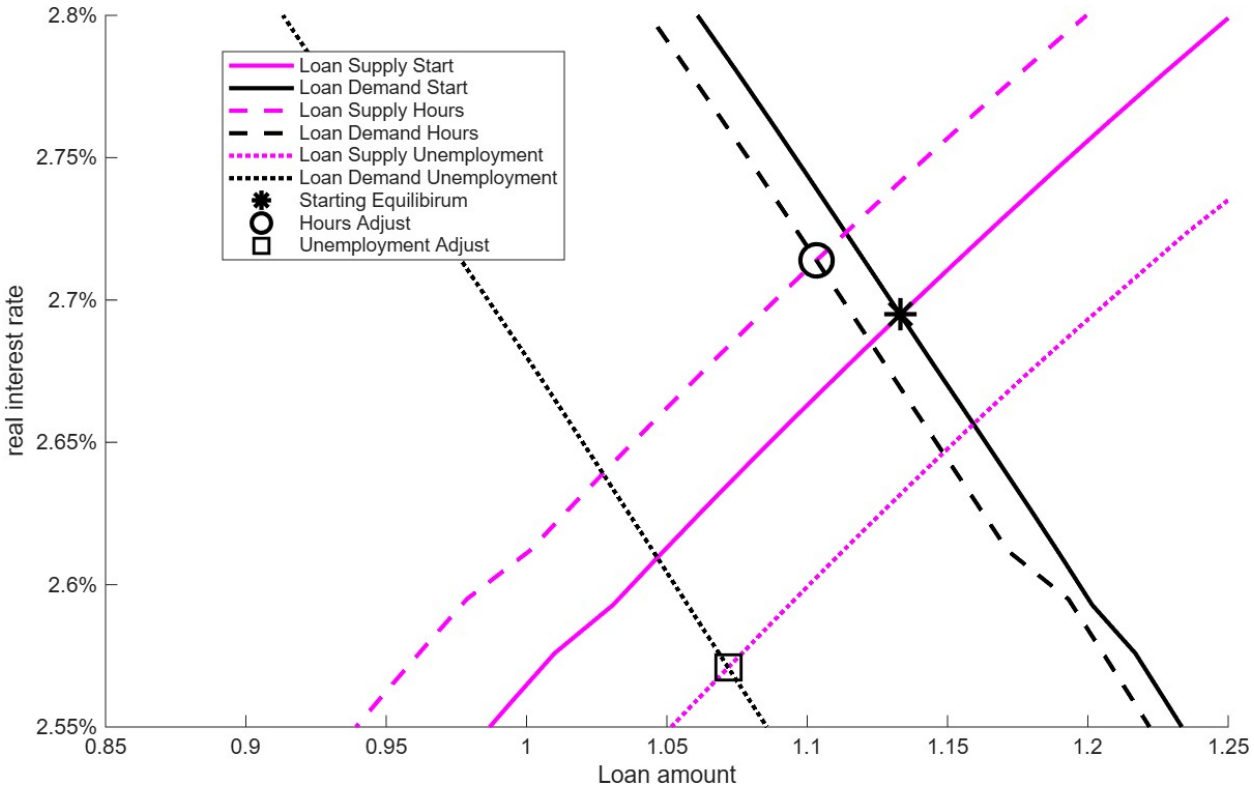


Figure 21: Heterogeneous Model Loan Demand and Loan Supply

Figure (22) outlines the real interest rate of each margin model as the productivity shock that generates unemployment in the extensive margin model varies. The black line displays the extensive margin real interest rates, which fall steeply. An increase in unemployment from 6% to 20% decreases real interest rates by a roughly 0.7 percentage points. The magenta line plots the intensive margin real interest rate evaluated at the same productivity levels that would cause the unemployment rate in the extensive margin model. Instead of the real interest rate dropping by 0.7 percentage points in response to the same productivity shock, it increases by roughly 0.1 percentage points. The resulting wedge reflects an economically meaningful difference. In the extensive margin, unemployed workers can no longer borrow, thereby contracting loan demand and compressing rates, whereas in the intensive margin, unemployment does not change, and agents continue to borrow while loan supply decreases faster than loan demand. Consistent with the comparative statics of loan demand and loan supply in Figure (23), this divergence implies that the choice of adjustment margin materially affects the transmission of productivity shocks in the loanable funds market.

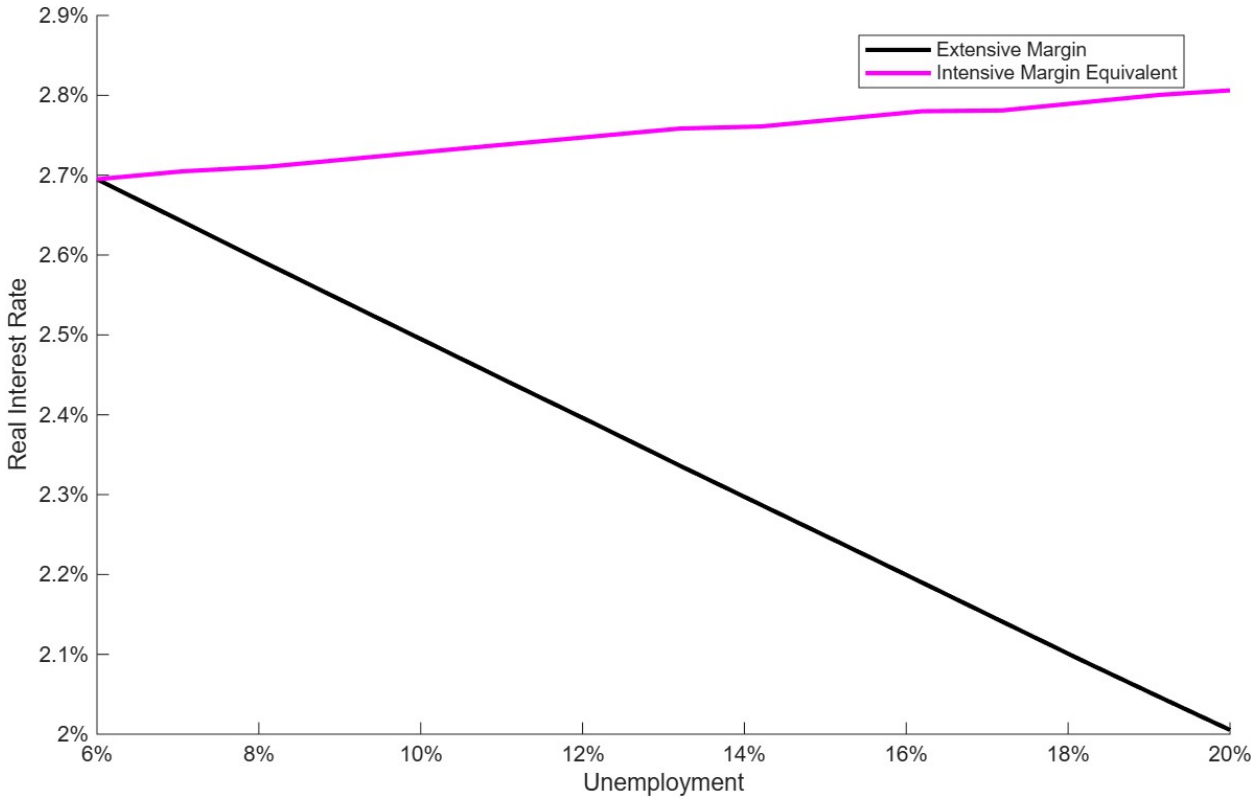


Figure 22: Heterogeneous Model Interest Rate

Figure (23) traces the response of real household debt to a productivity shock of increasing magnitude. The black line denotes the quantity of real household loans associated with the unemployment rate arising from the productivity shock in the extensive margin model. When the unemployment rate rises from 6% to 20%, household debt declines by roughly 29 percentage points. The magenta line plots the intensive margin real interest rate at the same productivity levels that generate the unemployment rates in the extensive margin model. In response to the identical shock, the real rate falls by only 16 percentage points. The smaller response is not only a difference in magnitude but also in mechanism. When labor adjusts along the extensive margin, unemployed workers lose access to borrowing, so loan demand contracts far more sharply than when hours worked adjust.

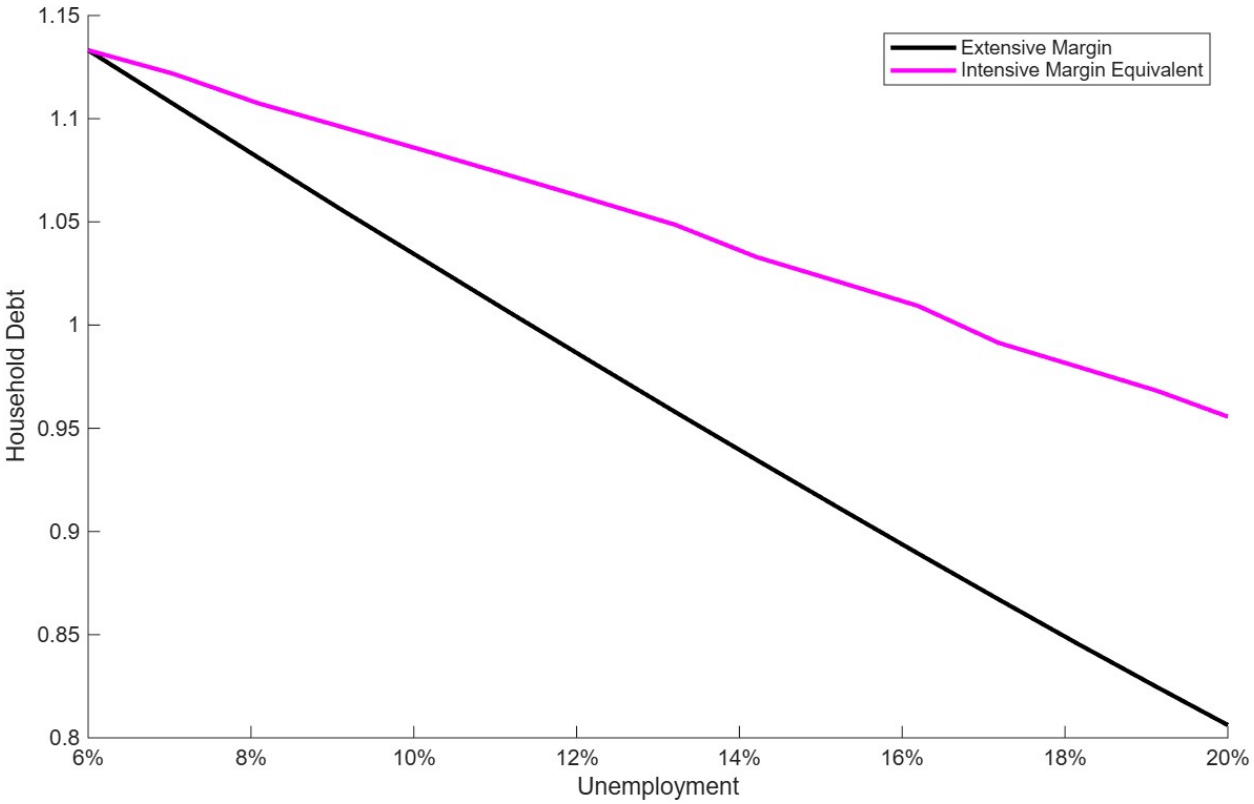


Figure 23: Heterogeneous Model Loan Quantity

Figure (24) illustrates the differences in saving behavior between the intensive and extensive margins for employed individuals. One of the most notable patterns is that indebted individuals aged 20 and 25 exhibit greater dissaving near their borrowing constraints in the extensive margin scenario than in the intensive margin scenario. Under the extensive margin, the lower real interest rate reduces the cost of servicing debt. Consumption and utility are therefore higher when at the debt limit in the extensive margin model, so the expected utility cost of hitting the limit falls. This weakens the precautionary motive to avoid the constraint and supports greater dissaving at ages 20 25.

Between ages 25 and 45, the largest divergence in saving behavior occurs at just above zero assets. Younger individuals who anticipate drawing down their savings do so more slowly at the extensive margin. This reduces the speed at which a zero asset balance is reached in the event of unemployment. In contrast, individuals aged 30 and older with negative asset levels have a higher savings rate, enabling them to move beyond the zero-asset threshold and smooth consumption if they become unemployed. At asset levels around 3, individuals with substantial positive savings tend to save less under the extensive margin. These agents are effectively self-insured against unemployment risk. This reduces the strength of the precautionary saving motive. Furthermore, in the extensive margin scenario with a lower real interest rate, the intertemporal trade-off begins

to dominate the effects of the precautionary savings motive. This makes current consumption relatively more attractive to these agents, thereby reducing the incentive to save further.

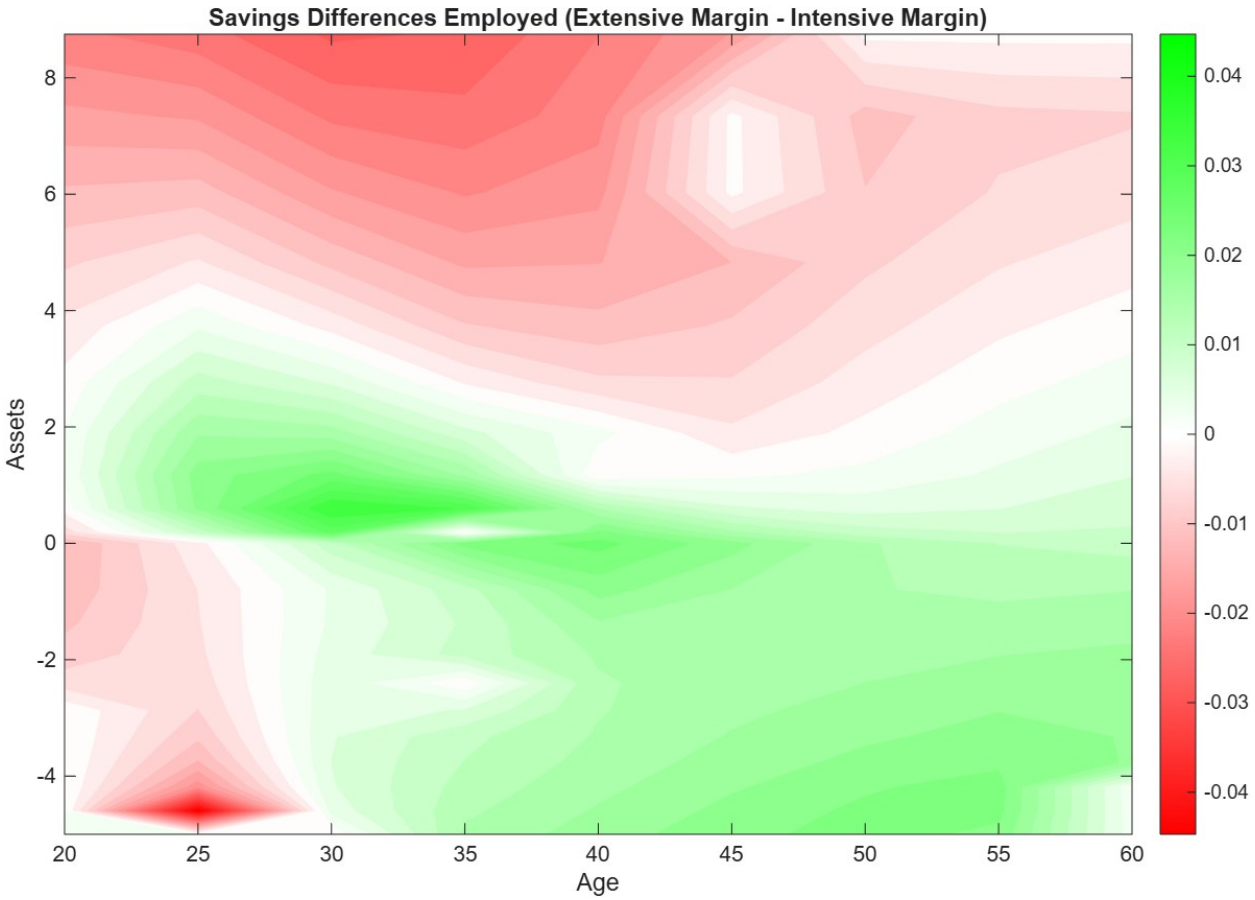


Figure 24: Savings Curve Difference Extensive vs Intensive Employed

Figure (25) illustrates the differences in saving rates (extensive minus intensive margin) by asset holdings and age, holding other model parameters fixed. Three regions stand out. First, for unemployed individuals with assets above roughly 1, saving is more negative under the extensive margin. Because the extensive margin is associated with a lower real interest rate, the motive to consume today is stronger, prompting these agents to draw down assets more rapidly. Second, for working age individuals with positive, but low asset holdings, the differential reflects a higher savings for the extensive margin. This is caused by a stronger precautionary saving motive due to higher unemployment transition risk and longer expected jobless spells. Agents are effectively self insuring against an extended unemployment spell. Third, in the debt constrained region, from roughly 5 in assets for 50 year olds to zero assets for 70 year olds, saving rates are higher under the extensive margin. In this range, the minimum consumption floor of 0.1 binds, and assets are devoted to debt repayment. Because the extensive margin features a lower interest rate, agents pay

less interest on their debt; with consumption pinned at the floor, the freed resources are channeled entirely into additional savings and principal repayment.

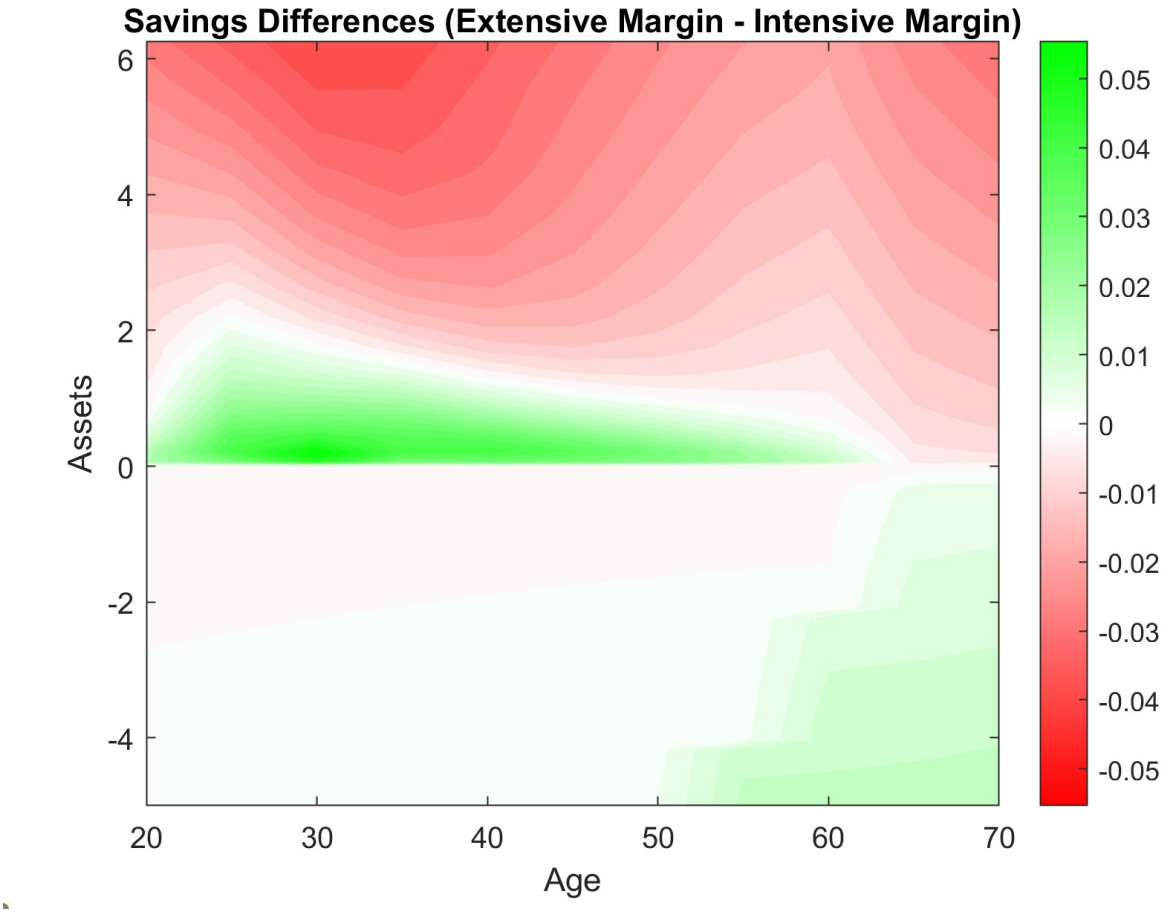


Figure 25: Savings Curve Difference Extensive vs Intensive Unemployed

Figure (26)) reports the welfare effects of a productivity shock that yields identical aggregate output under two alternative labor adjustment margins. Dashed lines plot the average utility across all agents. The solid lines plot utility for 25 year olds. Black lines show the extensive margin model, and magenta lines show the intensive margin model. The x-axis is the extensive margin unemployment rate. For the intensive margin economy, each model-implied employment outcome is mapped to the productivity shock equivalent unemployment rate, so both economies can be compared on a common scale. Under the intensive margin, households are able to smooth their consumption with the now lower level of income. Under the extensive margin, unemployment severs both labor income and borrowing capacity, so constrained households cannot smooth consumption. Because the two economies have identical aggregate output, the entire welfare gap in the figure is attributable to this constraint rather than to differences in output. The comparison is not made for unemployed 20 year olds because most received a recent intergenerational bequest,

inflating their assets and making the comparison of employment status not equivalent. The 25 year old comparison is clean because both employed and unemployed agents have a similar asset level. The comparison shows that the extensive margin welfare loss grows faster with the shock. The welfare loss for 25 year olds is roughly 4.2% larger than for the average agent when looking at the extensive margin, confirming that the young bear a disproportionate share of unemployment risk.

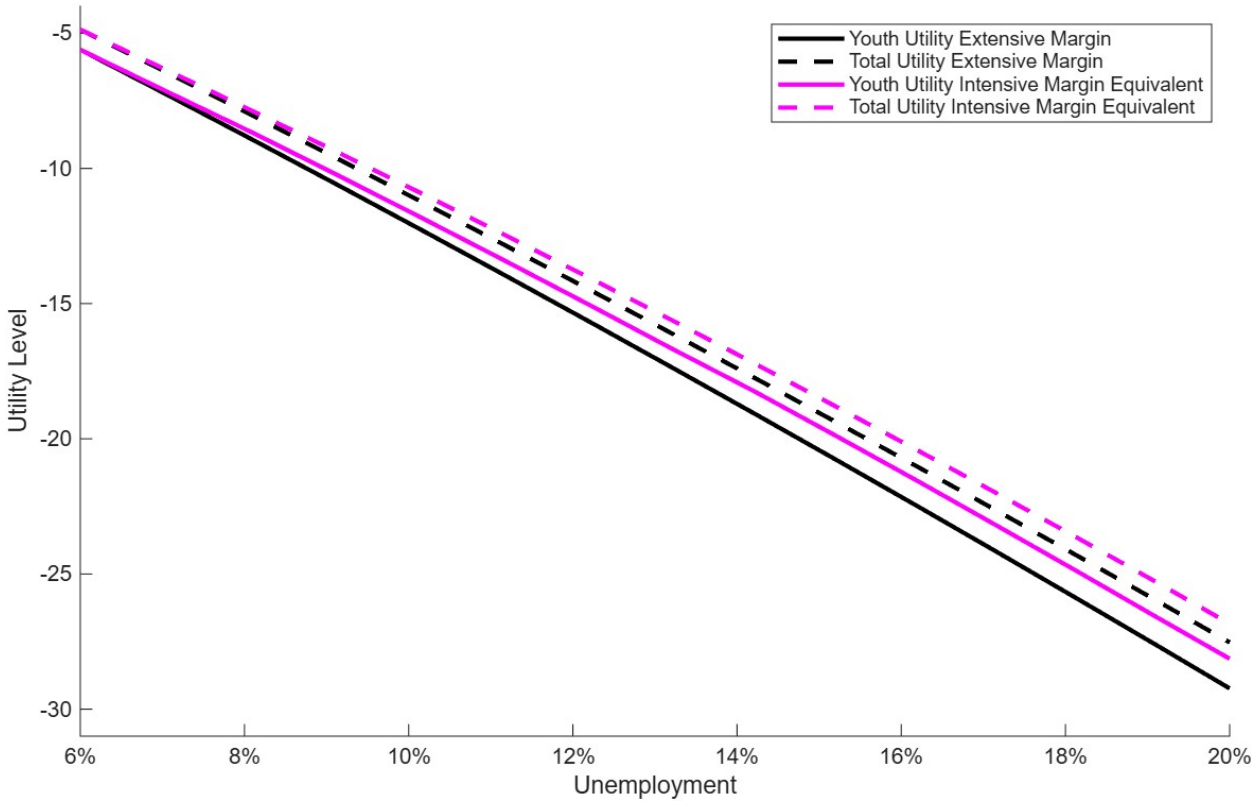


Figure 26: Utility

5.8 Huggett model Response

The Huggett model yields different outcomes from both HA life cycle models we estimated. To assess how the extensive margin mechanism differs from the existing literature, I solve the standard Huggett model as a benchmark. The graphs of loan demand and loan supply for the HA model show significant differences in interest rates and loan quantity, driven by the extensive margin. While this was nice to see, it would make sense to run a standard Huggett model to assess how the extensive model differs from the existing literature. Figure (27) highlights the difference in the Huggett model before and after a permanent productivity shock. The size of the graph is the same as Figure (21) to show the differences on the same scale. Two things should stand out. First, the lines are less steep, so loan demand and loan supply are much more elastic. This higher elasticity

is intuitive. In the HA model, young households with persistently low incomes exhibit interest inelastic loan demand, while middle-aged households save for retirement to smooth consumption, almost regardless of the interest rate. The Huggett model has neither feature, so its demand and supply curves respond much more strongly to the interest rate. The second is that the Figure (18) shows that there will be very little change in the labor each household provides in response to a permanent shock. Real income drops due to the productivity shock. Hours worked increase slightly for low income households around the borrowing limit. The borrowing limit relative to income rises only marginally. This results in such a small change in the loan supply and loan demand curves that the difference is hard to see in the graph. The black loan demand curve and the post shock dashed black line sit almost on top of each other. The same goes for the magenta loan supply and the dashed magenta line representing the loan supply post productivity shock.

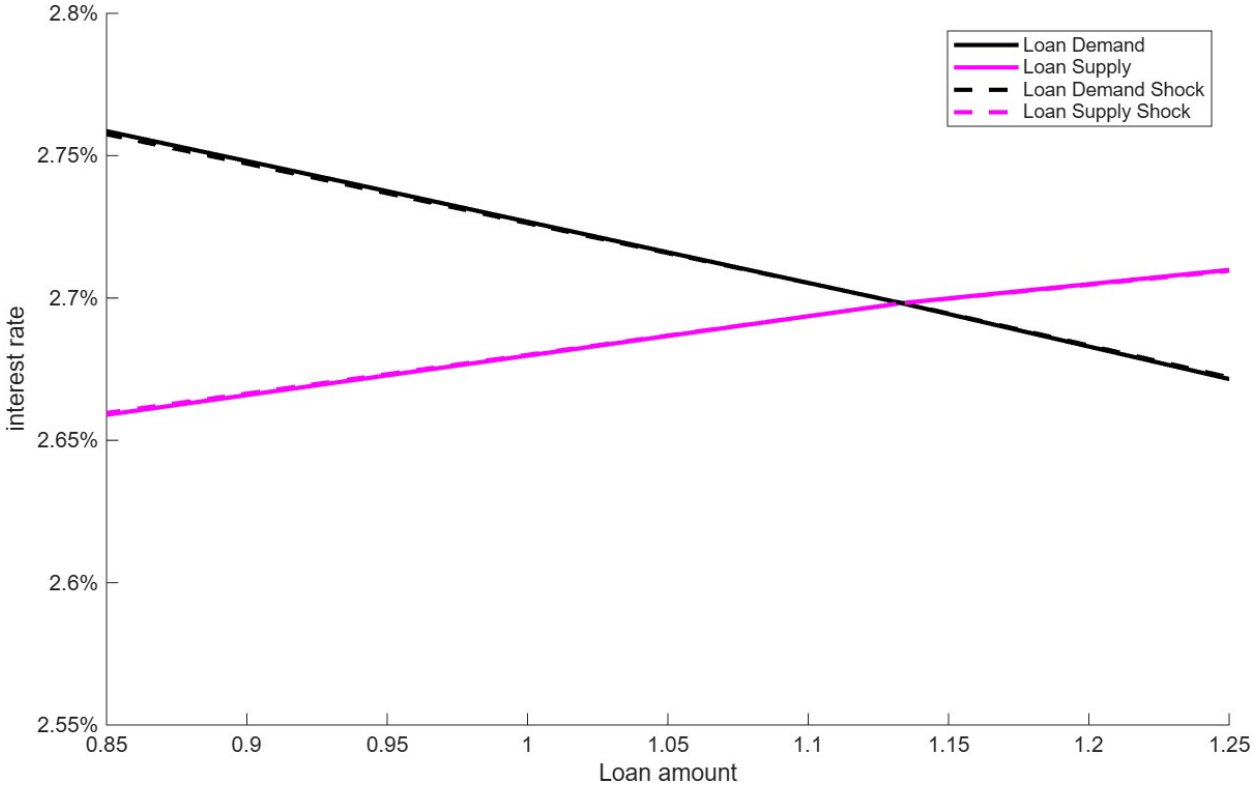


Figure 27: Huggett Model Loan Demand and Loan Supply

Consistent with the standard Huggett benchmark, the Modified Huggett Model shows that the loanable funds market changes only minimally in response to the productivity shock. This muted response extends to utility. Changes in households' welfare are small because their labor supply barely adjusts, and agents continue to work roughly the same number of hours. In both the intensive and extensive margin models, the firm responds by lowering the effective labor used. Because the Modified Huggett Model is not designed to capture how the lower employment level interacts with

life-cycle components and the extensive margin, this small response is unsurprising. The contrast underscores the ability of intensive margin and, especially, extensive margin models to highlight how income losses and restricted access to credit collapse together to shape household utility.

5.9 Policy Response

The first part of this section examines how increasing social security and unemployment transfer payments mitigate the decline in household utility during periods of reduced productivity and benchmarks these effects against the fixed transfer model. The analysis explores the policy's subsequent effects on the loanable funds market. It focuses exclusively on the extensive margin of labor adjustment, the dominant margin of adjustment, in order to isolate and quantify this channel. Specifically, transfers are set inversely proportional to aggregate output according to the countercyclical rule $T_{0,shock} = T_{0,ss} * \frac{Y_{ss}}{Y_{shock}}$, where $T_{0,ss}$ is the per period transfer in steady state, $Y_{ss}$is steady state output, and Y_{shock} is output along the unemployment path. An adverse productivity shock lowers output and raises both unemployment and unemployment transfers. This dampens the fall in income and utility while shifting saving and borrowing decisions in the loanable funds market. This section will proceed in three steps: utility/welfare, real interest rates, and quantity of loans.

The welfare implications of a productivity shock are presented in Figure (28). Dashed lines illustrate average utility across the agent population, whereas solid lines show utility for 25 year old agents. The baseline extensive margin model is represented by black lines, and the extensive margin model with higher transfer payments is represented by magenta lines. Both average agent utility and 25 year old utility fall by a smaller amount when transfer payments are increased. Although the effect is modest, the higher transfers do raise utility. In this model, transfer payments leave GDP unchanged. They nevertheless increase utility by offsetting some of the negative welfare effects of unemployment.

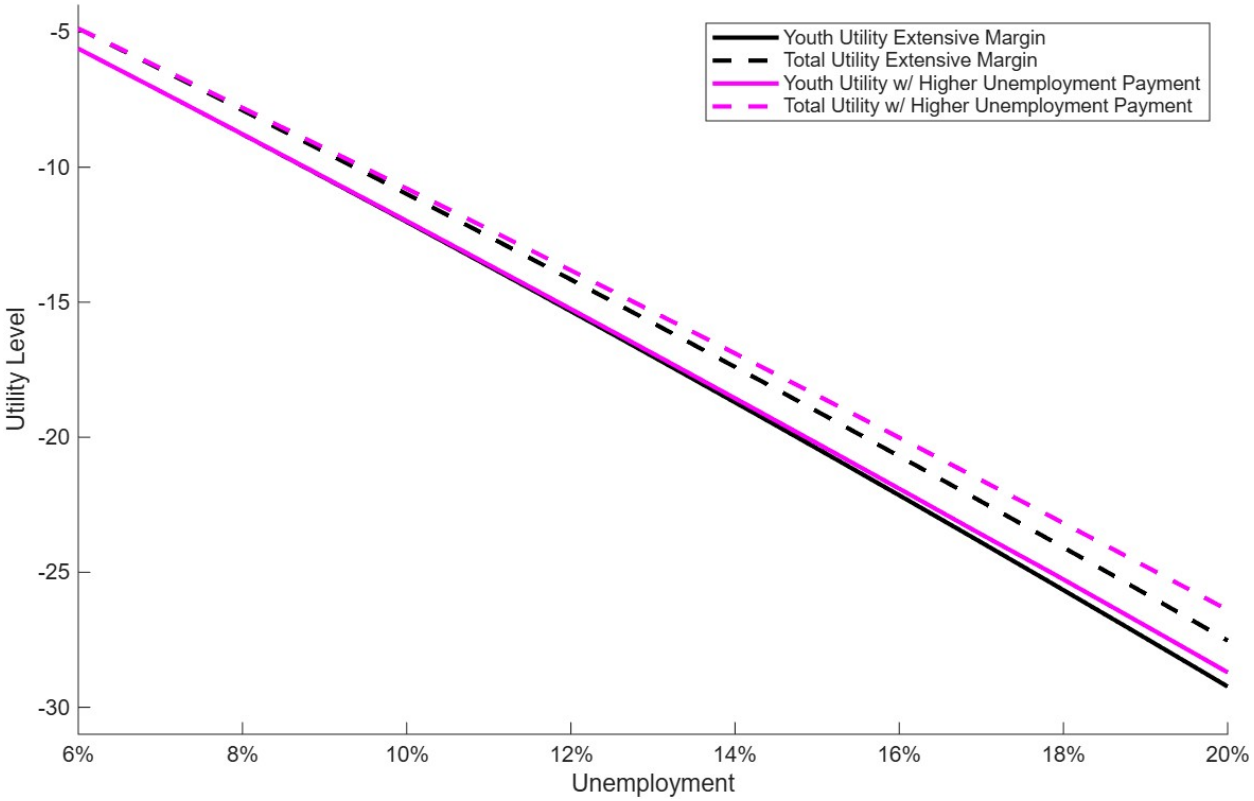


Figure 28: Extensive Margin Utility With Increased Transfer Payments

Figure (29) compares real interest rates in the two models. The black line represents the standard extensive margin model, while the magenta line represents the extensive margin model with transfer payments that increase as output falls. In the standard extensive margin model, real interest rates decline steeply and linearly. When transfer payments increase, real interest rates decline more slowly and begin to level off. The mechanism behind this is that when transfer payments rise, income rises not only for the young unemployed but also for the elderly. This increase in the elderly's retirement income reduces the incentive for middle aged households to save to smooth consumption in retirement, which slows the fall in real interest rates as output declines.

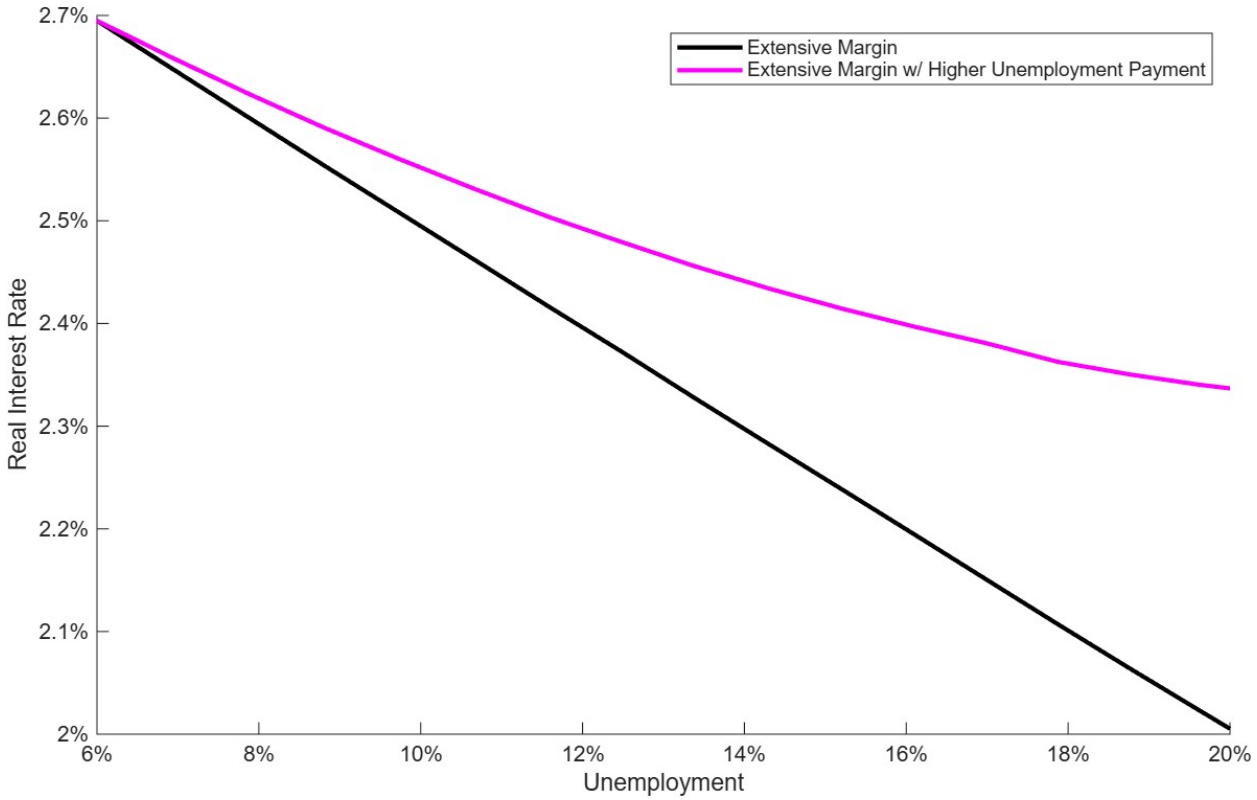


Figure 29: Extensive Margin real interest rates With Increased Transfer Payments

As shown in Figure (30), the black line represents the standard extensive margin model, while the magenta line illustrates the extensive margin model incorporating transfer payments that increase as output declines. Real household debt declines (roughly linearly) in both specifications, but the fall is larger in the model with countercyclical transfers. The difference operates through loan supply. Retirees now receive more transfer income, so they save less for retirement. The resulting contraction in loan supply pushes the equilibrium borrowing below the standard extensive margin path. Loan demand, by contrast, is fairly inelastic because it remains consumption maximizing for the young to borrow with little regard for the real interest rate. This causes the two models' debt paths to differ less than their real interest rates do. This asymmetry reflects the fact that the elasticity of loan supply is higher than that of loan demand. Middle aged saving effectively follows an Euler equation similar to Eq. (7) in the"Isolating the Demand Channel" section. The response of real household debt to the transfer shock is muted relative to the changes in real interest rates.

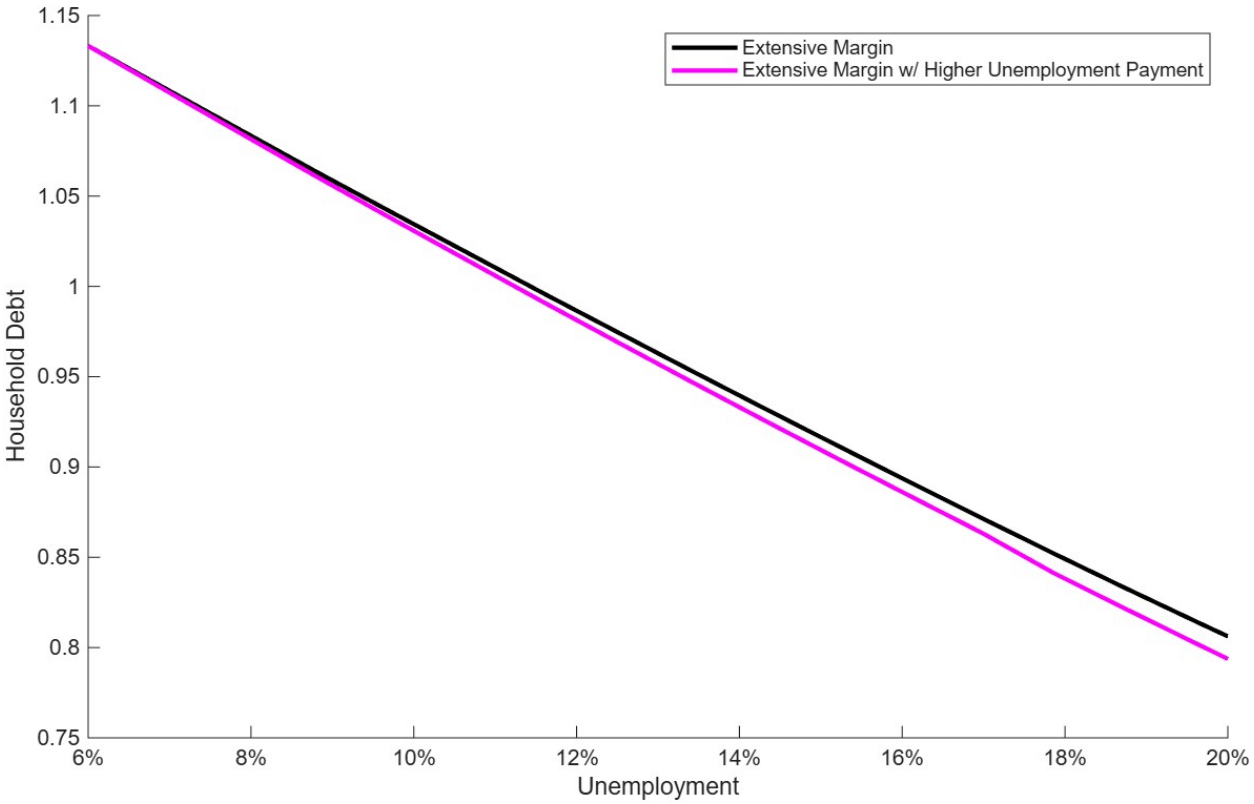


Figure 30: Extensive Margin real Quantity of Loans With Increased Transfer Payments

Taken together, Figures (28)–(30) reveal a cohesive narrative. The adverse productivity shock raises unemployment, which simultaneously lowers welfare, raises borrowing costs, and shrinks lending. The countercyclical transfer policy acts most directly by replacing a larger portion of lost labor income. The higher transfers slow the fall in consumption and yield a modest but consistent utility gain for both the average household and the young (Figure 28). Because retirees now treat the additional transfers as higher retirement income, they reduce their saving, and the resulting contraction in loan supply arrests the decline in real interest rates relative to the fixed transfer model (Figure 29) and pushes equilibrium household debt further below its standard extensive margin path (Figure 30). The policy, therefore, operates as a pure insurance mechanism. It smooths the utility cost of unemployment at the price of a tighter credit market in which borrowing is simultaneously more expensive and less abundant.

5.10 Section Conclusion

This section has demonstrated that incorporating extensive margin unemployment risk into a heterogeneous agent framework with a life cycle component and incomplete markets materially alters the transmission of productivity shocks through the loanable funds market and reduces utility.

The HA model departs decisively from the modified Huggett specification. It differs in both its steady state and its dynamic response to a negative productivity shock. Specifically, in its steady state, the HA model's differences are reflected in the distribution of assets, the shape of savings policy functions, and the concentration of mass near zero assets. The HA model that varies over the extensive margin generates an endogenous precautionary motive that intensifies with aggregate unemployment, making the extensive margin the key channel through which productivity shocks propagate to household borrowing, real interest rates, and welfare.

The comparative statics reveal stark differences between intensive and extensive margin adjustments. For a productivity shock that raises the unemployment rate from 6 to 20 percent, the extensive margin adjustment lowers real interest rates by roughly 0.7 percentage points and household debt by roughly 29 percentage points, whereas the intensive margin adjustment, evaluated at the same productivity levels, raises interest rates slightly (around 0.1 percentage points) and reduces debt by only about 16 percentage points. The mechanism driving this is the collapse in borrowing capacity that accompanies the reduction in income experienced by the unemployed. This strengthens the precautionary motive among employed households whose unemployment risk has risen. Welfare losses likewise grow faster along the extensive margin because young agents bear a disproportionate share of the burden of unemployment. The Modified Huggett benchmark's lack of response to the shock confirms that these outcomes depend crucially on the interaction between extensive-margin risk and life-cycle borrowing and saving, rather than on the productivity decline.

The policy analysis completes this narrative. A countercyclical transfer policy acts as a pure insurance mechanism that replaces a larger portion of lost labor income. It modestly raises utility for both average households and the young, without a constant GDP level. Yet the policy operates at the cost of a tighter credit market because retirees treat the additional transfers as higher retirement income and reduce saving, thereby halting the decline in real interest rates and pushing equilibrium household debt further below its standard extensive margin path. Taken together, these findings imply that macroeconomic models treating hours worked as the primary margin of risk will understate the sensitivity of credit markets and welfare to productivity shocks. Because the extensive margin is the empirically dominant margin of adjustment, incorporating it into quantitative models of household lending is essential for accurately characterizing how income losses and restricted access to credit collapse together to shape the loanable funds market, and utility.

6 Analyzing Dynamic Responses: The impact of productivity shocks in HA framework

6.1 Overview

This section advances the analysis beyond the descriptive comparative statics of the steady state by examining the impulse responses generated by the HA model. This section accomplishes two things. First, it separates the life cycle dynamics that the heterogeneous agent model generates on the intensive margin from those inherited from the Modified Huggett benchmark. Second, it isolates the impact of the intensive and extensive margins within the HA model. Five outcomes are tracked: the real interest rate, quantity of household loans outstanding, the share of constrained households, hiring transition probabilities, and the unemployment rate. The first two summarize conditions in the loanable funds market, while the latter two trace the labor market along the extensive margin. The share of constrained households is the variable of particular interest because it captures the joint effect of hitting the borrowing limit and being unemployed with nonpositive assets. Contrasting both margins against the benchmark not only documents the divergence between the models but also demonstrates why the extensive margin is essential to the model's predictions for credit and labor markets.

6.2 Computation

This paper uses computational methods from (Achdou et al. 2022) by solving the HJB value function backward from the steady state to the present, then solving the KFE for the distribution of agents forward in continuous time. Continuous-time models can often make intuition tricky because they require transition dynamics and model features to be considered more closely. A few assumptions are made for simplification. First, the firm unilaterally chooses the effective labor to maximize its utility based on profits less adjustment costs, discounting at a constant rate of time preference. Second, the government balances the budget in each period. This assumption will be relaxed in the appendix. There are relatively small differences in the resulting time paths. Because assuming no deficits is simpler and yields similar results.

The labor market transition dynamics are an important consideration, given the realistic constraint that employment does not adjust instantaneously. To ensure the resulting model adheres to real world economic behavior, quadratic labor adjustment costs were added to the firm's effective labor for both the intensive and extensive models to smooth labor dynamics and better match the real world. Two papers that had quadratic labor adjustment costs are (Janko 2008) and (Lechthaler and Snower 2013). They use adjustment cost values of 0.98 and 2, respectively. This approach

effectively parameterizes the inherent inertia in labor market adjustment, thereby avoiding discontinuous jumps in employment.

The sequence for solving the dynamic response for the HA models is as follows. First, the firm chooses the effective labor level based on the macroeconomic Frish elasticity reported in (Chetty, Guren, Manoli, and Weber 2011) and the marginal productivity of labor following a productivity shock, while accounting for adjustment costs to effective labor productivity. None of this is affected by real interest rates because firms maximize utility using a consistent rate of time preference, and they do not borrow or lend. This simplification yields the effective labor level in IRF by adjusting the hiring rate in the extensive margin model and reducing all employed agents' hours equally in the intensive margin model. Second, tax rates need to be adjusted. We will know the aggregate and the number of unemployed individuals at each stage. The tax rate can be adjusted to ensure tax inflows match tax outflows based on the no deficit assumption. This is relaxed in the appendix, where similar results are obtained. Third, we need to find the market clearing interest rate. We know employment dynamics and tax responses over the cycle, so we can keep those inputs constant and focus on clearing the asset market by adjusting interest rates during the IRF. It is possible to have all three variables solved at the same time, but it isn't necessary.

The Firm chooses effective labor, N_t, to maximize Eq.(38), the firm's utility throughout all time periods in response to an AR(1) productivity shock. The utility function is the integral of all profits and labor adjustment costs discounted by the rate of time preference. For simplicity, the labor adjustment cost is an implicit cost, so it does not affect profitability only firm utility; it only affects the firm's decision making as the firm is slow to adjust labor.

$$V = \max_{N_t} \int_0^\infty e^{-\rho t}(A_t(N_t)^{1-\alpha} - w_t N_t - \frac{\theta}{2}\dot{N_t}^2)\,dt \tag{38}$$

It is important to note that the wage, w_t, is based on the Frisch elasticity of the macroeconomy. Because of this, the wage curve is Eq. (39). This can be substituted into the firm's profitability equation, allowing the firm to maximize discounted utility by adjusting N_t across different time frames.

$$w_t = (1-\alpha) * N_t^{\phi} \tag{39}$$

Figure (31) shows the path that the effective labor takes both with and without adjustment costs. Without adjustment costs, the production shock is mirrored. With adjustment costs, the response is slower, and the optimal labor response is dampened. This also removes a large, instantaneous jump shock in effective labor, making hours and employment react more smoothly. The lower panel plots profit over time. The profit flows to households as dividend income.

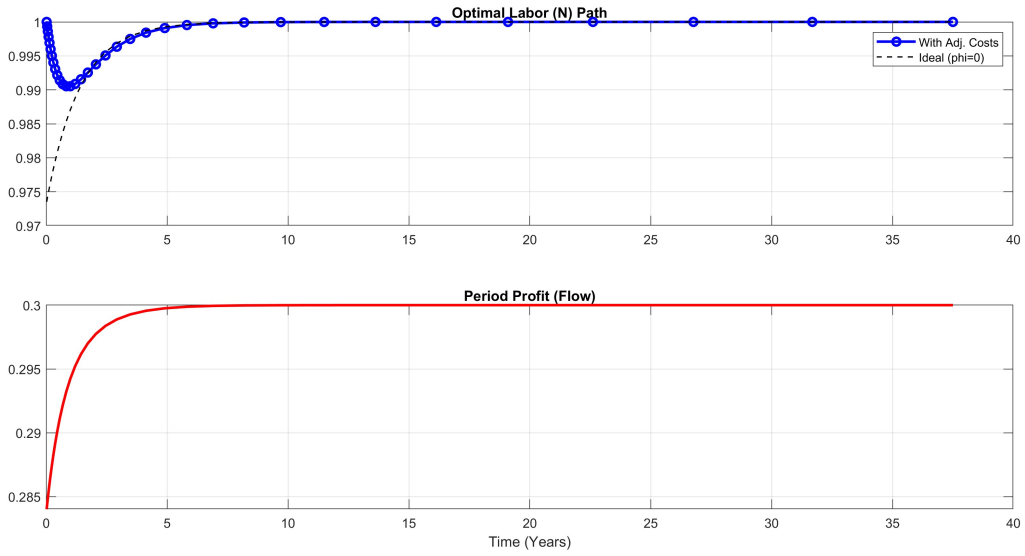


Figure 31: Labor IRF

From the optimal labor, hours in the intensive model can be adjusted by setting all agents' hours equal to the optimal labor path value. No additional calculations are needed. For the extensive margin model, the unemployed to employed transition probability must be solved. This probability will be adjusted across all time periods via scalar descent until the model's labor path matches the optimal labor path. This completes the employment transition dynamics and allows for movement to taxation.

The next market clearing condition is the government budget constraint eq. (23) for each individual time period. For the intensive margin model, we have hours worked, profit in each time period, and the same unemployment and unemployment payments throughout the shock. Because the tax base is total income net of interest payments, wage income plus dividend income, the budget constraint does not depend on interest rates, so the tax-rate path can be solved before the interest-rate path. This means that interest rates are not needed to solve for the tax rates. We know the unemployment and retirement expenses, so the tax rate for each period can be determined using Broyden's method across all periods and tax rates. In the extensive margin model, hours worked remain at 1. The same profit is earned in each time period as compared to the intensive margin adjustment, and we now know the unemployment rates and total transfer payments. From here, we can use the same Broyden's method to find the tax rate.

Given the dynamics of labor and tax rates, all real interest rates can be calculated. Using standard Broyden's methods yielded quite jagged interest rate responses that do not converge, so Gauss-Newton methods were used instead. The first was a smoothing penalty, and the Levenberg-Marquardt method was used to avoid overly large steps and those that would make the interest rate

more jagged. The smoothing penalty is given in eq. (40). The penalty penalizes the curvature, the second difference, of the interest rates. Equation (41) is the smoothing Jacobian that is seen from changing the interest rate of one of the variables. It is important to note that changing the rate in a single time period not only affects the smoothness penalty for that period but also the time periods directly before and after it.

$$p_i = \frac{x_{i+1} - x_i}{\Delta grid_f} - \frac{(x(i) - x(i-1))}{\Delta grid_b} \tag{40}$$

$$\begin{aligned} R(i,i-1) &= \frac{1}{\Delta grid_b} \\ R(i,i) &= -\frac{1}{\Delta grid_b} - \frac{1}{\Delta grid_f}; \\ R(i,i+1) &= \frac{1}{\Delta grid_f}; \end{aligned} \tag{41}$$

From here, the Augmented F and Augmented J matrices will be updated. The F matrix is the residuals from the asset clearing equation. Equation (42) shows that the augmented F is the original F with the smoothness penalty score added below it. If the F was 5x1 and the Ps was 5x1 F_aug would be 10x1.

$$F_{aug} = [F; sqrt(\lambda_s) * Ps]; \tag{42}$$

The augmented J matrix, eq (43), is the standard Jacobian stacked on top of the penalty Jacobian. The Jacobian, J, would be 5x5, and the R would be 5x5. This means when they are stacked, J_{aug} is 10x5.

$$J_{aug} = [J; sqrt(\lambda_s) * R]; \tag{43}$$

The update uses the regularized Hessian Eq (44), with 0. This avoids computing second derivatives of the residuals and adds a positive ridge term that shrinks the step size. If J_{aug} is 10x5 and I was 5x5, the resulting output is 5x5, the same as the original J, but now we have damping and smoothing built in compared to a standard Hessian.

$$H = J'_{aug} * J_{aug} + \mu * I; \tag{44}$$

Equation (45) represents the gradient. This is calculated using the transpose of J_{aug} times f_{aug}. This gives us the dissent rate. If J_{aug} was 10x5 and F_{aug} was 10x1 this would yield a 5x1 matrix.

$$G = J'_{aug} * F_{aug}; \tag{45}$$

Equation (46) shows the change in the interest rate variables based on the Hessian and Gradient. If the H was 5x5 and the G was 5x1, this would yield a change in interest rate of 5x1, matching the number of interest rates and residuals.

$$dx = -H^{-1} * G; \tag{46}$$

The modified Huggett model uses the same methods to solve for the interest rates. The biggest difference between these models is that the interest rate does not directly control consumption. It is set to clear the asset market and responds to intertemporal consumption and the marginal benefit of consumption at the aggregate level.

6.3 Results Comparison

From here, I will show the responses to productivity shocks with different levels of persistence on the extensive, intensive, and standard Huggett model. The different persistence levels will highlight how agents respond differently to shocks of different sizes. These graphs will show the shock size with no government debt, no government spending, and a zero budget deficit in all time periods. This yields a result similar to including government debt, government spending, and an unbalanced government budget constraint, but is much simpler and easier to understand. Figure (48) in the appendix displays the irf if government purchases of goods and services are 17% of GDP, government debt is 70% of GDP, and the government is not required to balance its budget each time period, but will bring the government debt back in line when output recovers.

Figure (32) shows what happens with a productivity shock that has low persistence. First, let's look at the labor market dynamics. The unemployment rate stays the same for the intensive and Modified Huggett model as hours are forced lower over the intensive margin, and in the Huggett model, agents actually choose to work slightly more, as the income effect dominates the substitution effect, and especially lower income agents work more hours. The unemployment transition dynamics also stand out: the intensive margin model maintains unemployment, whereas the Huggett model has none. The extensive margin model reduces the hiring rate before returning to equilibrium levels. Let's look at the borrowing constrained households. The intensive and extensive models start and end with the same level of constrained households. Constrained households are agents that cannot borrow. This includes agents that have hit the age based borrowing limit or agents that are unemployed and have zero or fewer assets. They either can't borrow because they are at their borrowing limit, or they can't get a loan because they're unemployed. The exten-

sive margin model shows an increase in the number of constrained households as a larger share of agents are unemployed and unable to borrow. Youth unemployment will increase the most in this model, and because youth have negative asset levels, this will increase the number of constrained households. There is a mild dip in constrained agents in the intensive margin model as there is no change in unemployment, but there is a total drop in wages as hours worked and dividends drop. Agents take lower life cycle income into account and end up borrowing less. The borrowing limit does not move, as doing so would cause problems with computational methods. There is a mild increase in the constrained households in the Huggett Model as low income agents with low wealth have a higher marginal utility of consumption when their income drops and partially make up for this by increasing their borrowing and having a higher chance of hitting their borrowing limit. This differs with the HA intensive margin model because the intensive model has consumption smoothing and life cycle dependent income, whereas the Huggett model does not.

Let's look at the interest rate. It is important to note that the Intensive and Extensive Models have a quadratic adjustment cost on the rate of change of labor. This means that labor adjusts more slowly as depicted in Figure (31). The Huggett model adjusts instantly. That means the Huggett model doesn't show a negative interest rate because the decrease in earnings is instantaneous, and from that point on income only increases, so the interest rate is above the steady state level and then drops back to the steady state as the shock dissipates. Both the extensive and intensive margin models show that interest rates drop when the shock hits due to intertemporal substitution. From there, output bottoms out and starts increasing. Rates increase and even overshoot the steady state, as agents have perfect foresight and can see their future income rising. As the shock approaches zero, the interest rate returns to its steady state value. It is important to note that the magnitude of the interest rate IRF is larger in the extensive model because it exhibits greater variation in precautionary motives.

There is also a divergence in household loans. The Huggett Model shows that total loans increase before converging to steady state levels. Both the Intensive and Extensive Models show agents borrowing less, reflecting a decline in expected overall income. As agents expect their incomes to increase, households resume borrowing. This corresponds to interest rates above the steady state value. It is important to note that the extensive margin model shows a larger decrease in loans as the number of constrained households increases. When the unemployment rate falls again, agents don't take out loans quickly enough to offset the shortfall in borrowing, and household loans don't reach the same peak that the intensive margin generates. The point at which loans increase past the steady state does not align with the empirical findings later in the paper. This is likely because agents do not have perfect foresight and will hold on to negative beliefs about the future, based on a past precautionary motive that is stronger than what they will see in the future. This can also be seen as agents being less likely to lend to agents with high debt levels. This is not included

in the model.

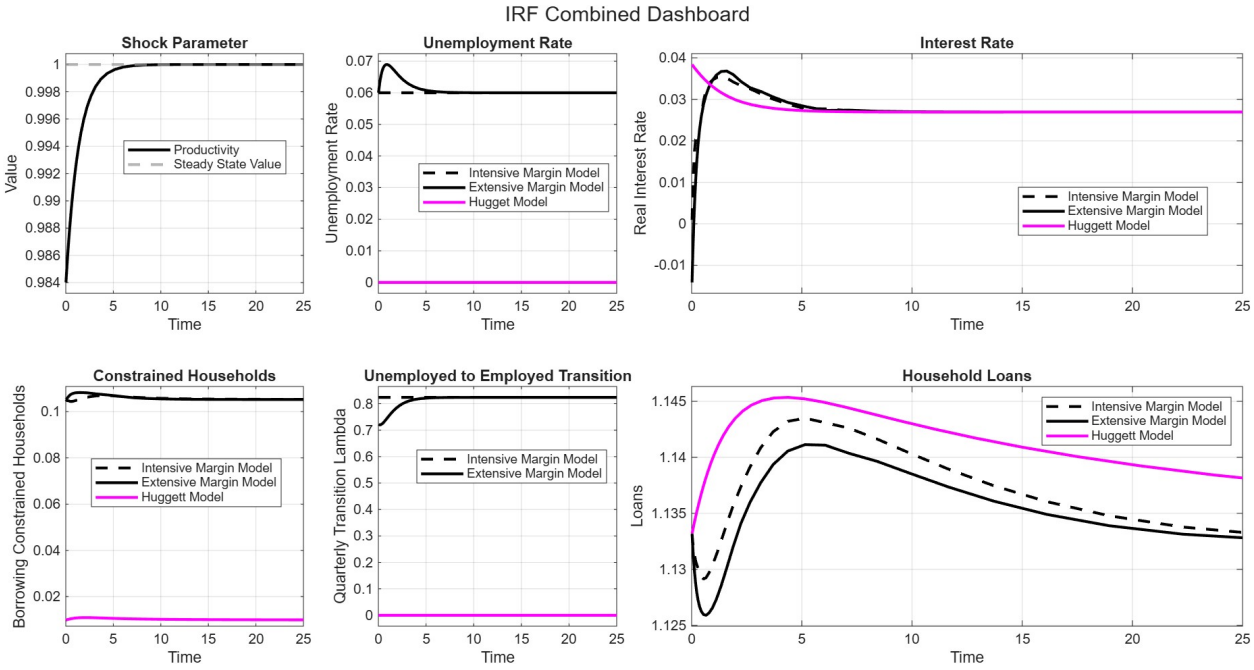


Figure 32: IRF Shock Low Persistence

Figure (33) shows what happens with a productivity shock that has a higher persistence. The first thing to notice is that the shock is the same magnitude; it just lasts longer. Now that the shock persists longer, both the Intensive and Extensive models adjust labor more quickly and for a longer period to address the more prolonged shock. This is evident in the unemployment rate going higher and taking longer to drop to steady state. Interest rates follow the same general path, but drop lower for the intensive and extensive models. When the overshoot is above the steady state, they don't jump up as high, but stay higher for longer. The constrained households and Unemployment transitions match the same theme or a larger, more persistent move. What really stands out is the Household Loans. Here, the Extensive Model not only deleverages further, but it doesn't overshoot the steady state and looks much more similar to the empirical findings later in this paper. In this model the difference between the Huggett and both HA models becomes stark as both interest rates and household loans become much different.

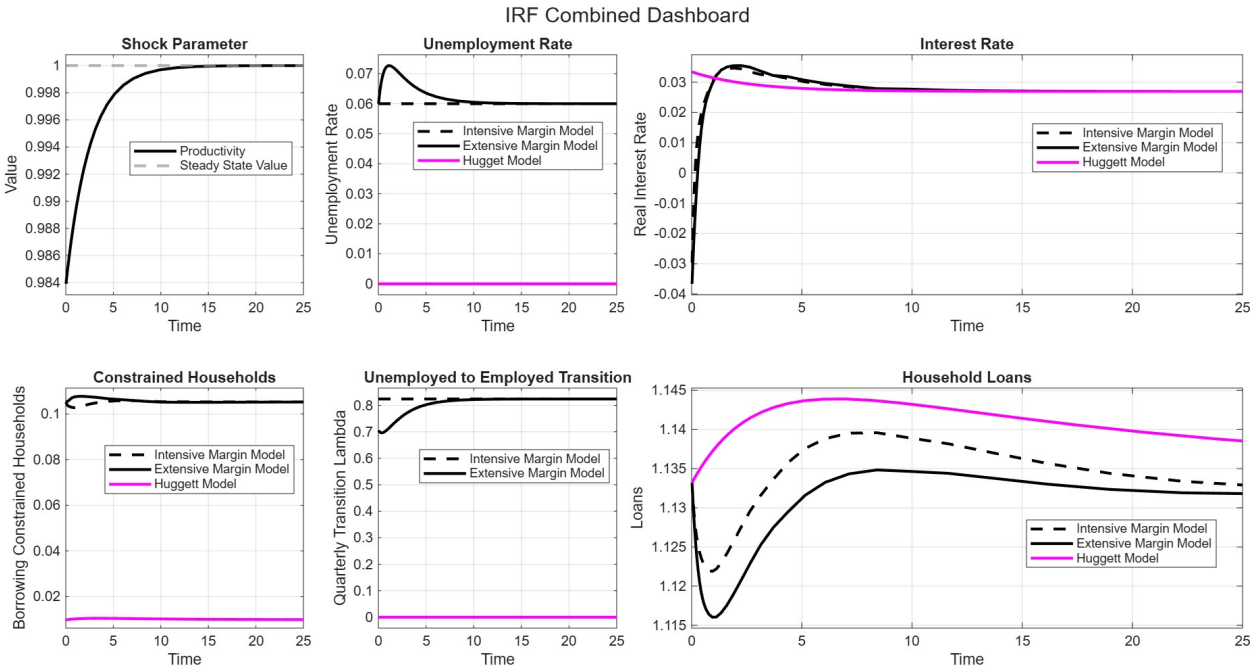


Figure 33: IRF Shock High Persistence

Figure (34) shows what happens if the market at equilibrium faces a permanent shock and the models transition to their respective steady state as seen in the comparative statics. First, the extensive margin unemployment rate increases and stays permanently elevated as the transition rate of unemployed ot employed drops. Both intensive and extensive margin models have interest rates drop before increasing to their permanent level. The intensive margin has a higher interest rate than the steady state, whereas the extensive margin has a lower value than the steady state. The Hugget Model's interest rate is roughly unchanged. Constrained households increase for the Extensive Model as unemployment goes up, go down for the Intensive margin as borrowing limits stay the same as income goes down, causing agents to borrow less, and Constrained stays roughly the same for the Huggett Model. Household deleveraging has significant differences. The Huggett Model has minimal changes to household loans and both the Intensive and Extensive Models have drops in leverage. The Extensive margin model is more pronounced. Once again, we are seeing significant differences in the two models.

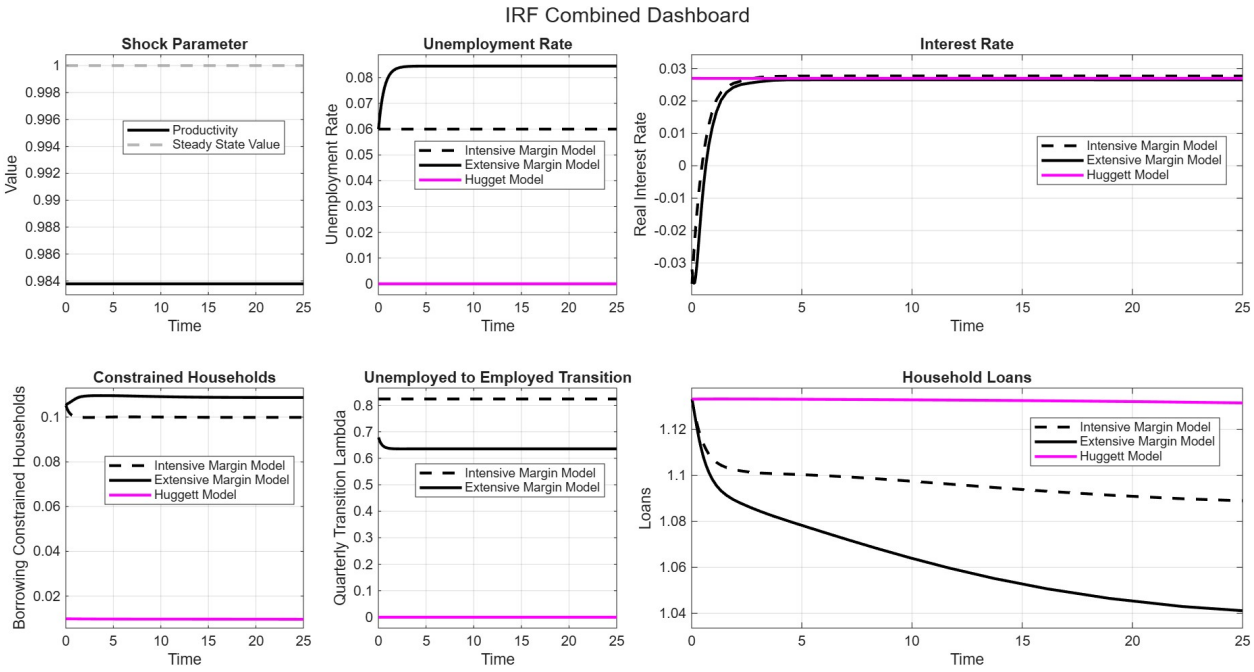


Figure 34: IRF Permanent SHock

6.4 Section Conclusion

This section demonstrates that the dynamic responses of the heterogeneous agent economy to productivity shocks are shaped by the extensive margin. Across all shock persistences, the intensive and extensive margin models diverge sharply from the Huggett model, which cannot generate unemployment and therefore misses the central channel linking credit and labor markets. The extensive margin model consistently produces increases in the share of constrained households, deeper household deleveraging, and interest rate responses that reflect stronger precautionary motives, whereas the intensive margin adjustment yields milder movements in constraints and household loans. The permanent shock results confirm these divergences in the new steady state, and the high-persistence case is particularly important, as only the extensive margin model generates household loan dynamics that align with the empirical patterns documented later in this paper. Taken together, these results show that the extensive margin is essential to the model's predictions for credit and labor markets, and that heterogeneity in life cycle income and wealth.

7 Bridging Theory and Data: A VAR/LP investigation into the labor market and household debt

7.1 Overview

In this section, the dynamic responses of weekly hours worked, the unemployment rate, and real household debt based on labor productivity shocks will be analyzed. The methods used will be VAR and LP This section will be much more informative than the empirical motivation alone because it directly evaluates the channels the HA models are designed to explain. Once again, European data will be used in addition to US data because there is not ample state level data on US household debt and weekly hours worked. The European panel provides substantial cross country variation that is unavailable in aggregate U.S. data. Specifically, European countries will be split into groups and analyzed for variation in weekly hours worked, unemployment, real household debt, and the household debt-to-GDP ratio. This variation provides a richer evaluation of the HA models' predictions and directly assesses the channels the HA models are designed to explain.

7.2 Data

The models use European quarterly macroeconomic data from 1998 to the end of 2019, collected from several sources. This includes the earliest date across all data sources and ends just before the COVID outbreak. Labor productivity was obtained from Eurostat. Labor productivity was used because multi-factor productivity was annual data, and the HA and Huggett model uses only labor, so it is more appropriate. Eurostat also provided the consumer price data, which is a harmonized measure. Average weekly hours worked also came from Eurostat and was used to distinguish fluctuations in labor input arising from changes in hours worked versus changes in employment. Unemployment data are obtained from the OECD database. Household debt data are collected from the Bank for International Settlements (BIS) and measured as credit to the household sector relative to GDP. The US data is collected from 1952 to 2019 from several sources. BLS for labor productivity, weekly hours worked, the unemployment rate, and consumer prices. Household debt data came from the Board of Governors of the Federal Reserve System. The resulting panel combines labor productivity, weekly hours worked, the unemployment rate, and real household debt across European economies at quarterly frequency, allowing for a deeper dive into the relationship between employment dynamics and household debt amid productivity shocks in the business cycle.

7.3 Empirical Methods

The methods in this section will include detrending the data using a Hamilton Filter, followed by analysis with Vector Autoregression (VAR) and Local Projection (LP).

The filter method used was (Hamilton 2018). The Hamilton filter was used for a few reasons. First, it removes the endpoint bias because the Hamilton Filter does not include smoothing that looks forward at the starting point and smoothing that looks backward for the end point. Second, the filter is much more interpretable because the cyclical component is the deviation from the predicted path. This will allow the filter to do a better job in low frequency cycles as well as avoid serial correlation, persistence, and co-movement between unrelated variables. Lastly, there is no artificial smoothing parameter, as a Hamilton filter uses a simple regression eq. (47)

$$y_{t+h} = y_t + y_{t-1} + y_{t-2} + y_{t-3} + \nu_{t+h} \tag{47}$$

The Hamilton Filter traditionally looks ahead 8 quarterly periods or two years. (Jordà, Schularick, and Taylor 2017) argued for looking ahead 5 years when examining debt cycles. Because of this, both the 8-period 2-year and 20-period 5-year look-ahead will be used to cover all bases.

To ensure that the estimated dynamic relationships are not driven by a particular econometric specification, the analysis employs both VARs and LPs. The VARs follow (Sims 1980) and provide a framework for estimating impulse response functions under a structured dynamic specification. In contrast, LPs follow (Jordà 2005) estimate responses at each forecast horizon separately, making them more robust to model misspecification and better suited for exploring nonlinear and heterogeneous effects. The empirical evidence is strengthened by the fact that both methods yield economically meaningful dynamic relationships among the variables.

The IRFs are computed from combined data across all countries. The countries include: Austria, Belgium, Czechia, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Poland, Portugal, Spain, Sweden, and Norway. It is important to note that several countries were excluded from the original empirical model because they lacked household debt data, and Switzerland was specifically excluded due to very limited cyclical data from 2015 to 2019.

The VAR will be setup is eq (48). This is a VAR with 4 quarters of lag. The variable setup in eq (49) is as follows: Labor Productivity is first, followed by weekly hours worked, the unemployment rate, and real household debt. There are alternative measures of household debt that can be used and are used in the appendix. These include nominal household debt and household debt-to-GDP. Both the VAR Figure (51) and LP Figure (52) in the appendix show similar results. Figure (53) and Figure (54) show that Debt to GDP increases following the negative shock because GDP drops more quickly than the household debt. As time proceeds, the debt-to-GDP ratio drops slightly below GDP before returning to normal.

There is a built-in AR(1) process for Labor Productivity to make it easier to model the shock to the structural model. This effectively puts zeros in all the first row of Φ, excluding the a_{11}. This can be seen in eq. (50) and shows that Labor productivity is an AR(1) and all of the other variables follow a traditional lagged VAR structure.

$$(I - \Phi_1 L - \Phi_2 L^2 - \Phi_3 L^3 - \Phi_4 L^4) X_t = e_t \tag{48}$$

$$X_t = \begin{bmatrix} A_t : \text{Labor Productivity} \\ H_t : \text{Weekly Hours Worked} \\ U_t : \text{Unemployment Rate} \\ B_t : \text{Real Household Debt} \end{bmatrix} \tag{49}$$

$$\Phi_1 \text{ layout } \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \quad \Phi_{2,3,4} \text{ layout } \begin{bmatrix} 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \tag{50}$$

The error terms used follow the Cholesky decomposition. The upper triangle follows eq.(51). This order makes sense, as a negative productivity shock would also show up as a drop in weekly hours worked, then as an increase in unemployment as hiring slows and layoffs increase, and finally as a reduction in borrowing and lending. Weekly hours worked increased unemployment as firms under pressure reduced headcount and borrowing, while households became more nervous about the economy. Unemployment rates affect borrowing and lending as agents are less willing and able to take out loans.

$$\begin{bmatrix} e_{1t} \\ e_{2t} \\ e_{3t} \\ e_{4t} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ b_{21} & 1 & 0 & 0 \\ b_{31} & b_{32} & 1 & 0 \\ b_{41} & b_{42} & b_{43} & 1 \end{bmatrix} \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \\ u_{4t} \end{bmatrix} \tag{51}$$

Recursive residual bootstrapping is used to create the confidence intervals. Recursive residual bootstrapping begins with fitting the VAR model and finding the residuals. From there, the residuals are resampled randomly to create synthetic data. The model is then refit with the synthetic data. This process is repeated a handful of times before an IRF is generated using the standard model, along with the 5th and 95th percentile values of the synthetic IRF for each time period.

Impulse response functions by running a sequence of OLS regressions for different horizons (h). The outcome in the future is based on a constant α_h plus the coefficient β_h of the shock to labor productivity, plus the covariates of the control variables $\gamma_h x_t$, plus the error terms v_{t+h} as seen

in eq. (52). This unstructured method allows for flexibility without formally structuring the model. It is important to point out that the error term is structured v_{t+h} as a time series moving average of each of the errors that occurred before it as seen in eq. (53). This structure is based on the idea that a time series can be expressed in its infinite MA form, which helps maintain independence across horizons. Conditional Mean independence is preserved because the error term is assumed to be independent of the structural shock, conditional on control variables. The conditional mean independence ensures that the OLS regression of yields a consistent estimate. So, the MA structure for the errors helps maintain this independence across horizons, because MA processes have no persistent correlation with past shocks beyond their own lag structure.

$$y_{t+h} = \alpha_h + \beta_h s_t + \gamma_h x_t + v_{t+h} \tag{52}$$

$$v_{t+h} = u_{t+h} + \psi_1 u_{t+h1} + ... + \psi_h ut \tag{53}$$

7.4 Empirical Results

The VAR is computed for the combined data from all European countries. The countries include: Austria, Belgium, Czechia, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Poland, Portugal, Spain, Sweden, and Norway. It is important to note that several countries were excluded from the original empirical model because they lacked household debt data, and Switzerland was specifically excluded due to very limited cyclical data from 2015 to 2019. The IRF in Figure (35) has a negative productivity shock of roughly -1.6% representing the standard deviation. It is visibly clear that the shock follows an AR(1) based on the restrictions that were imposed earlier. The decrease in hours is not significant and is limited to roughly a 0.015 drop in weekly hours worked in the 3rd quarter after the shock. The cycle unemployment rate increases significantly by 0.25pp during the 3rd through 5th periods after the shock, then slowly declines. The real debt shape mirrors the move in the unemployment rate, but instead shows a decrease in real debt. This shows that real debt drops by roughly 0.5% relative to trend debt after a shock. This margin is significant and rather persistent. This VAR has the 5 year Hamilton filter. The results of the 2-year filter show a similar framework, but are not as clean. This can be seen in the appendix as Figure (49)

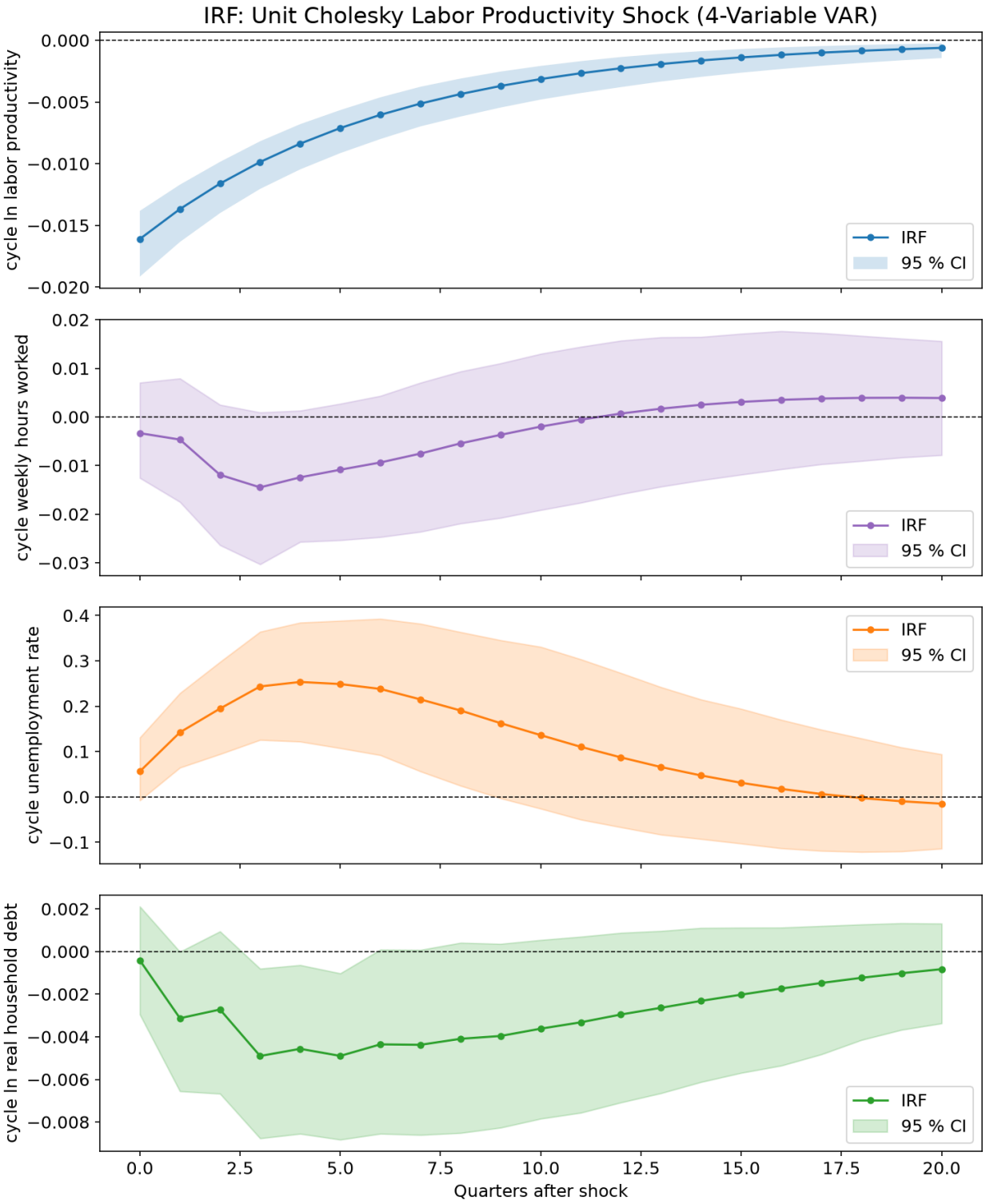


Figure 35: VAR All Countries

Figure (36) represents the LP's IRF to the productivity shock. As you can see, overall variable trends show similar directions and significance in the variables' movements, but with more chop.

The labor productivity shock response is now flexible and shows a linear increase back to the starting level. Weekly hours worked show a small, rather insignificant decrease, except for the 4th quarter after the shock. The unemployment rate increases by roughly .25%, a similar amount as the VAR, before dropping back to the starting point. Real household debt decreases linearly until peaking at roughly 0.7% in the 10th quarter after the shock, and does so significantly. It is important to note that the two year Hamilton filter was executed and posted in the appendix as Figure (50). It shows the same general movements across all variables, but now real household debt decreases significantly only in the 8th period.

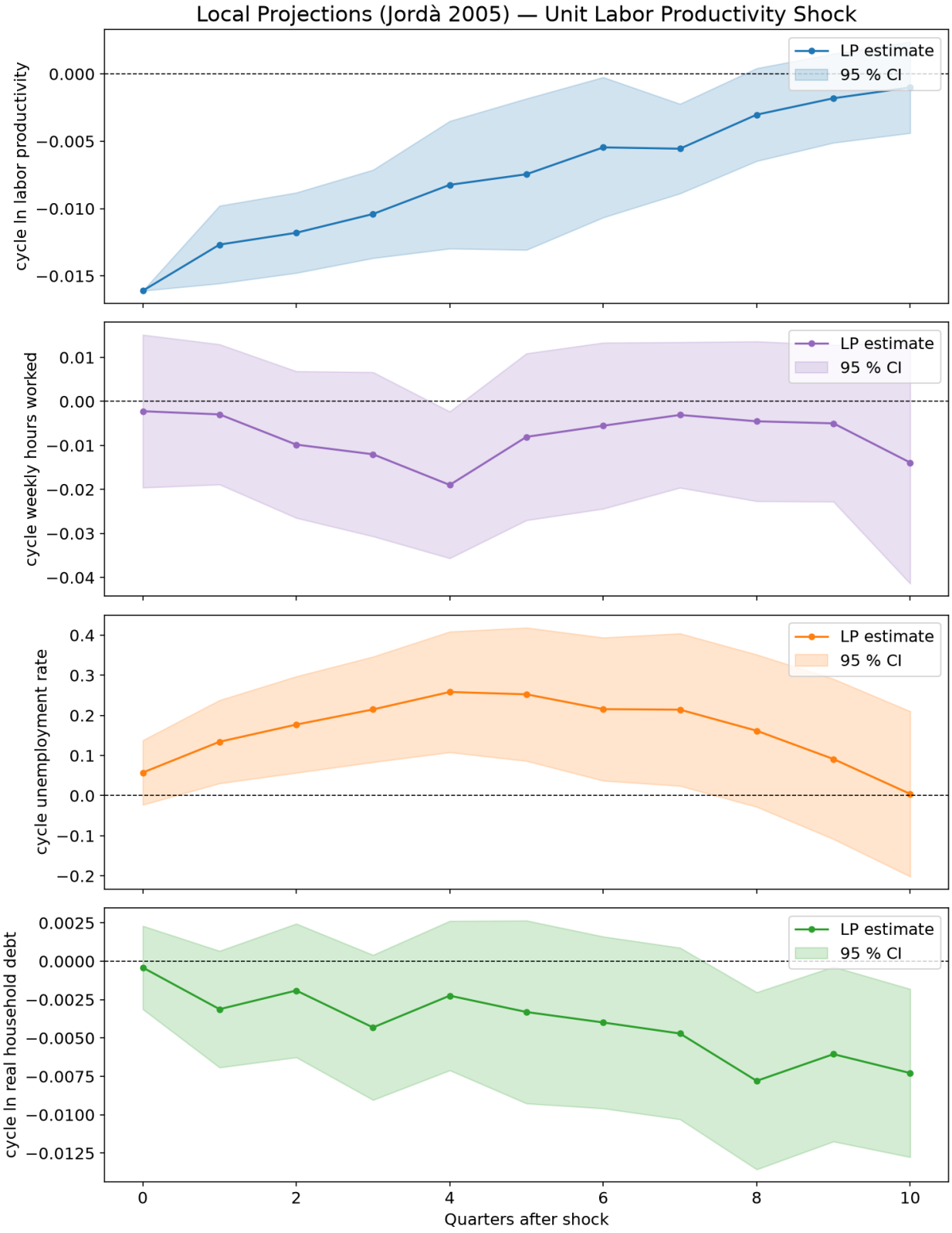


Figure 36: LP All Countries

A VAR and LP are again computed from only US data. The IRF in Figure (37) has a negative productivity shock of roughly -1.2% representing the standard deviation. It is visibly clear that the shock follows an AR(1) based on the restrictions that were imposed earlier. The decrease in hours is not significant and is limited to roughly a 0.015 drop in weekly hours worked in the 3rd quarter after the shock. The cyclical unemployment rate increases significantly by 0.19 pp after the shock, then slowly declines. The real debt shape mirrors the move in the unemployment rate, but instead shows a decrease in real debt. This shows that real debt drops by roughly 1.2% relative to trend debt after a shock. This margin is significant and rather persistent. This VAR has the 5 year Hamilton filter.

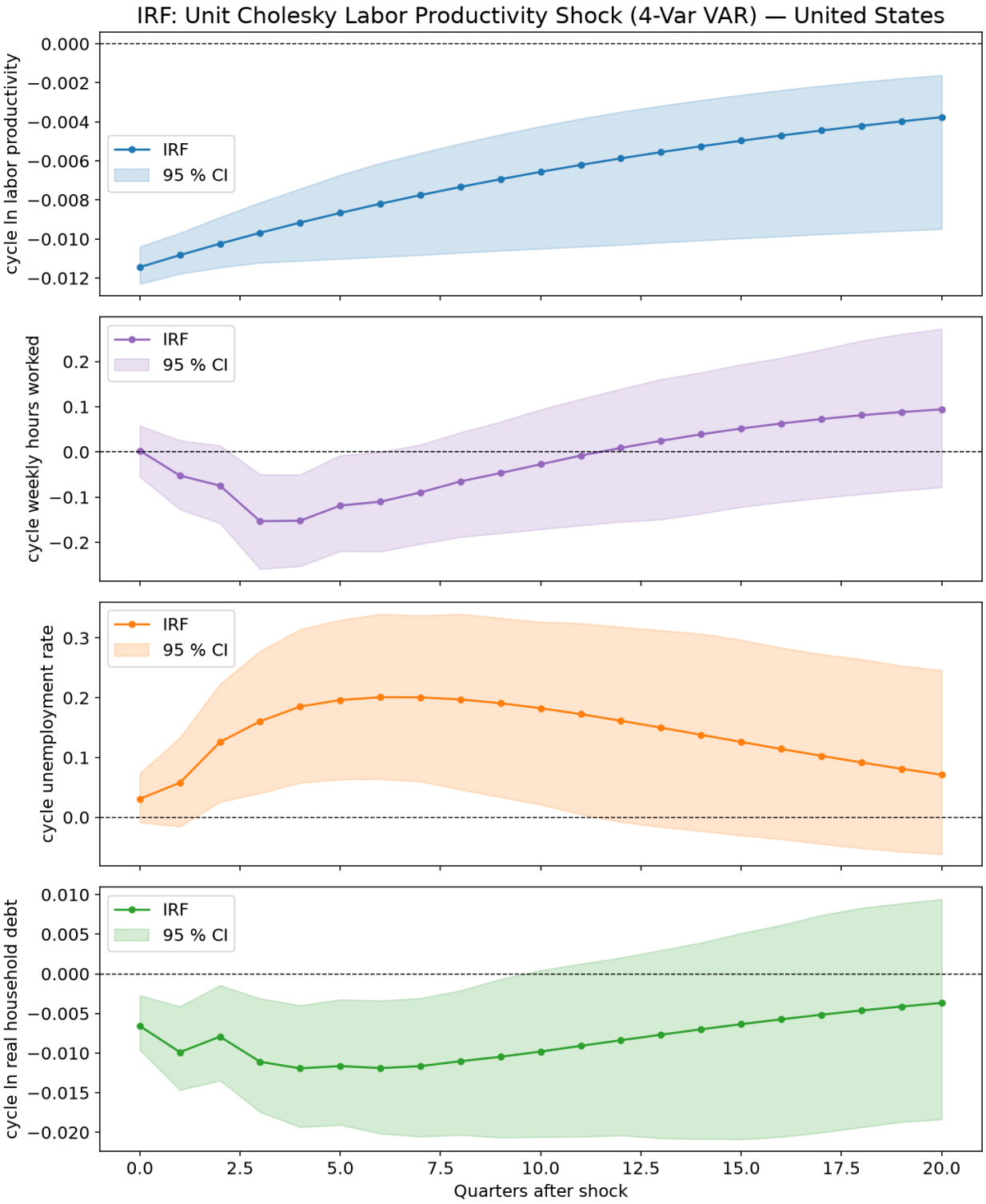


Figure 37: LP US

Figure (38) represents the LP's IRF for the US. As you can see, overall variable trends show similar directions and significance in the variables' movements, but with more chop. The labor

productivity shock response is now flexible and shows a linear increase back to the starting level. Weekly hours worked show a small, albeit insignificant, decrease, except in the 3rd, 4th, and 6th quarter after the shock. The unemployment rate increases by roughly 0.2 pp, a similar amount to the VAR, but does not return to the starting point. Real household debt decreases linearly until peaking at roughly 1.5% in the 8th to 17th quarter after the shock, and does so significantly.

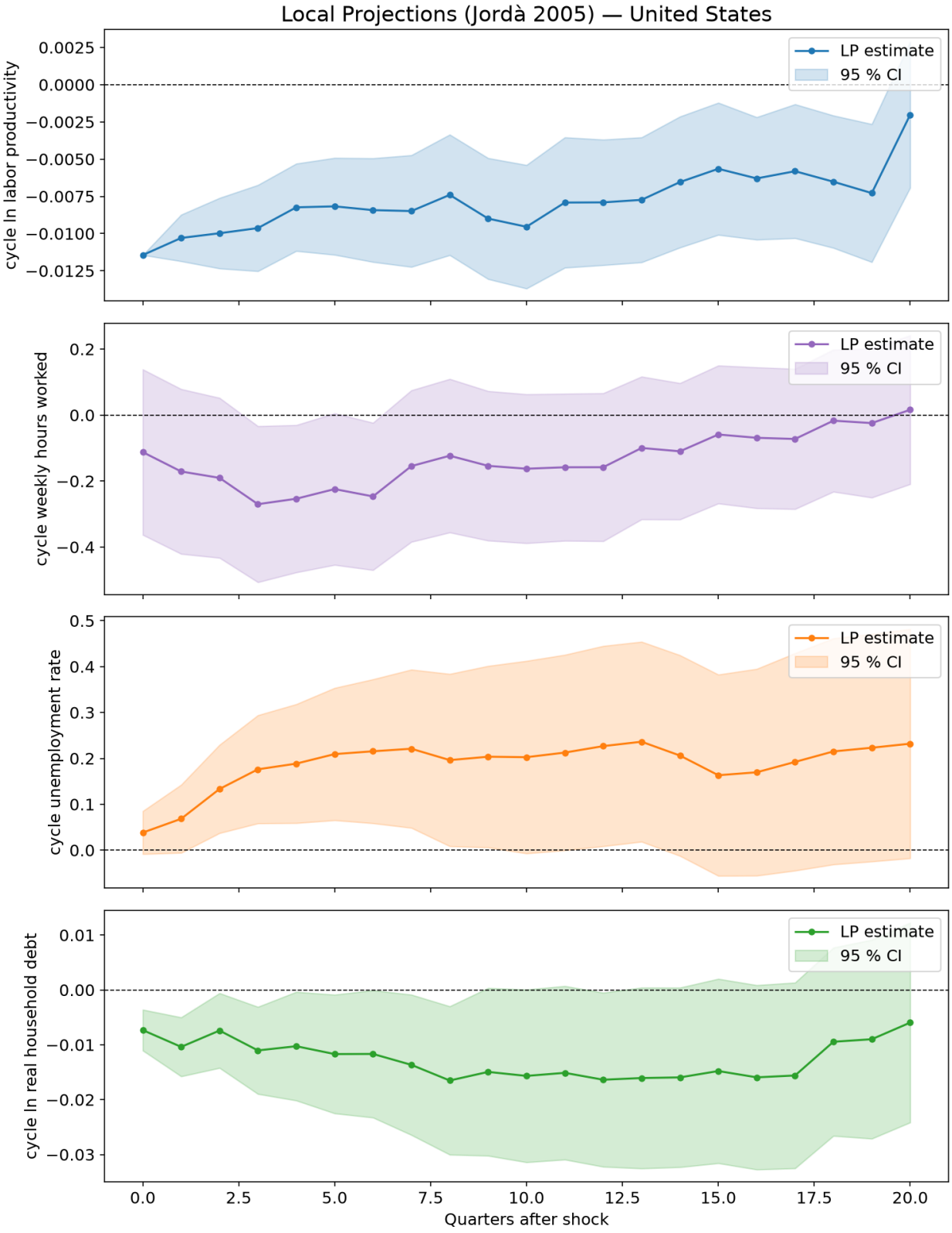


Figure 38: LP US

Both the European aggregates VAR and LP paint a picture similar to that of the US VAR and LP generated from US data. This strengthens the evidence of the unemployment rate moving more significantly than weekly hours worked and that real household debt dropped significantly in both cases.

From here on out, the VAR will be run for different splits based on high vs. low levels of variation in weekly hours worked, unemployment, real household debt, and the household debt amoung European countries. This will show how countries with different dynamics have their debt perform differently. This can be seen in Figure (39). The countries are listed alphabetically and the standard deviation and average household debt to GDP values are displayed. The values are bolded if the countries value is greater than the median value.

Country Summary: Debt/GDP, Debt, Hours & Unemployment Volatility
(Bold = high group for that column; country names never bold)

Country	Hours Cycle Std Dev	Unemployment Std Dev	Debt Cycle Std Dev	Avg Debt/GDP
Austria	**0.8664**	0.5812	0.0381	50.1036
Belgium	0.2472	0.8772	0.0524	49.8333
Czechia	**0.3640**	**1.9170**	**0.1540**	21.4750
Denmark	**0.3955**	1.4211	0.0464	**107.8405**
Finland	0.1295	0.8903	**0.0716**	50.5102
France	**0.3063**	0.5749	0.0300	**51.5750**
Germany	0.1755	0.2436	0.0193	**56.9417**
Greece	0.2386	**6.4267**	**0.0993**	45.8839
Hungary	0.2096	**1.6171**	**0.1390**	22.6417
Ireland	**0.4253**	**2.6542**	0.0437	**78.5722**
Italy	**0.4108**	**1.4799**	**0.0648**	34.8807
Norway	0.1519	0.6012	0.0188	**78.4737**
Poland	0.1902	**2.3402**	**0.1121**	26.6934
Portugal	**0.2484**	**3.0464**	**0.0974**	**74.2989**
Spain	**0.2476**	**4.9537**	**0.1034**	**65.5795**
Sweden	0.1211	0.8527	0.0258	**71.6132**

Figure 39: Country Split Overview

Weekly hours worked were split between countries with high and low variation. The countries with high volatility in weekly hours are listed with their standard deviation: Austria (0.8664), Czechia (0.3640), Denmark (0.3955), France (0.3063), Ireland (0.4253), Italy (0.4108), Portugal (0.2484), and Spain (0.2476). The countries with low weekly variation are: Belgium (0.2472), Finland (0.1295), Germany (0.1755), Greece (0.2386), Hungary (0.2096), Norway (0.1519), Poland (0.1902), and Sweden (0.1211).

Figure (40) displays the VAR split. Both high and low volatility countries experience a similarly sized labor productivity shock. The low weekly hours worked volatility in countries has a more persistent shock. Looking at weekly hours, high volatility shows a large decrease of roughly 0.03 hours, while low volatility in weekly hours in countries has almost no effect. This is driven by the split itself. Going to the unemployment rate, we start to see differences. Both splits exhibit a similar path from the shock to 2 quarters after the shock. From there, the low hours volatility countries continue to rise before slowly decreasing. This is likely due to the larger persistence of the labor productivity shock. In countries with high hours volatility, the unemployment rate drops and even overshoots to lower unemployment before approaching the steady state. This leads to an interesting IRF for real household debt. Both splits drop to roughly the same.006% decrease in real debt, but the variance split does it sooner at the 4th quarter after the shock vs the 5th quarter of the low volatility shock. From here, the highly volatile country's debt returns to its original level fairly rapidly, while the low hours volatility countries take longer to recover. This is likely due to the unemployment rate staying higher for longer. This is a great finding, as the group of countries with high weekly hours volatility includes Portugal and Spain, just above the cutoff. They have a very high unemployment rate, high volatility, and volatile household debt. This means that the high weekly hours volatility includes many countries with high overall unemployment volatility, while the low hours volatility countries include those with a range of unemployment volatility and low hours volatility. This does a good job of highlighting that weekly hours volatility across countries does not meaningfully affect on real household leverage.

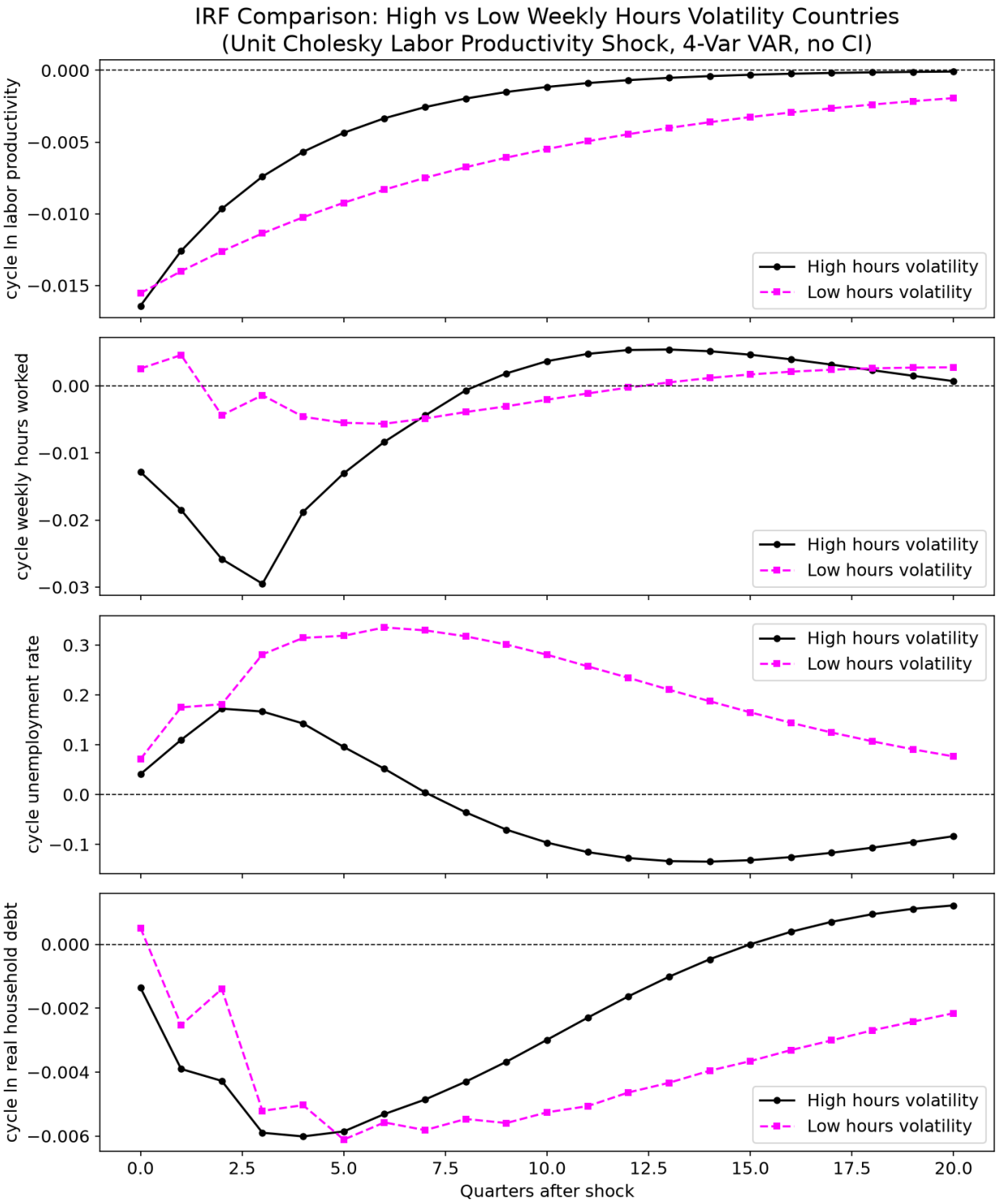


Figure 40: VAR High vs Low Weekly Hours Volatility Split

Next, we move on to the high vs low employment volatility split. The splits can be seen in Figure (41) The countries with high unemployment volatility are listed along with their standard

deviations: Czechia (1.9170), Greece (6.4267), Hungary (1.6171), Ireland (2.6541), Italy (1.4799), Poland (2.3402), Portugal (3.0464), and Spain (4.9537). The low unemployment volatility countries include: Austria (0.5812), Belgium (0.8773), Denmark (1.4211), Finland (0.8903), France (0.5748), Germany (0.2435), Norway (0.6012), and Sweden (0.8527).

The shock to labor productivity is roughly twice the size for the high unemployment volatility countries, and the persistence of the high unemployment countries is less. Hours worked drop for the high unemployment volatility split and a return to normal, a standard feature. The low-volatility feature leaves weekly hours worked roughly unaffected until around 5 quarters after the shock, after which they begin to increase. This is a bit odd, as hours worked should not increase like this. Unemployment response has a similar shape for both the high and low splits. The main difference is that the magnitude of the response is roughly twice the size for the high unemployment volatility split. Real household debt decreases by a larger margin under the high-unemployment volatility benchmark and peaks at roughly 0.6% 3 periods after the shock. The low unemployment volatility split bottoms out 9 quarters after the shock, with a decrease of roughly 0.55%. This is extremely interesting, as it is happening with higher hours worked and a similar unemployment rate to that seen in the high split.

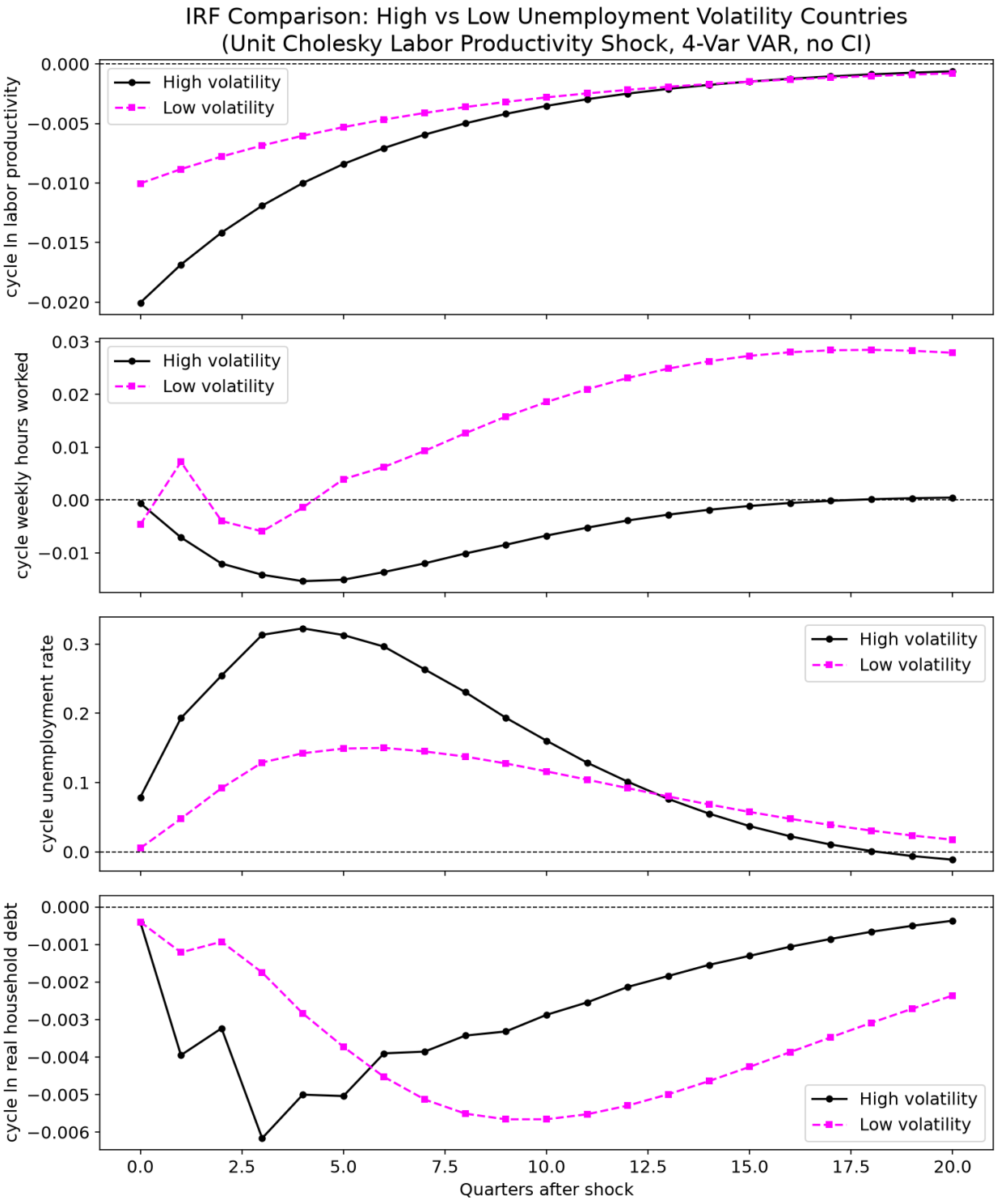


Figure 41: VAR High vs Low Unemployment Volatility Split

The next split of focus is the high vs low debt volatility splits. The countries with high volatility in real debt are listed with their standard deviation: Czechia (0.1540), Finland (0.0716),

Greece (0.0993), Hungary (0.1390), Italy (0.0648), Poland (0.1121), Portugal (0.0974), and Spain (0.1034). The countries with low volatility in real debt are listed with their standard deviation: Austria (0.0381), Belgium (0.0524), Denmark (0.0464), France (0.0299), Germany (0.0193), Ireland (0.0437), Norway (0.0188), and Sweden (0.0258).

Figure (42) presents the splits finding. This split specifically tends to tell a better story of what is happening and the underlying mechanics. First off, the labor productivity shock sizes are similar in magnitude, but the low debt volatility shock is slightly larger. The difference is that countries with high debt volatility have much longer persistence in their shocks. Weekly hours worked have a small response from both splits. The low-debt split initially drops but then increases to a positive response after the 5th period, following the shock. The high debt split starts positively in the first two periods, then drops slightly below the equilibrium. The unemployment response starts to paint the picture of why labor productivity persistence matters. The high debt volatility split has its unemployment rate increase by roughly 3x the magnitude of the low debt split, and one period later. The unemployment rate returns to its starting value 12 periods after the shock for the low-debt-volatility split and 20 quarters after the shock for the high debt volatility split. This mirrors what happens to the labor response when persistence of the shock increases in the HA model. The response to real debt differs significantly due to the split in real debt volatility. The response is fairly similar for both splits, starting at the shock point until 2 quarters after the shock. From here, the low debt volatility group has debt volatility increase to the starting value 5 periods after the shock and roughly stays there. The high debt split decreases further, to roughly a 0.7% decrease in leverage. Debt does not return to the steady state in the time frame and, in fact, is roughly half the maximum shock 20 quarters after the shock.

Recapping this finding, the shock sizes are fairly similar, but the high-debt variation split has much higher persistence. This doesn't show much change in hours worked, but we start to see large changes in unemployment and real household debt. This fits the story that firms face some kind of adjustment cost and countries that have higher persistence to their shocks tend to have their unemployment rate adjust by a larger magnitude and households within those countries make adjustments to their borrowing and saving because not only is the shock bigger, but they themselves see a higher and more persistent unemployment rate.

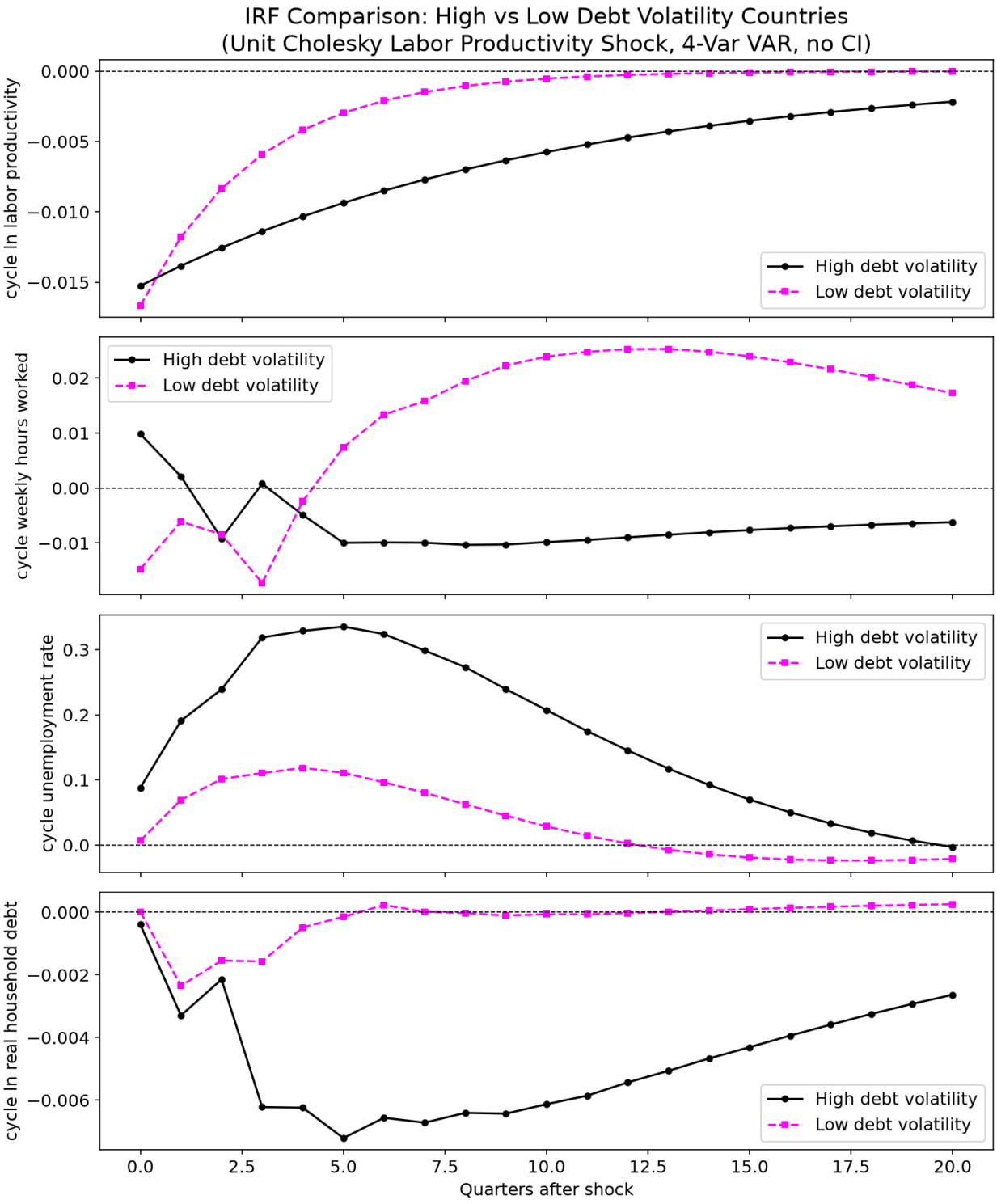


Figure 42: VAR High vs Low Debt Volatility Split

The countries were then split based on having high or low average household debt to GDP. High average household debt to GDP countries are listed with mean debt/gdp: Denmark

(107.8), France (51.6), Germany (56.9), Ireland (78.6), Norway (78.5), Portugal (74.3), Spain (65.6), and Sweden (71.6). The low debt to GDP countries include: Austria (50.1), Belgium (49.8), Czechia (21.5), Finland (50.5), Greece (45.9), Hungary (22.6), Italy (34.9), and Poland (26.7).

Figure (43). Impulse Response Analysis by average debt levels. The findings in this section support much of what was found for the split on high vs low debt volatility. First, labor productivity shocks are of comparable magnitudes across the two splits; however, high debt to gdp countries exhibit marginally larger initial impacts. Notably, high debt environments are characterized by greater persistence of productivity shocks, a critical observation made during the last split. In terms of weekly hours worked, responses remain relatively muted in both regimes. The high debt level countries' weekly hours initially contract, then recover to positive territory 4 periods after the initial shock. Conversely, the low debt split shows a rather persistent decrease in hours.

The unemployment response elucidates why labor productivity persistence warrants special attention. Low debt to GDP countries experience approximately double the increase unemployment relative to High debt counterparts, with peak effects occurring one year sooner and dropping back to the original unemployment level 5 periods after the shock. The low debt countries employment level never recovers 20 periods after the shock.

Real debt responses further illuminate regime-dependent outcomes following the debt level split. Both groups exhibit similar behavior during impact and two subsequent quarters before diverging. The lhigh debt group returns to baseline, maintaining stability thereafter. In contrast, the high-debt split experiences an additional leverage reduction to approximately -1%, with debt levels failing to revert within the observation window, persisting at roughly one-half of the maximum shock magnitude twenty quarters after impact.

These patterns replicate key features observed in the high debt volatility split. An interpretation of the debt level split is that countries with lower household debt to GDP have more persistent shocks. The more persistent the socks, the more volatility in labor productivity. Unemployment's movements are more volatile and persistent. This causes changes in household borrowing and saving to decrease leverage by a larger amount with persistence. This unemployment volatility that households feel is causing them to initially choose less leverage. Another way of saying this is that households in countries with greater volatility in labor productivity and unemployment not only experience larger changes in leverage but also choose to take on less debt. While this cannot be directly observed in HA model because the level of unemployment is fixed, by increasing the level of unemployment at the same output level and looking at how debt levels decrease to increased risk backs up these findings.

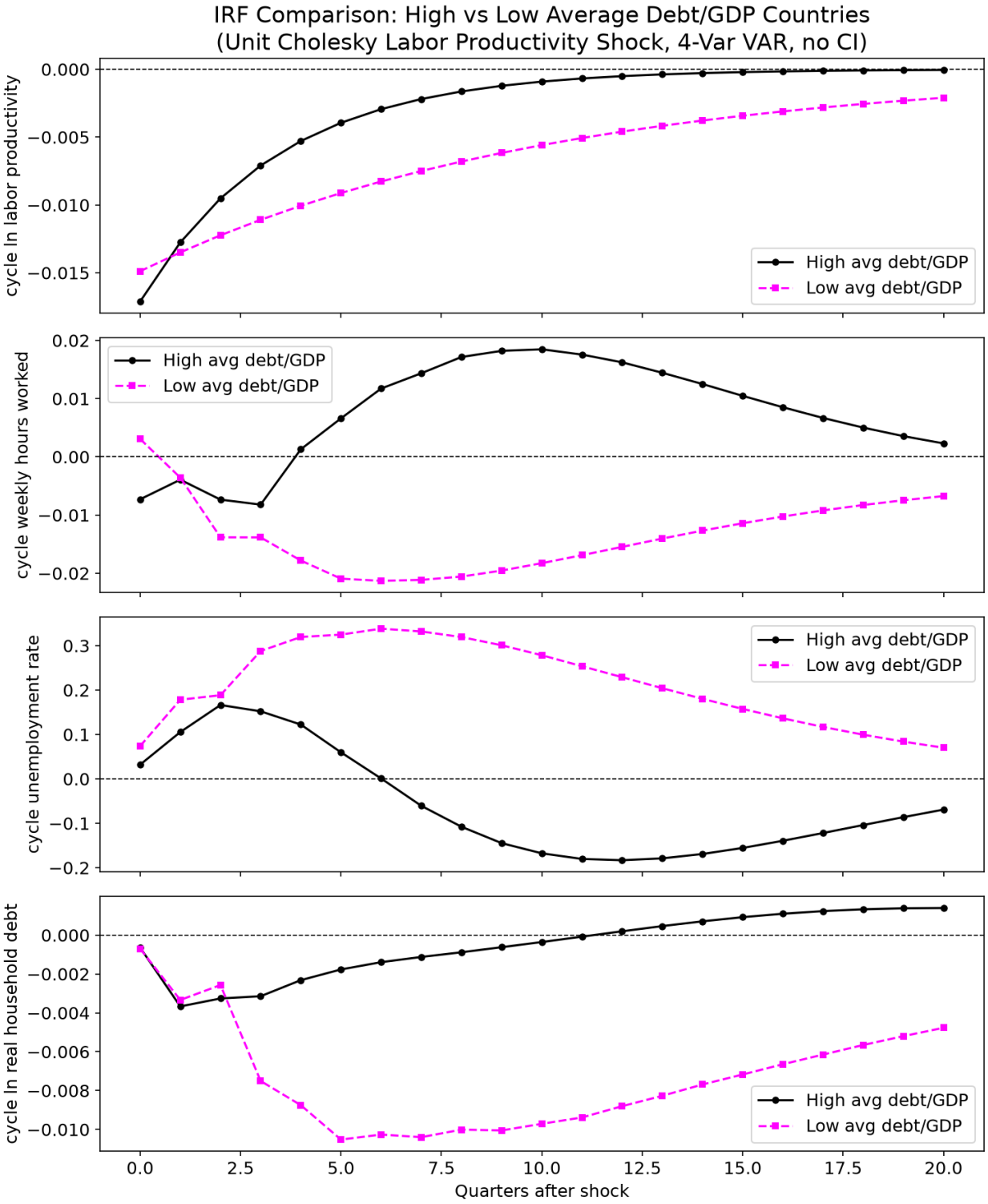


Figure 43: Var High vs Low Debt Level Split

7.5 Conclusion

Overall, the HA IRF does a decent job of capturing some of the dynamics. The aggregate VAR and LP show similar trends with a decrease in weekly hours worked, an increase in unemployment, and a decrease in real household debt. The key splits were high vs low real debt volatility and high vs low average household debt, which indicate that countries that split with a more persistent labor productivity shock experience larger and more prolonged increases in unemployment and a larger decrease in real household debt. These findings are significant and align with what occurs when the persistence of the labor productivity shock is increased in the HA model.

8 Conclusion

Most structural models rely on intensive margin adjustments (hours worked per agent) despite overwhelming empirical evidence for extensive-margin mechanisms. This paper resolves that gap by analyzing a three-period OLG model with exogenous youth unemployment and an endogenous Heterogeneous Agent Life cycle framework in which labor productivity drives transitions in employment status. Our comparative statics show that variations in the extensive margin, not just hours per worker, substantially reshape credit markets by lowering household debt levels. Impulse response function analysis of the Heterogeneous Agent (HA) model reveals that incorporating extensive margin adjustments yields economically meaningful deviations from conventional predictions. Specifically, variations in employment status yield distinct responses that lower both real interest rates and aggregate household debt. These dynamics are frequently overlooked by intensive margin focused models. Vector autoregressive and Local projection empirical tests validate these theoretical implications across countries, confirming that persistent labor productivity shocks generate larger, more prolonged increases in unemployment, leading to sharper and more prolonged declines in real household leverage.

Within the three period Overlapping Generations (OLG) framework, introducing youth unemployment exposes the loan demand transmission channel previously neglected by intensive margin models. Youth unemployment faces stricter borrowing constraints, as agents are less willing to lend to the unemployed. When the unemployment rate is changed exogenously, comparative statics show that the constraints depress aggregate loan demand, leading to lower real interest rates and a lower quantity of loans.

The Heterogeneous Agent framework enables a more nuanced characterization of household behavioral dynamics through several key extensions: age-dependent unemployment rate distributions, employment transition probabilities, dividend income, and precautionary savings motivations, all grounded in consumption savings decisions under employment uncertainty. Comparative

statics are exercises that exogenously reduce labor productivity. This yields two distinct findings. First, there is a decrease in loan demand similar to the 3 period OLG model. Second, loan supply increases among agents with around zero assets, who exhibit precautionary saving behavior to buffer against heightened unemployment risk. The HA model is further analyzed in terms of its impulse response to a negative productivity shock. The response makes it clear that changes in unemployment rates can cause persistent household deleveraging dynamics. These findings contrast with canonical intensive margin models, which fail to capture heterogeneity in borrowing constraints and the heightened precautionary-saving motive arising from incomplete markets.

The dynamic responses of weekly hours worked, unemployment rates, and real household debt to labor productivity shocks is empirically analyzed using VAR and Local Projection (LP) methods. European panel data are employed due to limited U.S. state-level observations for these variables; this cross-country variation enhances identification beyond the limitations of aggregate U.S. data. Countries are grouped to assess heterogeneity in responses across weekly hours worked, unemployment, real household debt, and the household debt-to-GDP ratio. These analyses provide a direct evaluation of HA model channels and their predictive validity by demonstrating that labor productivity shocks drive aggregate hours worked primarily by raising unemployment rates. Additionally, country-grouping analysis reveals that nations exhibiting more persistent productivity shocks experience larger increases in both the duration and persistence of unemployment, which subsequently induce greater household debt deleveraging

However, the model presented has certain limitations. In particular, it does not include capital. In reality, increased loan supply (used for self-insurance) would likely lead to higher investment and capital accumulation, raising output beyond what the current HA steady state predicts. Still, this is complicated by adjustment costs, as seen in two-asset HA models, which discourage the use of illiquid assets, such as capital, for consumption smoothing. As a result, households are more likely to save as a precaution in liquid assets and to reduce investment in illiquid assets when faced with income risk. Furthermore, the model does not account for shopping externalities, which could further exacerbate downturns. In a setting where consumers can accept a disutility of shopping effort to purchase goods at lower prices, higher unemployment might raise the wage-to-price ratio because the unemployed have lower income and more time (lower disutility) to search for lower prices. This would depress prices. It is important to note that this would increase unemployment and cause further deleveraging, as firms receiving lower prices for goods would reduce labor demand further. This shift could lower the capital share of income and reduce investment, thereby exacerbating output declines.

This paper establishes the empirical validity of modeling unemployment as an endogenous channel. The Heterogeneous Agent model with the extensive margin has substantive implications for the evolution of both real household debt and real interest rates.

References

Achdou, Yves, Jiequn Han, Jean-Michel Lasry, Pierre-Louis Lions, and Benjamin Moll. 2022. "Income and wealth distribution in macroeconomics: A continuous-time approach." *The review of economic studies* 89 (1): 45–86.

Aiyagari, S Rao. 1994. "Uninsured idiosyncratic risk and aggregate saving." *The Quarterly Journal of Economics* 109 (3): 659–684.

Attanasio, O, P Levell, H Low, and V Sánchez-Marcos. 2015. *Aggregating Elasticities: Intensive and Extensive Margins of Female Labour Supply. University College London.* Technical report. mimeo.

Auclert, Adrien. 2019. "Monetary policy and the redistribution channel." *American Economic Review* 109 (6): 2333–2367.

Bewley, Truman F. 1986. "Interest-bearing Money and the Equilibrium Stock of Capital." Unpublished manuscript, Yale University, written 1977–1986.

Challe, Edouard, Julien Matheron, Xavier Ragot, and Juan F Rubio-Ramirez. 2017. "Precautionary saving and aggregate demand." *Quantitative Economics* 8 (2): 435–478.

Chetty, Raj, Adam Guren, Day Manoli, and Andrea Weber. 2011. "Are micro and macro labor supply elasticities consistent? A review of evidence on the intensive and extensive margins." *American Economic Review* 101 (3): 471–475.

———. 2013. "Does indivisible labor explain the difference between micro and macro elasticities? A meta-analysis of extensive margin elasticities." *NBER macroeconomics Annual* 27 (1): 1–56.

Clarida, Richard, Jordi Gali, and Mark Gertler. 1999. "The science of monetary policy: a new Keynesian perspective." *Journal of economic literature* 37 (4): 1661–1707.

Deaton, Angus. 1991. "Saving and liquidity constraints." *Econometrica* 59 (5): 1221–1248.

Eggertsson, Gauti B, Neil R Mehrotra, and Jacob A Robbins. 2019. "A model of secular stagnation: Theory and quantitative evaluation." *American Economic Journal: Macroeconomics* 11 (1): 1–48.

Gagliardone, Luca, and Mark Gertler. 2023. "Oil prices, monetary policy and inflation surges."

Galí, Jordi, Frank Smets, and Rafael Wouters. 2012. "Unemployment in an estimated New Keynesian model." *NBER macroeconomics annual* 26 (1): 329–360.

Gourinchas, Pierre-Olivier, and Jonathan A Parker. 2002. "Consumption over the life cycle." *Econometrica* 70 (1): 47–89.

Guerrieri, Veronica, and Guido Lorenzoni. 2017. "Credit crises, precautionary savings, and the liquidity trap." *The Quarterly Journal of Economics* 132 (3): 1427–1467.

Hamilton, James D. 2018. "Why you should never use the Hodrick-Prescott filter." *Review of Economics and Statistics* 100 (5): 831–843.

Hansen, Gary D. 1985. "Indivisible labor and the business cycle." *Journal of monetary Economics* 16 (3): 309–327.

Haughwout, Andrew, Donghoon Lee, Daniel Mangrum, Joelle Scally, Wilbert Van der Klaauw, and Crystal Wang. 2024. "Delinquency Is Increasingly in the Cards for Maxed-Out Borrowers." *Liberty Street Economics.*

Heckman, James. 1984. "Comments on the Ashenfelter and Kydland papers." In *Carnegie-Rochester Conference Series on Public Policy,* 21:209–224. Elsevier.

Heckman, James J, and Thomas E MaCurdy. 1980. "A life cycle model of female labour supply." *The Review of Economic Studies* 47 (1): 47–74.

Huggett, Mark. 1993. "The risk-free rate in heterogeneous-agent incomplete-insurance economies." *Journal of economic Dynamics and Control* 17 (5-6): 953–969.

Imrohoroğlu, Ayşe. 1989. "Cost of business cycles with indivisibilities and liquidity constraints." *Journal of Political economy* 97 (6): 1364–1383.

Janko, Zuzana. 2008. "Nominal wage contracts, labor adjustment costs and the business cycle." *Review of Economic Dynamics* 11 (2): 434–448.

Jordà, Òscar. 2005. "Estimation and inference of impulse responses by local projections." *American economic review* 95 (1): 161–182.

Jordà, Òscar, Moritz Schularick, and Alan M Taylor. 2017. "Macrofinancial history and the new business cycle facts." *NBER macroeconomics annual* 31 (1): 213–263.

Kaplan, Greg, Kurt Mitman, and Giovanni L Violante. 2020. "The housing boom and bust: Model meets evidence." *Journal of Political Economy* 128 (9): 3285–3345.

Kaplan, Greg, and Giovanni L Violante. 2014. "A model of the consumption response to fiscal stimulus payments." *Econometrica* 82 (4): 1199–1239.

Kaplan, Greg, and Giovanni L Violante. 2018. "Microeconomic heterogeneity and macroeconomic shocks." *Journal of Economic Perspectives* 32 (3): 167–194.

Katz, Lawrence F. 1986. "Efficiency wage theories: A partial evaluation." *NBER macroeconomics annual* 1:235–276.

Krueger, Dirk, Kurt Mitman, and Fabrizio Perri. 2016. *On the distribution of the welfare losses of large recessions.* Technical report. National Bureau of Economic Research.

Krusell, Per, and Anthony A Smith Jr. 1998. "Income and wealth heterogeneity in the macroeconomy." *Journal of political Economy* 106 (5): 867–896.

Kydland, Finn E, and Edward C Prescott. 1982. "Time to build and aggregate fluctuations." *Econometrica: Journal of the Econometric Society,* 1345–1370.

Lechthaler, Wolfgang, and Dennis J Snower. 2013. "Quadratic labor adjustment costs, business cycle dynamics, and optimal monetary policy." *Macroeconomic Dynamics* 17 (2): 464–475.

McGrath, Maggie. 2016. "63% Of Americans Don't Have Enough Savings To Cover A $500 Emergency,"." *Forbes Magazine.*

McKay, Alisdair, and Ricardo Reis. 2016. "The role of automatic stabilizers in the US business cycle." *Econometrica* 84 (1): 141–194.

Meyer, Bruce D. 2002. "Labor supply at the extensive and intensive margins: The EITC, welfare, and hours worked." *American economic review* 92 (2): 373–379.

Quercia, Roberto G, George W McCarthy, and Susan M Wachter. 2003. "The impacts of affordable lending efforts on homeownership rates." *Journal of Housing Economics* 12 (1): 29–59.

Ravn, Morten O, and Vincent Sterk. 2017. "Job uncertainty and deep recessions." *Journal of Monetary Economics* 90:125–141.

Rendahl, Pontus. 2016. "Fiscal policy in an unemployment crisis." *The Review of Economic Studies* 83 (3): 1189–1224.

Rotemberg, Julio J, and Michael Woodford. 1997. "An optimization-based econometric framework for the evaluation of monetary policy." *NBER macroeconomics annual* 12:297–346.

Shimer, Robert. 2012. "Reassessing the ins and outs of unemployment." *Review of Economic Dynamics* 15 (2): 127–148.

Sims, Christopher A. 1980. "Macroeconomics and reality." *Econometrica: journal of the Econometric Society,* 1–48.

9 Appendix

9.1 Adding Labor Force Participation to Employment Variance Decomp

To address this question, I use data from Eurostat and the OECD from 2002 to 2019. Eurostat provides weekly hours worked, total hours worked within in country, and household debt. The OECD has data on nominal GDP and the unemployment rate. These datasets are well suited to the question because they offer consistent, cross-country data across European nations. The data will be used to decompose the variance in total hours worked into the percent of variance caused by the employment rate, weekly hours worked, labor force participation, and the covariances between terms. The data will also be useful for analyzing labor market fluctuations and their relationship to household leverage. The focus on Europe is deliberate. The presence of a shared central bank and common monetary policy, along with high capital mobility across EU countries, helps isolate the role of loan demand caused by fluctuations in employment rather than interest rate fluctuations in driving household debt-to-GDP dynamics. This setup allows for a clearer identification of how employment outcomes relate to household borrowing behavior in a coordinated financial environment.

The variance decomposition of total hours worked follows the methodology outlined in (Hansen 1985). To extract cyclical components, all series are detrended using the Hodrick-Prescott (HP) filter with a smoothing parameter of 100, which is appropriate for the annual frequency data. The detrended value is then calculated as the ratio of the detrended (cyclical) component to the trend, plus one, a formulation that facilitates interpretation in log-linear terms. The variances and covariances are computed using the natural log of the detrended value of each term. It is important to note that covariances can be negative if the underlying variables are negatively correlated. The variance decomposition following equation (1). T stands for total hours worked in an economy over the past year. E stands for the employment rate. LF is the labor force size. H stands for weekly hours worked. The decomposition is performed in an equal-weighted manner across all data points to construct aggregate measures. Additionally, a within-country variance decomposition is conducted for each country in the sample.

$$\sigma^2_{\ln T} = \sigma^2_{\ln E} + \sigma^2_{\ln LF} + \sigma^2_{\ln H} + 2\,\sigma_{\ln E,\ln LF} + 2\,\sigma_{\ln E,\ln H} + 2\,\sigma_{\ln H,\ln LF}.$$

To examine the relationship between labor market dynamics and household indebtedness, detrended employment and detrended household debt-to-GDP ratios were analyzed across European countries. A two-year lag in detrended employment was selected based on empirical fit, reflecting the realistic delay with which households respond to labor market conditions by adjusting their

borrowing behavior, both in leveraging during periods of high employment and deleveraging during periods of lower employment. The correlation is then conducted in an equal-weighted fashion across all available country-year observations to construct aggregate patterns. Within-country was assessed to capture the heterogeneity in the employment to household debt to GDP correlation within each country.

The aggregate variance decomposition was computed in an equal-weighted manner across all data points to ensure that no single country disproportionately influences the aggregate results. This provides a more balanced view of overall labor market dynamics within Europe, as different countries dealt with different idiosyncratic shocks. This approach allows for the construction of aggregate measures that reflect the central tendencies of the countries as a whole, rather than being skewed by larger economies such as Germany. Additionally, within-country decompositions were done to show important heterogeneity across nations, enabling the identification of distinct labor market patterns that may be masked in the aggregate. Together, these two levels of analysis reinforce the robustness of the findings by showing that the significance of the extensive margin is not driven by outliers and holds consistently across multiple contexts.

To strengthen my analysis of the relationship between household debt-to-GDP and employment, I first examine detrended employment and detrended household debt-to-GDP at the country level. It indicates that there is a correlation, and the choice of using lagged employment deviation. Specifically, a two-year lag aligns empirically and intuitively with the time it takes for households to adjust their borrowing behavior in response to labor market changes. Once the country-level patterns are established, an aggregate correlation is constructed by pooling all countries in an equal-weighted manner, ensuring no single economy skews the results. This dual approach within the country and in the aggregate strengthens the robustness of the findings by capturing both aggregate and within-country dynamics.

In Figure (44), the aggregate variance decomposition of total hours worked reveals that fluctuations in the employment rate are the primary driver of variation across countries at over 60 %. The second largest contributor is the variance in the labor force size, followed by the covariance between the employment rate and average weekly hours worked. The variance in weekly hours ranks fourth. This is significant because not only is the employment rate, extensive margin, variation larger than the intensive margin of hours worked, but so is the often-ignored deviation in the labor force size. The covariance between the employment rate and labor force participation contributes a smaller, though still positive, portion. Lastly, the covariance between weekly hours and the labor force is negative, reflecting the negative correlation between these two components. This analysis underscores the pivotal role of the extensive margin, driven by fluctuations in employment rates, in explaining variations in total hours worked across European countries.

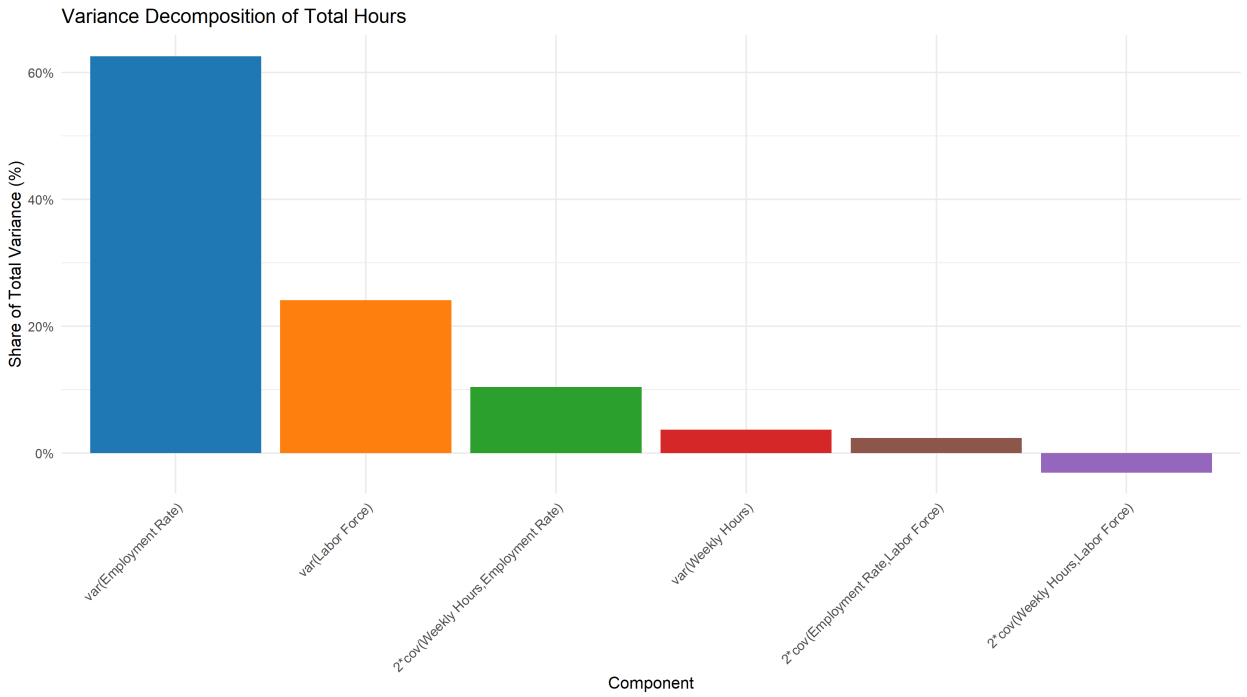


Figure 44: Aggregate Variance Decomposition

Austria, Denmark, France, Luxembourg, and Sweden stand out as exceptions to the broader trend observed across Europe, where the deviation in the employment rate is typically the dominant driver of total hours worked. Among these, Austria is unique in being primarily influenced by changes in weekly hours worked. This is due to Austria's relatively stable employment rate during the Great Recession, which only rose modestly from 4.4% in 2008 to 5.7% in 2009, and remained below 5.7% until 2014. This reflected a milder recessionary impact on the employment rate. Denmark, France, Luxembourg, and Sweden exhibit a variance in total hours worked that is predominantly driven by deviations in labor force size. These fluctuations stem from shifts in the native working-age population, migration patterns, and labor force participation rates. Despite these exceptions, the extensive margin of deviations in the employment rate was the primary contributor to labor market fluctuations in the majority of European countries, as illustrated in Figure (45).



Figure 45: Within Country Variance Decomposition

9.2 Heterogeneous Agent Alternative Specifications

Figure (46) shows the loan demand and loan supply if we add government debt (B gov) and government expenditures (G). Government Debt was set to 70% of GDP, and government expenditures were set to 18% of GDP; these are roughly in line with these values since the 1990s. The loan amount is roughly 1.4 because the debt is 0.7; that means household borrowing is roughly 0.7 household debt to GDP, and is much more in line with the actual data. As the interest rates rise, household borrowing is crowded out. The effects are roughly the same through both margins; they are just at higher levels of interest and total loan amount.

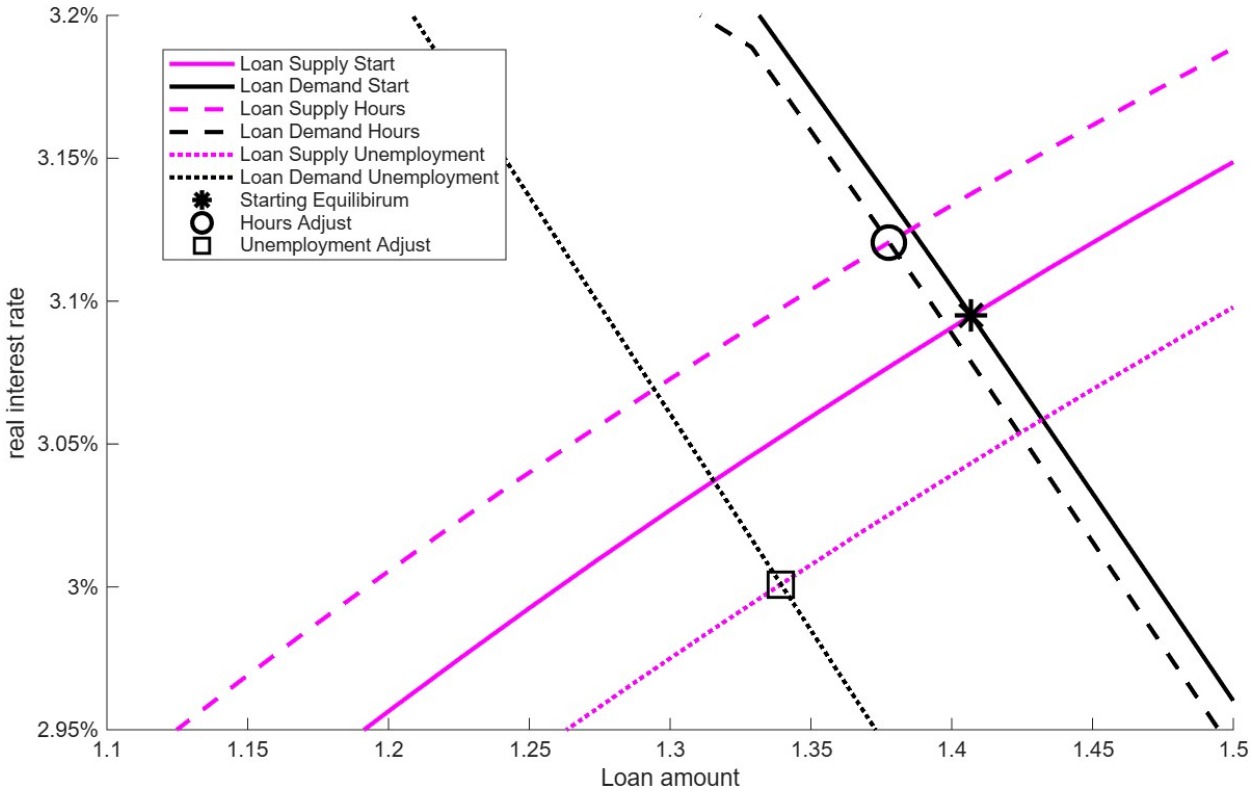


Figure 46: Het Loan Demand and Loan Supply with B gov and G

Figure (47) displays the intensive and extensive margin models with no smoothing. This produces very similar results to the smoothed version. The main difference is that the real interest rate jumps up and back down between 4 and 5 quarters after the shock. While this is slightly more erratic without smoothing, there are no significant differences between the unsmoothed and smoothed versions.

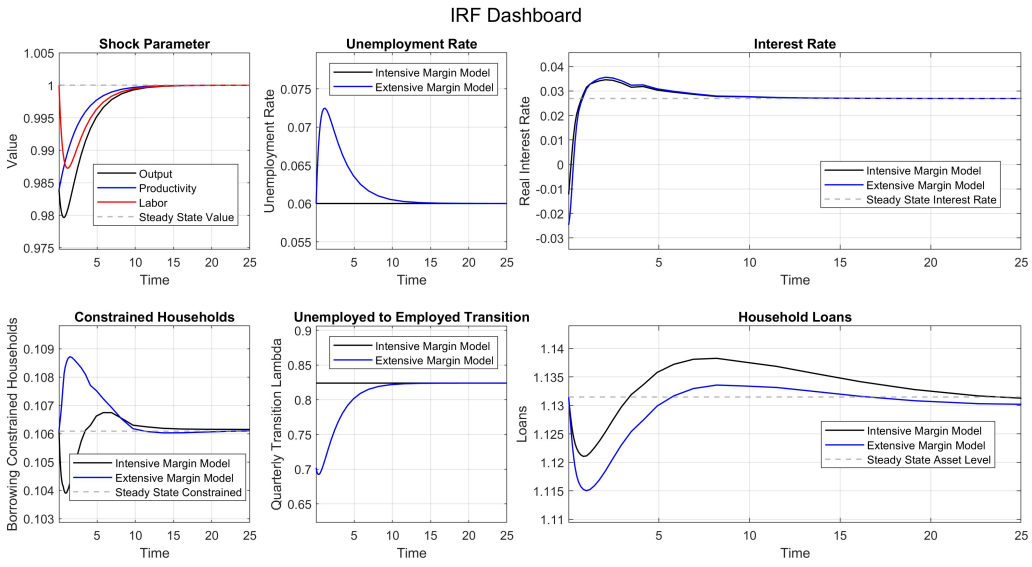


Figure 47: HA IRF with no smoothing

Figure (48) shows what would happen if you added government debt and deficits. One thing to note is that if the government has taken out debt, that means that the households are holding the debt and they have a higher asset level as a whole. A consequence of this is that the percent of constrained households is roughly 6.9% vs the 10.6 % when there is no government debt and government expenditures. Government debt was set to .7 or 70% of GDP. Government purchases of goods and services were set to 0.18, or 18%, of GDP. This is separate from interest payments and Unemployment/Social Security. Now the government market clearing condition is given by equation (54), which states that the revenue equals unemployment and social security transfers, plus interest payments, plus government expenditures, plus change in government debt. Government revenue will be increased by adjusting the tax rate. The change in government debt is governed by eq (55). The change in debt is interest expense, unemployment transfer payments, and government purchases of goods and services, less tax revenue that would be generated with the equilibrium interest rate, less the amount of tax revenue that would be collected if the tax rate was still the equilibrium rate, less a factor times the value of current government debt, less the equilibrium value. It is important to note that in order for the model to converge ϕ_g must be larger than the interest rate.

$$revIncome = expTransf + Bgov_t * r_t + G_t + B\dot{gov}_t \tag{54}$$

$$B_dot = Bgov_t * r_t + expTransf + G_t - \sum_e \int_l \int_a \tau_{equm}(w_t(z_l h_{e,t}) + \pi_{l,t}) - \phi_g * (Bgov_t - Bgov_{equm}) \tag{55}$$

The IRF response shows a few things. First, the interest rate shock is not immediate and, in fact, takes longer to decrease before increasing and overshooting the equilibrium rate. It is important to note that when the rate is lower, government interest payments decrease, and when the rate is higher, they increase. Constrained households increase similarly under the extensive margin model, but then decrease and remain lower for a much longer time. Household debt decreases more slowly, but there is greater deleveraging because the process takes longer. From here, household debt takes longer to return to its starting level. It is important to note that a factor that contributes to this is that more lending flows to the government, which increases interest rates and causes a crowding out of household borrowing.

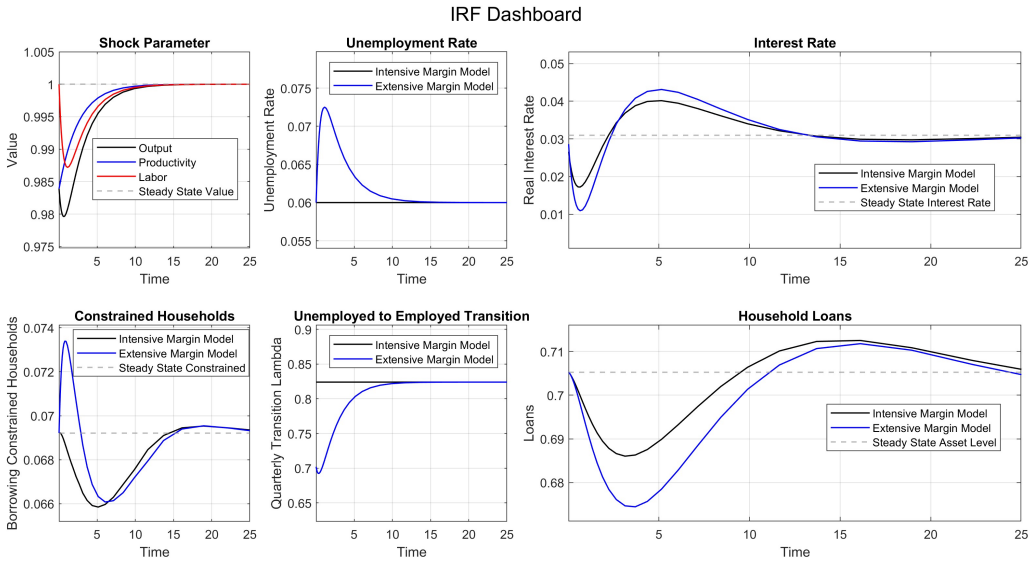


Figure 48: HA IRF with G Bgov and Deficits

9.3 Empirical IRF Additional Specifications

Figure (49) shows aggregate VAR for all of the countries with the Hamilton filter set to two. As you can see, the overall shapes are similar for the following: Labor productivity is a standard AR(1), weekly hours worked show no significant change, and unemployment increases by a statistically significant amount. The main difference here is that real household debt now increases over the first 3 periods before decreasing and returning to zero. There is now no significance to the real

in any period. While this is not promising, this went somewhat as expected, as (Jordà, Schularick, and Taylor 2017) argued for looking ahead 5 years when examining debt cycles.

Figure (50) shows the LP with the Hamilton filter set to 2 years. The two year filter gives a rather similar result. The productivity shock follows the same linear approach back to normal: there is no significant movement in weekly hours worked, unemployment increases significantly before returning to normal, and real household debt starts at an insignificant value before decreasing linearly. It is important to note that the decrease in household debt is statistically significant only 8 quarters after the shock. Overall, both VAR and LP with the 2 year Hamilton filter show similar results for labor productivity shock, weekly hours worked, and the unemployment rate. The main difference is that the decrease in household debt is no longer statistically significant, as other papers have pointed out the need to use a longer filter period when examining debt leverage.

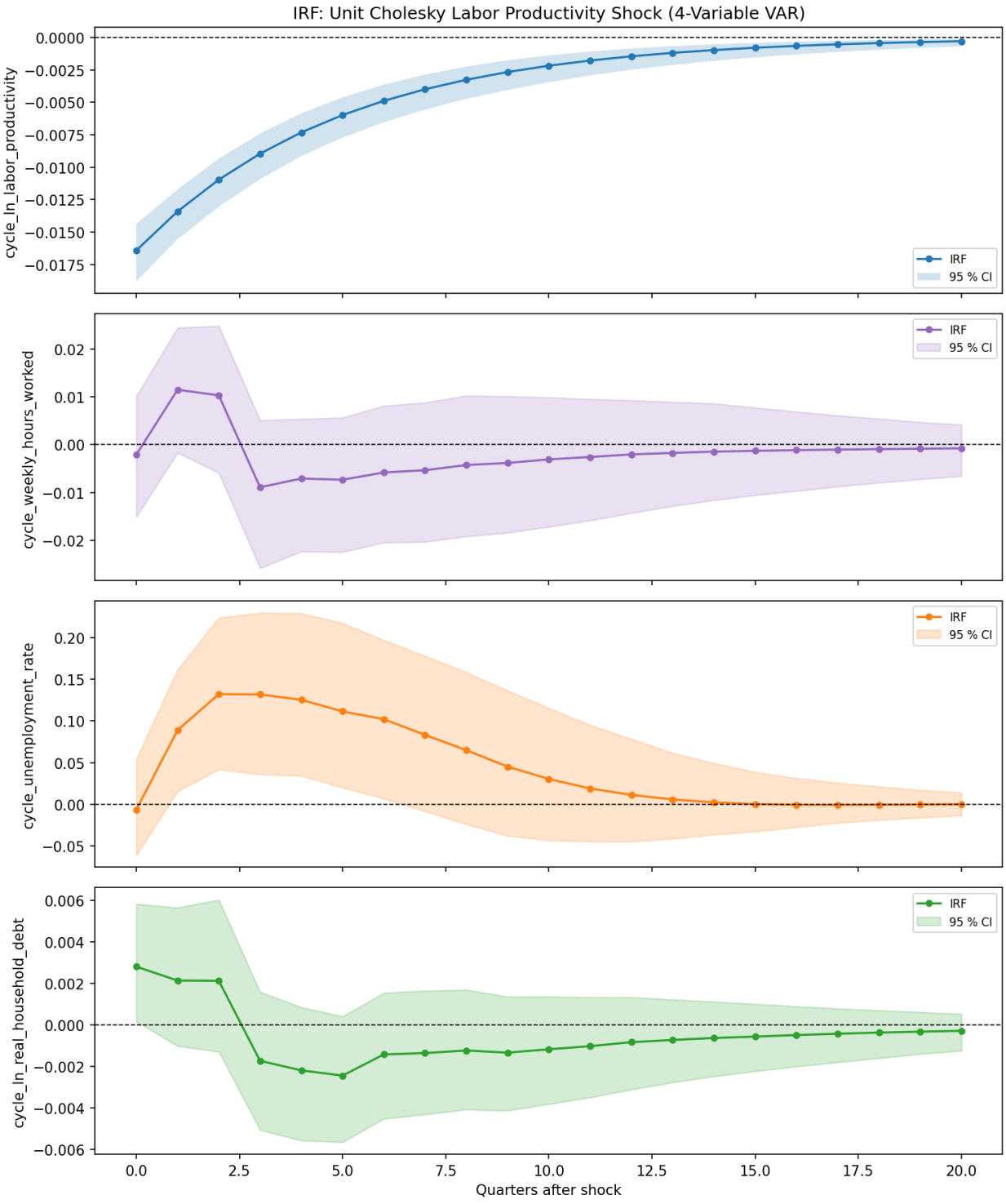


Figure 49: VAR All Countries 2 Year filter

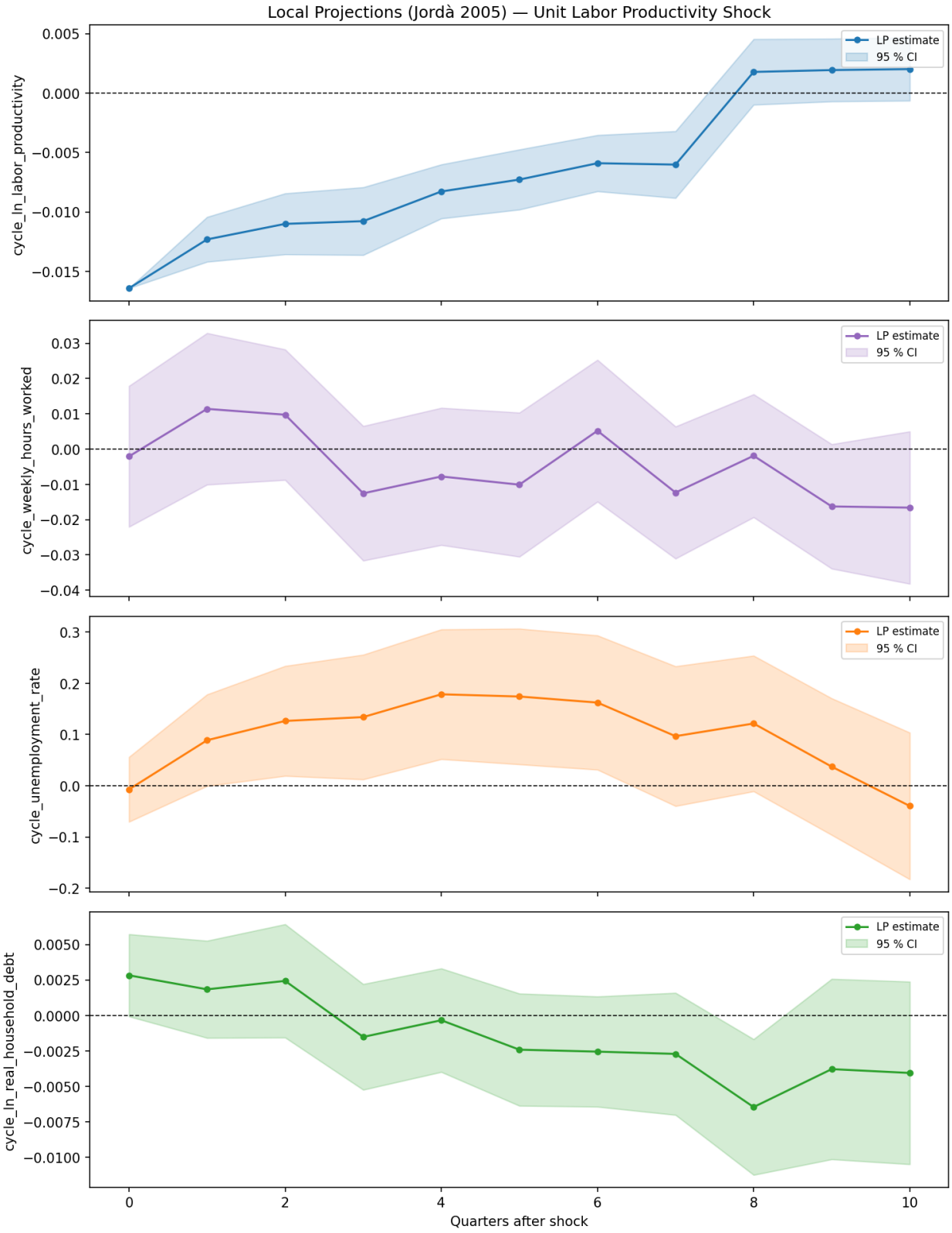


Figure 50: LP All Countries 2 Year filter

Figure (51) displays a VAR with the only difference being that the 4th variable is now the cyclical component of nominal household debt. Figure (52) displays a LP with the only difference being that the 4th variable is now the cyclical component of nominal household debt. In both these figures, there is little to no difference for all four variables. This was expected for the first three variables, as none of these were affected, and the fourth debt variable was somewhat unknown. The fact that the impulse response of nominal debt closely tracks that of real household debt suggests that productivity shocks operate through the extensive margin of employment, reducing labor demand, rather than via price effects. This distinction is more nuanced. (Gagliardone and Gertler 2023) shows that oil shocks can simultaneously affect employment and prices when oil is a complementary good for both household consumption and production. The fact that nominal debt responds as real debt implies that the labor productivity shock operates primarily through labor demand rather than through price changes. This is consistent with a productivity shock weakening credit channels via changes in employment status, rather than through inflationary price adjustments.

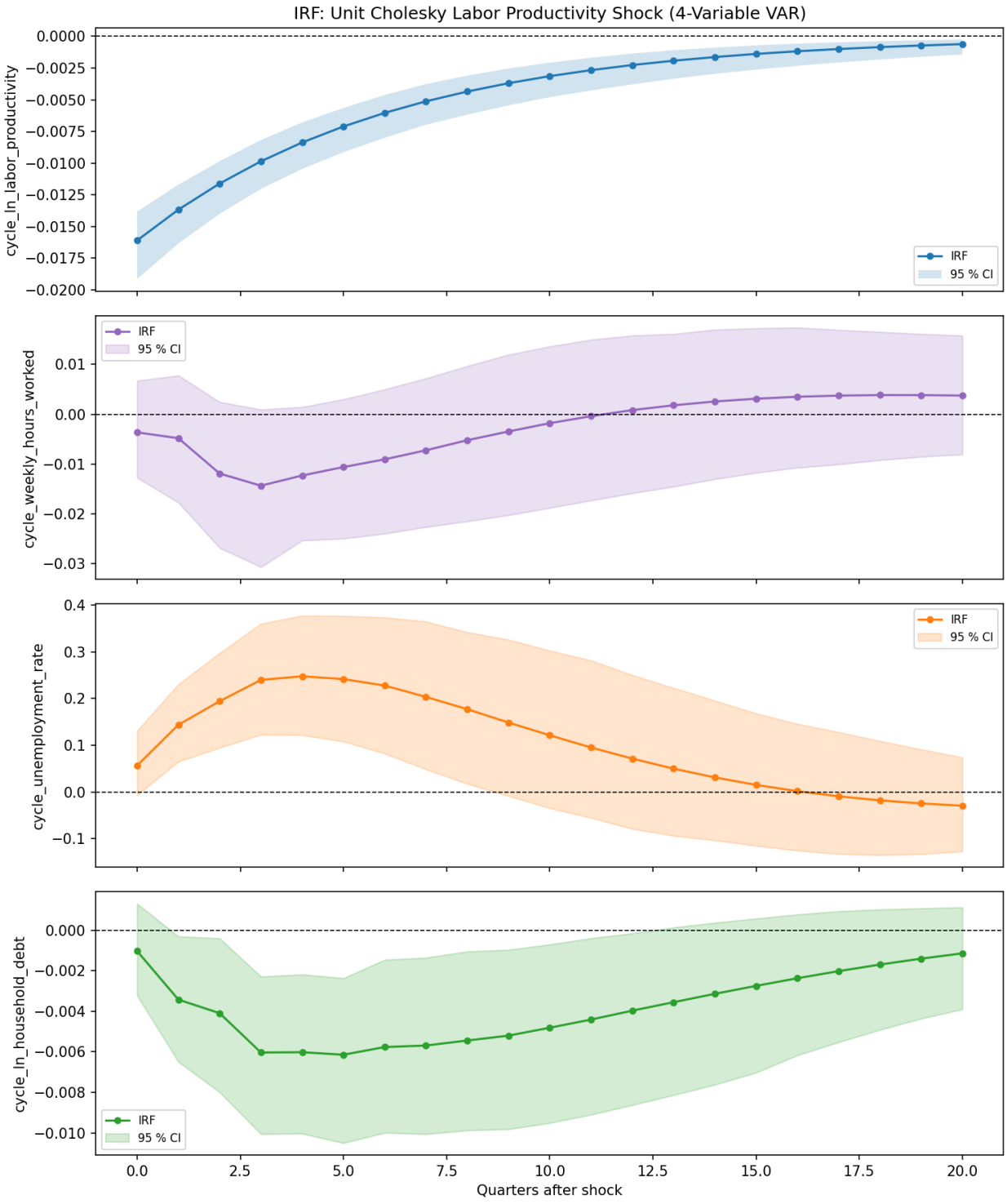


Figure 51: VAR All Countries Nominal Debt

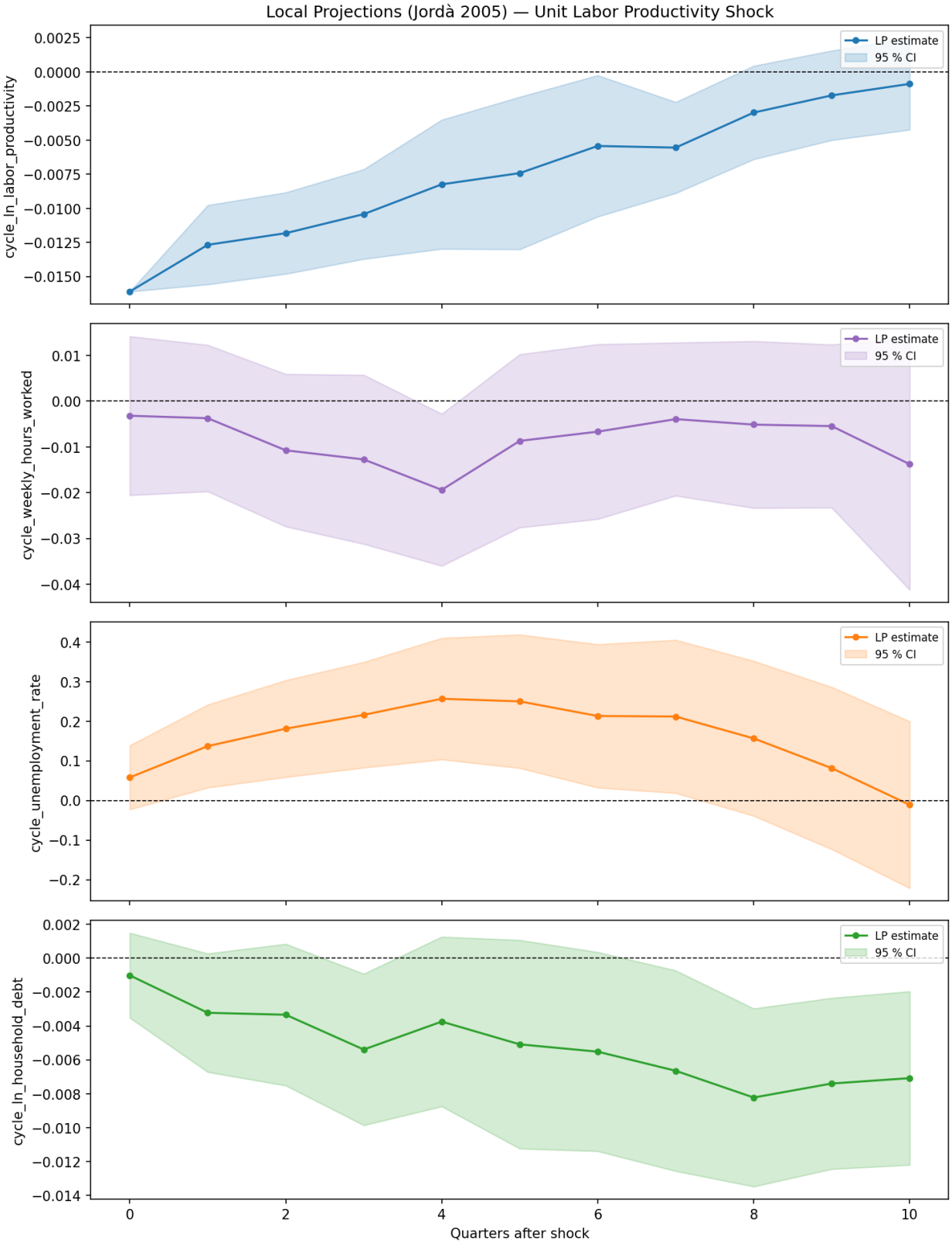


Figure 52: LP All Countries Nominal Debt

Figure (53) displays a VAR with the only difference being that the 4th variable is now the household debt to GDP. Figure (54) displays a LP with the only difference being that the 4th variable is now the household debt to GDP. The labor productivity, weekly hours worked, and unemployment show little to no difference, as expected. The difference is that the fourth variable, cycle household loans to GDP, increases right after the shock and decreases in both around 6 to 7 quarters after the shock. This makes sense, as the labor productivity shock reduces GDP more quickly than real household debt. This causes the initial increase in household debt to GDP, but as time goes on and labor productivity increases, so does GDP. This causes GDP to return to a value similar to the pre-shock level, and the decrease in household debt is evident in the nominal value of household debt, which is now roughly what it was at the start.

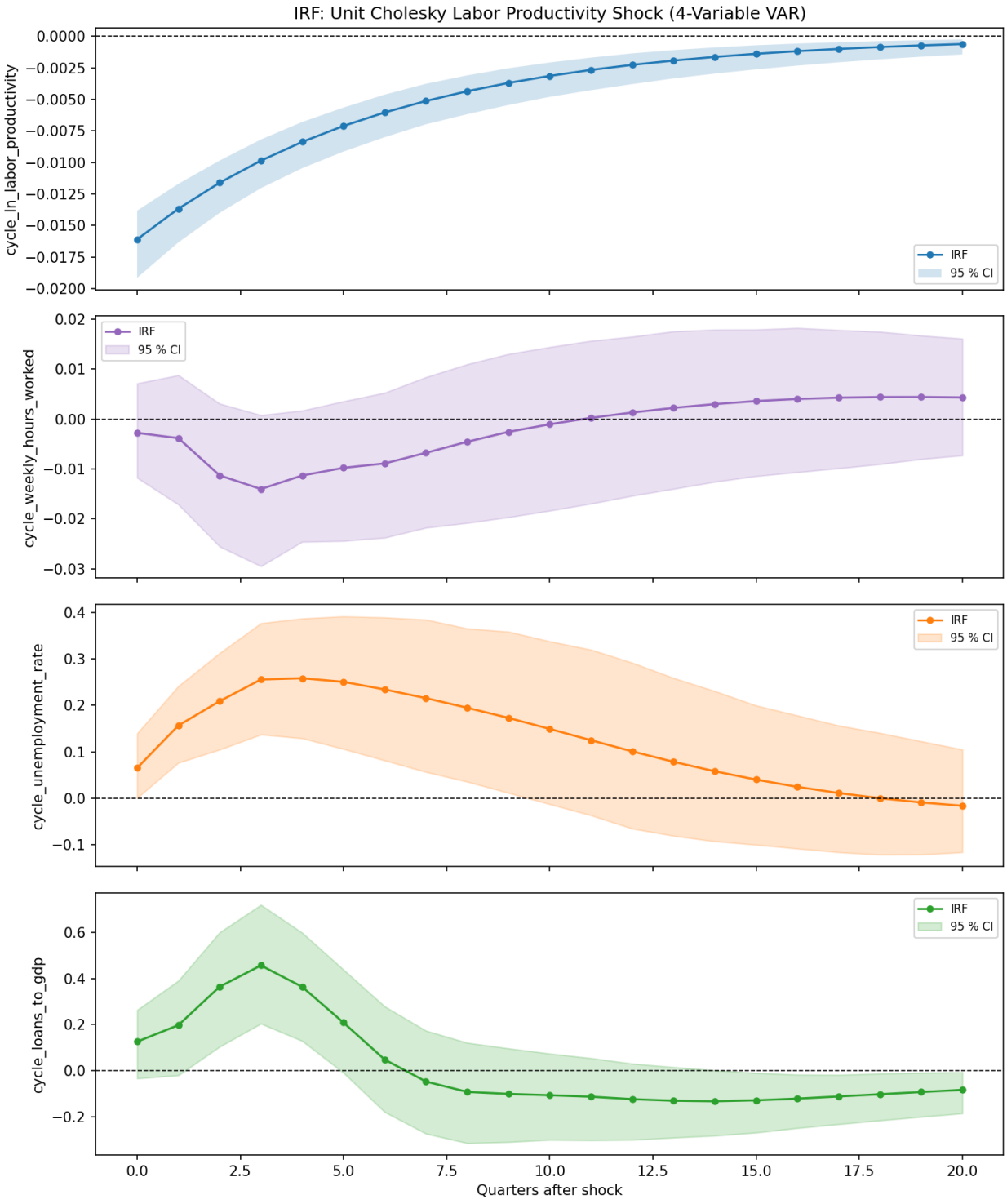


Figure 53: VAR All Countries Debt to GDP

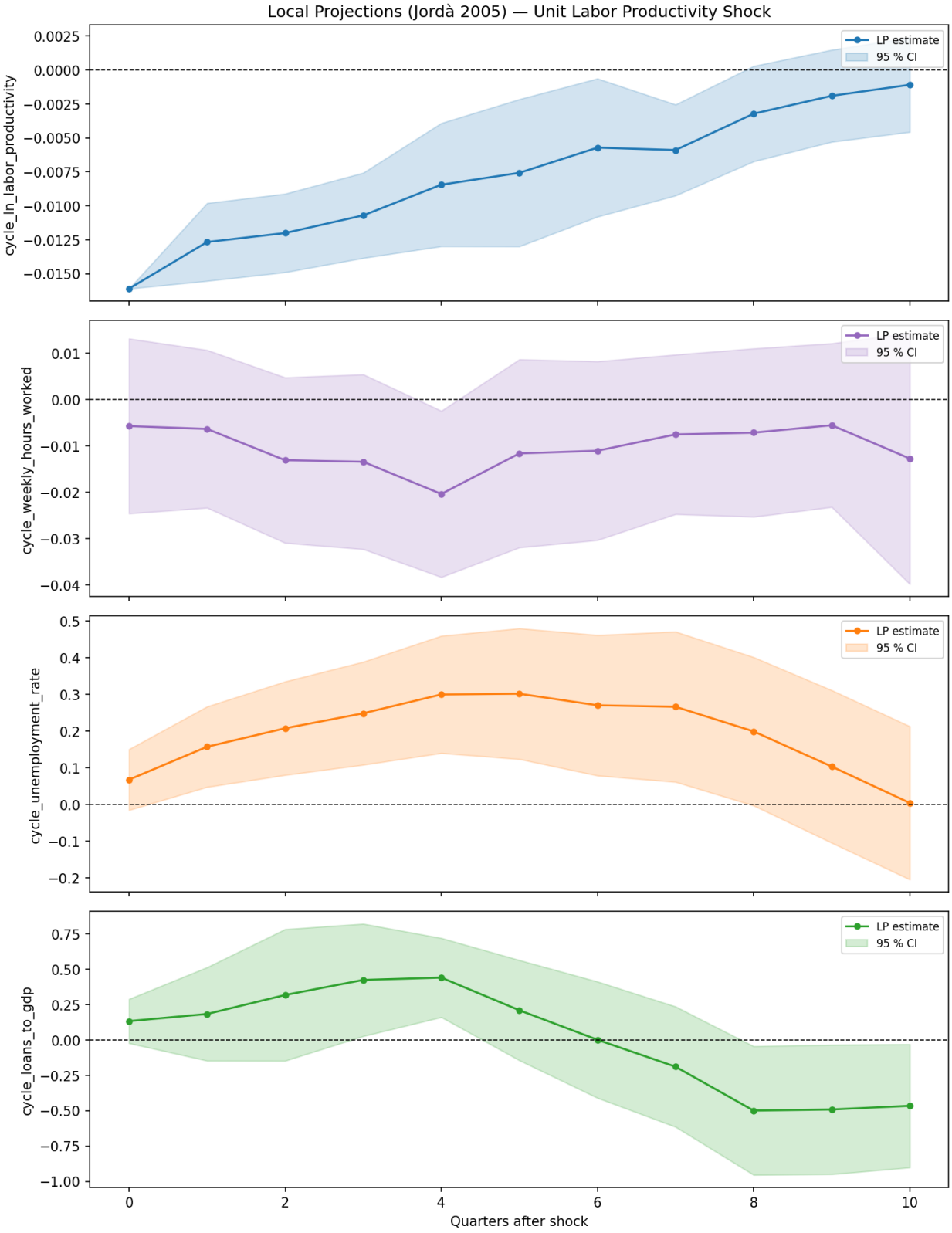


Figure 54: LP All Countries Debt to GDP